

UNIVERSITÀ DEGLI STUDI DI TORINO

Corso di Laurea Magistrale in Matematica



UNIVERSITÀ
DI TORINO

Tesi di Laurea Magistrale

Ideally Exact Categories

Beyond semiabelian categories,
a categorical approach to ideals

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Anno Accademico 2024-2025

Abstract

This thesis investigates non-pointed counterpart of the notion of semiabelian category. Rather than simply dropping pointedness, we aim to keep some of the categorical structure inherent to semiabelian categories. This leads us to the definition of ideally exact categories. This notion captures the categorical structure of unital rings. Indeed, at first glance, ideals of a unital ring seem somewhat unnatural in a categorical sense, since they are non-unital rings. This perspective allows for a deeper understanding of ideals and highlights how categorical tools can clarify and generalize classical algebraic notions. This study employs several categorical tools, such as the pullback functor, lifting properties in the Eilenberg-Moore category and effective descent morphisms.

Preface

In this thesis I explore the notion of ideally exact category, introduced by George Janelidze in 2024. The main goal is to provide an overview of the ideally exact categories with the necessary background needed to understand it, and present some of the most important results in this field.

A basic knowledge of category theory — such as limits, colimits, adjunctions, and monads — is assumed throughout this thesis. Some familiarity with Malt'sev, exact, regular, protomodular and semiabelian categories would provide a natural foundation; however the main constructions and definitions of these categories are reviewed in the first chapter. The reader who find it helpful, can also refer to the appendix, where some results about these categories — necessary for proving several results in the thesis — are presented.

Right after the preliminaries, the first chapter introduces the motivating example of this theory, which is the category of rings with unit. This example brings us to ideally exact categories, which capture the essential categorical features of ideals and rings.

The second chapter presents the pullback functor and its left adjoint, which are a key tool in the theory of ideally exact categories. In this chapter I also define the connected limits, since they have useful properties with the mentioned adjunction. Then, I will introduce the notion of nullary and essentially nullary monads and their results, which will be used to prove the main result of the ideally exact categories.

The third chapter is devoted to some results concerning the lifting of properties to the Eilenberg-Moore category of a monad. This part is quite technical and articulate. It will be useful to characterize the notion of ideal exact category.

The fourth chapter is dedicated to the study a monadicity theorem for the pullback functor in an exact category.

The fifth deals with an overview of the notion of ideal exact category, which is the main topic of this thesis. I will present the definition and the characterization of this notion, and some of its properties. Then, I will show the main theorem. This result has been central in motivating interest in this

topic.

In the sixth chapter I recall the notion of topos and its dual, a cotopos. Then, I will show that this last is an example of ideally exact category. Finally, I will analyze the more concrete example of the category of dual of sets and try to understand how the ideals work in this case.

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Preliminaries

In order to understand the notions introduced in this thesis is important to recall some definitions and constructions. We also take this opportunity to introduce and fix some notations.

First we define some categories that will be used throughout the thesis. Let $\mathcal{A}$ be a category, we defines the following categories.

- The **slice category**: given an object B in the category $\mathcal{A}$, we call $(\mathcal{A} \downarrow B)$ the category of objects over E , we denote an object by $(A, \alpha : A \rightarrow B)$. A morphism between two objects (A, α) and (A', α') is a morphism $f : A \rightarrow A'$ such that $\alpha' f = \alpha$.

$$\begin{array}{ccc} A & & A \xrightarrow{f} A' \\ \downarrow \alpha & & \downarrow \alpha \swarrow \alpha' \\ B & & B \end{array}$$

- The **coslice category**: given an object B in the category $\mathcal{A}$, we call $(B \downarrow \mathcal{A})$ the category of objects under B , we denote an object by $(\alpha : B \rightarrow A, A)$. A morphism between two objects (α, A) and (α', A') is a morphism $f : A \rightarrow A'$ such that $f\alpha = \alpha'$.

$$\begin{array}{ccc} B & & B \\ \downarrow \alpha & & \downarrow \alpha \searrow \alpha' \\ A & & A \xrightarrow{f} A' \end{array}$$

- The **category of points**: given an object B in the category $\mathcal{A}$, we call $\text{Pt}(B)$ the category of points of B , we denote an object by (A, B, p, s) which is a diagram of the type $A \begin{smallmatrix} \xrightarrow{p} \\ \xleftarrow{s} \end{smallmatrix} B$ with the condition $ps = 1_B$. Morphisms between two objects (A, B, p, s) and (A', B', p', s') are diagrams

of the type

$$\begin{array}{ccc} A & \xrightarrow{f} & A' \\ \swarrow p & & \nwarrow s' \\ & B & \end{array} \quad \begin{array}{ccc} & \nearrow s & \\ & \nwarrow p' & \end{array}$$

with the conditions $p'f = p$ and $fs = s'$

We recall the following definitions.

Definition 0.0.1. A finitely complete category $\mathcal{A}$ is said to be **regular** if

- Every morphism in $\mathcal{A}$ admits an (regular epi, mono) factorization;
- Regular epis in $\mathcal{A}$ are pullback stable.

Definition 0.0.2. A category $\mathcal{A}$ is said to be **exact** if

- It is regular;
- Any equivalence relation is the kernel pair of some arrow.

In order to define the next notion, we need the following construction. Suppose $\mathcal{A}$ has pullbacks and consider a morphism $f : B' \rightarrow B$ in the same category. We define the **change-of-base functor** of f .

$$f^* : \text{Pt}(B) \rightarrow \text{Pt}(B')$$

$$\begin{array}{ccc} \begin{array}{c} A \\ \uparrow s \downarrow p \\ B \end{array} & \xrightarrow{f^*} & \begin{array}{ccccc} & & B' & & \\ & & \searrow \langle 1_{B'}, sf \rangle & & \nearrow sf \\ & & B' \times_B A & \xrightarrow{\text{proj}_2} & A \\ & \nearrow 1_{B'} & \downarrow \text{proj}_1 & & \uparrow s \downarrow p \\ & B' & \xrightarrow{f} & B \end{array} \end{array}$$

where $\langle 1_{B'}, sf \rangle$ is the section of proj_1 . It holds that $(\text{proj}_1)\langle 1_{B'}, sf \rangle = 1_{B'}$. The action on a morphism $\alpha : A \rightarrow A'$ is given by

The diagram illustrates the relationship between a morphism $\alpha: A \rightarrow A'$ and its pullback $f^*(\alpha): B' \times_B A \rightarrow B' \times_B A'$ via the change-of-base functor $f^*: \text{Pt}(B) \rightarrow \text{Pt}(A)$. On the left, a square represents the pullback of α along a morphism $f: B \rightarrow B'$. The square has vertices A (top-left), A' (top-right), B (bottom-left), and B' (bottom-right). The edges are α (top), f (bottom), p (left), and p' (right). A diagonal arrow s goes from B to A , and another diagonal arrow s' goes from B' to A' . The pullback square is labeled f^* . On the right, a larger diagram shows the pullback of the pullback square. The top row is $B' \times_B A \xrightarrow{\text{proj}_2} A \xrightarrow{\alpha} A' \xleftarrow{\text{proj}'_2} B' \times_B A'$. The bottom row is $B' \times_B A \xrightarrow{\langle 1_{B'}, sf \rangle} B' \xleftarrow{\langle 1_{B'}, s'f \rangle} B' \times_B A'$. The middle row is $A \xrightarrow{p} B \xrightarrow{p'} A'$. The edges are labeled proj_1 , proj'_1 , proj_2 , proj'_2 , p , p' , s , s' , and f . A dashed arrow $f^*(\alpha)$ connects the top-left and top-right vertices of the pullback square.

Definition 0.0.3. A finitely complete category $\mathcal{A}$ is said to be **protomodular** if for every morphism $f: A \rightarrow B$ in $\mathcal{A}$, the change-of-base functor $f^*: \text{Pt}(B) \rightarrow \text{Pt}(A)$ is conservative.

We conclude recalling the following.

Definition 0.0.4. A category finitely complete category $\mathcal{A}$ is **Mal'tsev** if every internal reflexive relation in $\mathcal{A}$ is an equivalence relation.

Definition 0.0.5. A category $\mathcal{A}$ is said to be **semiabelian** if

- It is pointed;
- It is exact;
- It is protomodular;
- It has binary coproducts.

Chapter 1

The categorical structure of Rings

In this chapter, we aim to introduce a motivating example for the notions of ideally exact categories. This is based on an article by George Janelidze [Jan24].

Consider the category $\mathbf{Ring}$ of rings with unit. The notion of ideal is, at first glance, somewhat unnatural in the context of categories. In fact, in general, an ideal of a ring with unit is not an object in the category $\mathbf{Ring}$: if an ideal contains the unit, then it coincides with the whole ring. However, we observe that it is an object of the category $\mathbf{Ring}_*$ of rings without unit.

It is a standard fact that $\mathbf{Ring}$ and $\mathbf{Ring}_*$ are finitely complete, protomodular and exact categories. They both have binary coproducts, even if they are not the same. The coproduct in the category $\mathbf{Ring}_*$ is the free product, while in $\mathbf{Ring}$ it is the free product with amalgamation, in order to ensure that there is a unit in the coproduct.

That said, they have a crucial difference: $\mathbf{Ring}$ is not pointed, whereas $\mathbf{Ring}_*$ is. In fact, in $\mathbf{Ring}_*$ the initial and terminal objects are both the zero ring 0 , while in $\mathbf{Ring}$ the initial object is the zero ring $0 = \mathbb{Z}$ of integers, but the terminal object is the ring $1 = \{0\}$. Thus, we conclude the $\mathbf{Ring}_*$ is semiabelian, while $\mathbf{Ring}$ fails to be.

So we have the forgetful functor that forgets the unit

$$U : \mathbf{Ring} \rightarrow \mathbf{Ring}_*$$

with $\mathbf{Ring}$ protomodular and exact, and $\mathbf{Ring}_*$. This functor has a left adjoint: the unitalization. This functor, also known as the Dorroh extension, is given by

$$\begin{aligned} F : \mathbf{Ring}_* &\rightarrow \mathbf{Ring} \\ R &\mapsto \mathbb{Z} \oplus R \end{aligned}$$

where the sum $\mathbb{Z} \oplus R$ is the sum by components, while the product $\cdot$ is defined as follows

$$\begin{aligned} \cdot : (\mathbb{Z} \oplus R) \times (\mathbb{Z} \oplus R) &\rightarrow \mathbb{Z} \oplus R \\ ((m, r), (n, s)) &\mapsto (mn, rs + ms + nr) \end{aligned}$$

and with unit of $(\mathbb{Z} \oplus R)$ given by $(1, 0)$.

The unit and the counit of the adjunction are given by

$$\begin{aligned} \eta_R : R &\rightarrow UF(R) = \mathbb{Z} \oplus R \\ r &\mapsto (0, r) \end{aligned}$$

$$\begin{aligned} \epsilon_R : FU(R) = \mathbb{Z} \oplus R &\rightarrow R \\ (n, r) &\mapsto (n1_R + r) \end{aligned}$$

The functor U turns out to be monadic, simply by Beck's theorem, since coequalizers in $\mathbf{Ring}$ and $\mathbf{Ring}_*$ are the same. In fact, given f, g morphisms, the coequalizer of (f, g) is the canonical projection

$$\pi : B \rightarrow B/I \quad I = (f(r) - g(r) \mid r \in R)$$

This adjunction will lead us to the definition of the ideally exact categories. We will see that the definition involves a generic monadic functor in a semiabelian category. Sometimes, as we will see, such functor is, up to category equivalences, the pullback functor of $! : 0 \rightarrow 1$. That's the case for the functor U above. As we will see, this is intrinsically due to the fact that the unit η of the adjunction is cartesian. In fact, one can easily prove that, given a morphism $f : R \rightarrow R'$ in $\mathbf{Ring}_*$, the pullback in $\mathbf{Ring}_*$ of $\eta_{R'}$ and $UF(f)$

$$\begin{aligned} R' \times_{UF(R')} UF(R) &= R' \times_{\mathbb{Z} \oplus R'} (\mathbb{Z} \oplus R) = \\ &= \{(r', (n, r)) \in R' \times (\mathbb{Z} \oplus R) \mid UF(f)(x) = \eta_{R'}(y)\} \\ &= \{(r', (n, r)) \in R' \times (\mathbb{Z} \oplus R) \mid n = 0 \wedge f(r) = r'\} \\ &= \{(f(r), (0, r)) \mid r \in R\} \end{aligned}$$

is isomorphic to R .

$$\begin{array}{ccccc} & & & \text{proj}_2 & \\ & & & \curvearrowright & \\ R' \times_{\mathbb{Z} \oplus R'} (\mathbb{Z} \oplus R) & & & & \mathbb{Z} \oplus R \\ & \searrow \text{dashed} & & & \downarrow UF(f) \\ & & R & \xrightarrow{\eta_R} & \mathbb{Z} \oplus R \\ & & \downarrow f & & \downarrow UF(f) \\ & & R' & \xrightarrow{\eta_{R'}} & \mathbb{Z} \oplus R' \\ & \swarrow \text{proj}_1 & & & \end{array}$$

We now want to deep into the mentioned category equivalence. First we consider the pullback functor of $! : 0 \rightarrow 1$

$$!^* : (\mathbf{Ring} \downarrow 1) \rightarrow (\mathbf{Ring} \downarrow 0)$$

Let R be an object in $\mathbf{Ring}$, then $!^*(R) = (0 \times_1 R, \text{proj}_1)$

$$\begin{array}{ccc} 0 \times_1 R & \xrightarrow{\text{proj}_2} & R \\ \downarrow \text{proj}_1 & & \downarrow !^R \\ 0 & \xrightarrow{!} & 1 \end{array} \quad (1.1)$$

Observe that

$$(\mathbf{Ring} \downarrow 1) \cong \mathbf{Ring}$$

in fact, we can always provide an object in $\mathbf{Ring}$ with the unique morphism to 1. Furthermore

$$(\mathbf{Ring} \downarrow 0) = (\mathbf{Ring} \downarrow \mathbb{Z}) \cong \mathbf{Ring}_*$$

where the equivalence of category is given by the unitalization functor

$$\begin{aligned} (F, \pi_{\mathbb{Z}}) : \mathbf{Ring}_* &\rightarrow \mathbf{Ring} \downarrow \mathbb{Z} \\ R &\mapsto (\mathbb{Z} \oplus R, \pi_{\mathbb{Z}} : \mathbb{Z} \oplus R \rightarrow \mathbb{Z}) \end{aligned}$$

and its quasi inverse Ker^1 .

$$\begin{aligned} \text{Ker} : (\mathbf{Ring} \downarrow \mathbb{Z}) &\rightarrow \mathbf{Ring}_* \\ (R, \phi : R \rightarrow \mathbb{Z}) &\mapsto \text{Ker}(\phi) \end{aligned}$$

The following diagram, which is commutative up to natural isomorphism, summarizes the situation and shows that the forgetful functor U is, up to equivalence of categories, the pullback functor $!^*$.

$$\begin{array}{ccc} (\mathbf{Ring} \downarrow 1) & \xrightarrow{\cong} & \mathbf{Ring} \\ \uparrow \text{!} \downarrow \text{!}^* & & \uparrow F \downarrow U \\ (\mathbf{Ring} \downarrow \mathbb{Z}) & \xrightleftharpoons[\text{Ker}]{(F, \pi_{\mathbb{Z}})} & \mathbf{Ring}_* \end{array}$$

¹We are using the algebraic definition of Ker , in fact we cannot use the categorical definition since $\mathbf{Ring}$ is not pointed. However, one can still induce an action of Ker in the morphisms. Alternatively, one may consider the categorical definition of Ker , but it is the Ker in $\mathbf{Ring}_*$, which is pointed, via the forgetful functor U . This perspective will be used when we treat the corresponding theorem in an ideally exact category, where the existence of the associated categorical equivalence is established.

Observe that the downward-right directed square commutes up to natural isomorphism, since, referring to the diagram 1.1, given an object R in $\mathbf{Ring}$, $\text{Ker}(\text{proj}_1)$ is isomorphic to R in $\mathbf{Ring}_*$. On the other hand, the upward-left directed square also commutes up to natural isomorphism, because in both cases we applied the unitalization functor F .

As already noticed, ideals in $\mathbf{Ring}$ are not objects in $\mathbf{Ring}$. However they are the normal subobject, ie the kernel of an arrow, in $\mathbf{Ring}_*$, which is semiabelian. Our aim is to study the notion of ideally exact category, that better captures the categorical structure of rings and ideals.

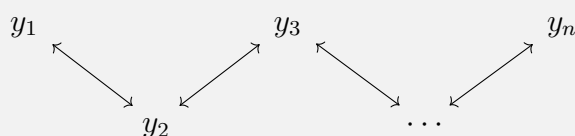
Remark that this anomaly does not occur in the category $\mathbf{Grp}$. In fact normal subgroups are objects in the category of groups and, since $\mathbf{Grp}$ is semiabelian, they are exactly the normal subobjects in the semiabelian sense.

Chapter 2

Pullback functor

2.1 Connected limits

Definition 2.1.1. A **connected limit** is a limit over a connected category. A category is connected if it is inhabited and any two objects can be connected by a zigzag of morphisms



where each symbol $\leftrightarrow$ is a morphism $y_i \rightarrow y_{i+1}$ or $y_{i+1} \rightarrow y_i$.

Example 2.1.2. Some examples

- The pullback is a connected limit.

$$\begin{array}{ccc}
 & & 2 \\
 & & \downarrow j \\
 1 & \xrightarrow{i} & 3
 \end{array}$$

- The equalizer is a connected limit.

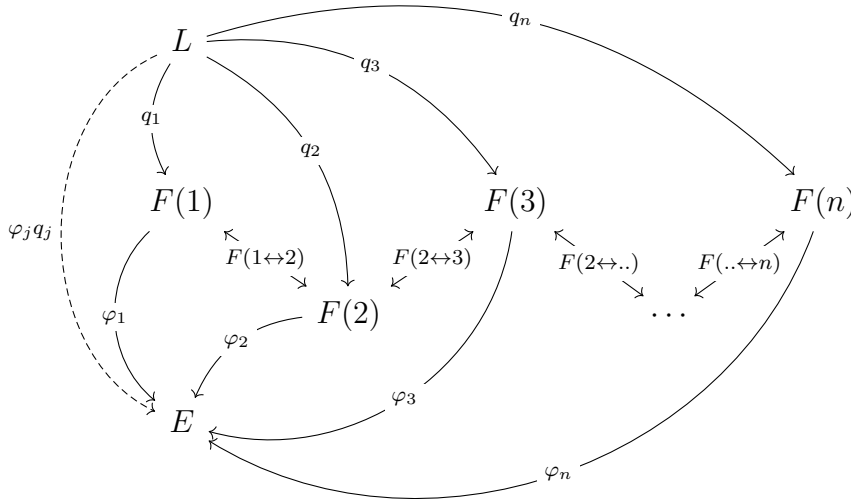
$$1 \begin{array}{c} \xrightarrow{i} \\ \xrightarrow{j} \end{array} 2$$

The following holds.

Proposition 2.1.3. *Let E be an object in $\mathcal{A}$. Then the forgetful functor $U : (\mathcal{A} \downarrow E) \rightarrow \mathcal{A}$ creates*

1. *all the connected limits that $\mathcal{A}$ has;*
2. *all the colimits that $\mathcal{A}$ has.*

Proof. 1. The connected limits in the category $(\mathcal{A} \downarrow E)$ can be computed in $\mathcal{A}$. Consider a connected limit cone $(L, (q_i)_{i \in \mathcal{I}})$. Denote each object in $(\mathcal{A} \downarrow E)$ associated to $i \in \mathcal{I}$ as $(F(i), \varphi_i)$. Since the limit is connected, it is the limit over a connected, hence inhabited, category. Thus we can pick $j \in \mathcal{I}$. The limit in $(\mathcal{A} \downarrow E)$ is $((L, \varphi_j q_j), (q_i)_{i \in \mathcal{I}})$. The connection in the zigzag guarantees that all the morphisms $q_1, \dots, q_n$ involved in the limit in $\mathcal{A}$ are also morphisms in $(\mathcal{A} \downarrow E)$, ie $\varphi_i q_i = \varphi_j q_j \forall i$.



2. The colimits in the category $(\mathcal{A} \downarrow E)$ can be computed in $\mathcal{A}$. Consider a colimit cocone $(L, (q_i)_{i \in \mathcal{I}})$. Denote each object in $(\mathcal{A} \downarrow E)$ associated to $i \in \mathcal{I}$ as $(F(i), \varphi_i)$. Observe that $(E, (\varphi_i)_{i \in \mathcal{I}})$ is a cocone over the diagram F . By the universal property, there exists a unique $\lambda : L \rightarrow E$ such that $\lambda q_i = \varphi_i$ for all $i \in \mathcal{I}$. The colimit in $(\mathcal{A} \downarrow E)$ is

$$((L, \lambda), (q_i)_{i \in \mathcal{I}}).$$

$$\begin{array}{ccc} F(i) & \xrightarrow{q_i} & L \\ \varphi_i \downarrow & \swarrow \lambda & \\ E & & \end{array}$$

□

Note that this result does not hold for all the limits. This is extensively shown in the following example.

Example 2.1.4. *The product is not a connected limit, in fact as we can observe the following category is not connected.*

1 2

We try to compute the product directly in $(\mathcal{A} \downarrow E)$. Let (D, δ) and (D', δ') be two objects in $(\mathcal{A} \downarrow E)$, the product is given by the following pullback

$$\begin{array}{ccccc} & & D \times_E D' & \longrightarrow & D' \\ & \swarrow & & \searrow \delta' & \\ D & \xrightarrow{\delta} & E & & \\ & \searrow \delta & \downarrow 1_E & \swarrow \delta' & \\ & & E & & \end{array}$$

which doesn't correspond to the product $D \times D'$ in $\mathcal{A}$.

2.2 Pullback functor's adjunction

Given a category $\mathcal{A}$ with pullbacks and a morphism $p : E \rightarrow B$ in $\mathcal{A}$, one can consider functors pullback and shrink

$$\begin{array}{ccc} & p^* & \\ (\mathcal{A} \downarrow B) & & (\mathcal{A} \downarrow E) \\ & p_! & \end{array}$$

defined as follows:

- $p^* : (\mathcal{A} \downarrow B) \rightarrow (\mathcal{A} \downarrow E)$ is the **pullback functor**

$$\begin{array}{ccc}
\begin{array}{c} A \\ \downarrow \alpha \\ B \end{array} & \xrightarrow{p^*} & \begin{array}{ccc} E \times_B A & \xrightarrow{\text{proj}_2} & A \\ \downarrow \text{proj}_1 & & \downarrow \alpha \\ E & \xrightarrow{p} & B \end{array} \\
\\
\begin{array}{ccc} A & \xrightarrow{f} & A' \\ \downarrow \alpha & \swarrow \alpha' & \\ B & & \end{array} & \xrightarrow{p^*} & \begin{array}{ccccc} & & p^*(f) = \tilde{f} & & \\ & \swarrow & \text{dashed} & \searrow & \\ E \times_B A & \xrightarrow{\text{proj}_2} & A & \xrightarrow{f} & A' & \xleftarrow{\text{proj}'_2} & E \times_B A' \\ & \searrow \alpha & & \swarrow \alpha' & & & \\ & & B & & & & \\ \text{proj}_1 \swarrow & & \uparrow p & & \searrow \text{proj}'_1 & & \\ & E & & & & & \end{array}
\end{array}$$

- $p_! : (\mathcal{A} \downarrow E) \rightarrow (\mathcal{A} \downarrow B)$ is the **shrink functor**

$$\begin{array}{ccc}
\begin{array}{c} D \\ \downarrow \delta \\ E \end{array} & \xrightarrow{p_!} & \begin{array}{ccc} D & & \\ \delta \downarrow & \searrow p\delta & \\ E & \xrightarrow{p} & B \end{array} \\
\\
\begin{array}{ccc} D & \xrightarrow{g} & D' \\ \delta \downarrow & \swarrow \delta' & \\ E & & \end{array} & \xrightarrow{p_!} & \begin{array}{ccc} D & \xrightarrow{g} & D' \\ & \searrow p\delta & \downarrow p\delta' \\ E & \xrightarrow{p} & B \end{array}
\end{array}$$

Proposition 2.2.1. *Let $p : E \rightarrow B$ be a morphism in $\mathcal{A}$ with pullbacks, then $p_!$ preserves connected limits.*

Proof. By Proposition 2.1.3 connected limits are computed in $\mathcal{A}$ and $p_!$ doesn't affect objects and morphism in $\mathcal{A}$. $\square$

Proposition 2.2.2. *Let $p : E \rightarrow B$ be a morphism in $\mathcal{A}$ with pullbacks, then*

$$\begin{array}{ccc}
& \xrightarrow{p^*} & \\
(\mathcal{A} \downarrow B) & & (\mathcal{A} \downarrow E) \\
& \xleftarrow{p_!} &
\end{array} \quad \text{with } p_! \dashv p^*$$

is an adjunction. The unit is given by

$$\begin{aligned}
& \eta : \text{Id}_{(\mathcal{A} \downarrow E)} \rightarrow p^* p_! \\
& \eta_{(D, \delta)} = \langle \delta, 1_D \rangle \\
& \langle \delta, 1_D \rangle : (D, \delta) \rightarrow (E \times_B D, \text{proj}_1)
\end{aligned}$$

and the counit is given by

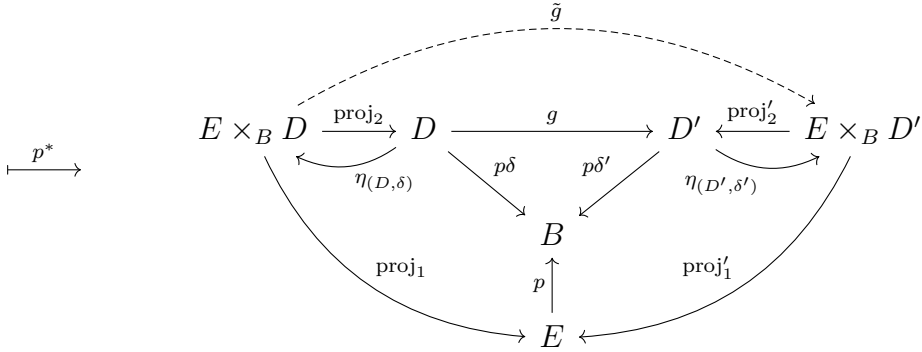
$$\begin{aligned}
& \epsilon : p_! p^* \rightarrow \text{Id}_{(\mathcal{A} \downarrow B)} \\
& \epsilon_{(A, \alpha)} = \text{proj}_2 \\
& \text{proj}_2 : (E \times_B A, p(\text{proj}_1)) \rightarrow (A, \alpha)
\end{aligned}$$

Proof. We want to prove this is adjunction. We start proving that the unit is a natural transformation. Let $g : (D, \delta) \rightarrow (D', \delta')$ be a morphism in $(\mathcal{A} \downarrow E)$, we want to prove that the following diagram commutes:

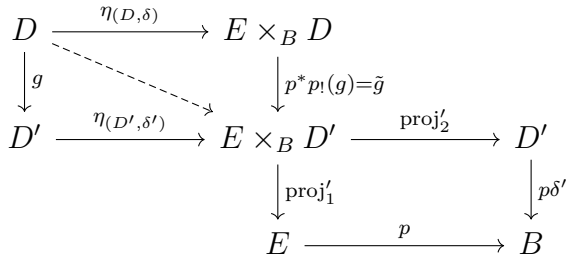
$$\begin{array}{ccc}
D & \xrightarrow{\eta_{(D, \delta)}} & E \times_B D \\
\downarrow g & & \downarrow p^* p_!(g) = \tilde{g} \\
D' & \xrightarrow{\eta_{(D', \delta')}} & E \times_B D'
\end{array}$$

In order to do that we need to expand the action of the functor $p^* p_!$ on the morphism g . We have:

$$\begin{array}{ccc}
D \xrightarrow{g} D' & & D \xrightarrow{g} D' \\
\downarrow \delta \quad \swarrow \delta' & \xrightarrow{p_!} & \searrow p\delta \quad \downarrow p\delta' \\
E & & B
\end{array}$$



Consider the following commutative diagram.



Since $(\text{proj}'_1, \text{proj}'_2)$ are jointly monic, by the following compositions

- $(\text{proj}_2')(\tilde{g})(\eta_{(D,\delta)}) = (g)(\text{proj}_2)(\eta_{(D,\delta)}) = (g)(1_D) = g$
- $(\text{proj}_2')(\eta_{(D',\delta')})(g) = (1_{D'})(g) = g$
- $(\text{proj}_1')(\tilde{g})(\eta_{(D,\delta)}) = (\text{proj}_1)(\eta_{(D,\delta)}) = \delta$
- $(\text{proj}_1')(\eta_{(D',\delta')})(g) = (\delta')(g) = \delta$

we conclude that $(\tilde{g})(\eta_{(D,\delta)}) = (\eta_{(D',\delta')})(g)$. Thus the naturality condition of η is satisfied.

Now we want to prove that the counit is a natural transformation. Consider $f : (A, \alpha) \rightarrow (A', \alpha')$ a morphism in $(\mathcal{A} \downarrow B)$, we want to prove that the following diagram commutes:

$$\begin{array}{ccc} E \times_B A & \xrightarrow{\epsilon_{(A, \alpha)}} & A \\ p!p^*(f) = \tilde{f} \downarrow & & \downarrow f \\ E \times_B A' & \xrightarrow{\epsilon_{(A', \alpha')}} & A' \end{array}$$

As similarly done above, we need to expand the action of the functor $p!p^*$ on the morphism f , we have:

The top diagram illustrates the pullback of the pair of maps $f: A \rightarrow B$ and $g: A' \rightarrow B$ along the map $p: B \rightarrow E$. The pullback is represented by the map $p^*: A \times_B A' \rightarrow E \times_B A$. The diagram shows the commutative squares and the induced map p^* .

The bottom diagram illustrates the pushout of the pair of maps $f: A \rightarrow B$ and $g: A' \rightarrow B$ along the map $p: B \rightarrow E$. The pushout is represented by the map $p!: E \times_B A \rightarrow B$. The diagram shows the commutative squares and the induced map $p!$.

Since $\epsilon_{(A,\alpha)} = \text{proj}_{2,E \times_B A}$, the composition $(f)(\epsilon_{(A,\alpha)}) = (\epsilon_{(A',\alpha')})(\tilde{f})$ holds by the definition of $\tilde{f}$. Thus also the naturality of ϵ is satisfied.

Now we can check that the unit and counit fulfil the first triangle identities. Let (A, α) be an object in $(\mathcal{A} \downarrow B)$.

$$((p^* \epsilon) \circ (\eta p^*))_{(A, \alpha)} = (p^* \epsilon)_{(A, \alpha)} (\eta p^*)_{(A, \alpha)} = p^* (\text{proj}_{2, E \times_B A}) (\eta_{(E \times_B A, \text{proj}_1)}) = \circledast$$

Considering first how p^* acts on the morphism $\mathrm{proj}_{1,E \times_B A}$

$$\begin{array}{ccc}
 E \times_B A & \xrightarrow{\text{proj}_2, E \times_B A} & A \\
 \searrow \alpha(\text{proj}_2, E \times_B A) & & \swarrow \alpha \\
 & B &
 \end{array}$$

$$\begin{array}{ccccc}
& & \text{proj}_{2, \tilde{E} \times_B A} & & \\
& \swarrow & & \searrow & \\
E \times_B (E \times_B A) & \xrightarrow{\text{proj}_{2, E \times_B (E \times_B A)}} & E \times_B A & \xrightarrow{\text{proj}_{2, E \times_B A}} & A \xleftarrow{\text{proj}_{2, E \times_B A}} E \times_B A \\
& \searrow & \downarrow (\alpha)(\text{proj}_{2, E \times_B A}) & \downarrow \alpha & \\
& & B & & \\
& \swarrow \text{proj}_{1, E \times_B (E \times_B A)} & \uparrow p & \swarrow \text{proj}_{1, E \times_B A} & \\
& & E & &
\end{array}$$

$$\circledast = (\tilde{\text{proj}}_{2, E \times_B A})(\eta_{(E \times_B A, \text{proj}_1)})$$

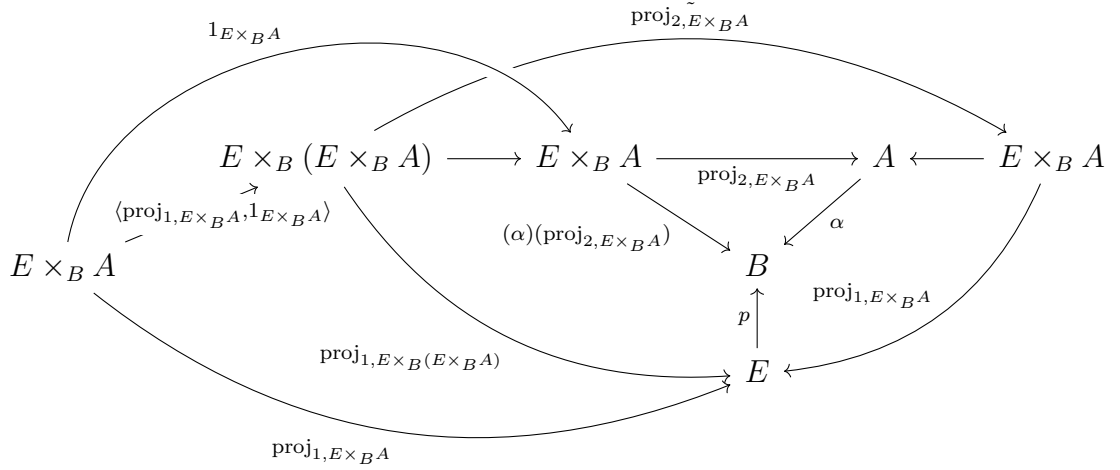
and then considering the definition of η , we obtain

$$\begin{array}{ccccc}
& & 1_{E \times_B A} & & \\
& \swarrow & & \searrow & \\
E \times_B A & \xrightarrow{\langle \text{proj}_{1, E \times_B A}, 1_{E \times_B A} \rangle} & E \times_B (E \times_B A) & \xrightarrow{\quad} & E \times_B A \\
& \searrow \text{proj}_{1, E \times_B A} & \downarrow & & \downarrow p(\text{proj}_{1, E \times_B A}) \\
& & E & \xrightarrow{\quad p \quad} & B
\end{array}$$

$$\circledast = (\tilde{\text{proj}}_{2, E \times_B A})\langle \text{proj}_{1, E \times_B A}, 1_{E \times_B A} \rangle$$

In order to conclude we need to check some compositions so that, by the universal property of the pullback $E \times_B A$ in the right of the following

diagram, we obtain the identity morphism.



We want to check that composition of the projections of $E \times_B A$ with $\circledast$ is the same of composing those projection with $1_{(E \times_B A, \text{proj}_1)}$. Then, since the projections are jointly monic, we have the result. Following the first composition

$$\begin{aligned}
 & (\text{proj}_{2, E \times_B A})(\tilde{\text{proj}}_{2, E \times_B A})(\text{proj}_{1, E \times_B A}, 1_{E \times_B A}) = \\
 & = (\text{proj}_{2, E \times_B A})(\text{proj}_{2, E \times_B (E \times_B A)})(\text{proj}_{1, E \times_B A}, 1_{E \times_B A}) \\
 & = (\text{proj}_{2, E \times_B A})(1_{E \times_B A}) = (\text{proj}_{2, E \times_B A})(1_{(E \times_B A, \text{proj}_1)})
 \end{aligned}$$

and the second composition

$$\begin{aligned}
 & (\text{proj}_{1, E \times_B A})(\tilde{\text{proj}}_{2, E \times_B A})(\text{proj}_{1, E \times_B A}, 1_{E \times_B A}) = \\
 & = (\text{proj}_{1, E \times_B (E \times_B A)})(\text{proj}_{1, E \times_B A}, 1_{E \times_B A}) \\
 & = \text{proj}_{1, E \times_B A} \\
 & = (\text{proj}_{1, E \times_B A})(1_{(E \times_B A, \text{proj}_1)})
 \end{aligned}$$

Hence

$$\circledast = (\tilde{\text{proj}}_{2, E \times_B A})(\text{proj}_{1, E \times_B A}, 1_{E \times_B A}) = 1_{(E \times_B A, \text{proj}_1)}$$

Finally we can check the second triangle identities. Let (D, δ) be an object in $(\mathcal{A} \downarrow E)$, we have:

$$((\epsilon p!) \circ (p! \eta))_{(D, \delta)} = ((\epsilon p!)_{(D, \delta)}(p! \eta))_{(D, \delta)} = (\epsilon_{p!(D, \delta)})p! \langle \delta, 1_D \rangle =$$

$$= (\epsilon_{(D,p\delta)})p_!(\langle \delta, 1_D \rangle) = \text{proj}_{2,E \times_B D} \langle \delta, 1_D \rangle = 1_{(D,p\delta)}$$

$$\begin{array}{ccc} E \times_B D & \xrightarrow{\text{proj}_{2,E \times_B D}} & D \\ \downarrow \text{proj}_{1,E \times_B D} & & \downarrow p\delta \\ E & \xrightarrow{p} & B \end{array}$$

So we can conclude that the adjunction holds. $\square$

Remark 2.2.3. p^* is a right adjoint functor, so it preserves limits. $p_!$ is a left adjoint functor, so it preserves colimits.

Property 2.2.4. Let $p : E \rightarrow B$ be a morphism in a category $\mathcal{A}$. The diagram constructed by p^* acting on a morphism $f : (A, \alpha) \rightarrow (A', \alpha')$ is a pullback.

$$\begin{array}{ccc} E \times_B A & \xrightarrow{p^*(f)=\tilde{f}} & E \times_B A' \\ \downarrow \text{proj}_2 & & \downarrow \text{proj}'_2 \\ A & \xrightarrow{f} & A' \end{array}$$

Proof. Since the right and the total diagrams below are pullback, then the left diagram is a pullback too.

$$\begin{array}{ccccc} & & \text{proj}_1 & & \\ & \nearrow & & \searrow & \\ E \times_B A & \xrightarrow{p^*(f)=\tilde{f}} & E \times_B A' & \xrightarrow{\text{proj}'_1} & E \\ \downarrow \text{proj}_2 & & \downarrow \text{proj}'_2 & & \downarrow p \\ A & \xrightarrow{f} & A' & \xrightarrow{\alpha'} & B \\ & \searrow & & \nearrow & \\ & & \alpha & & \end{array}$$

$\square$

Property 2.2.5. Let $p : E \rightarrow B$ be a morphism in $\mathcal{A}$ with pullbacks. Then the unit η and the counit ϵ of the adjunction associated to p^* are cartesian.

Proof. First we prove that η is cartesian. Let $g : (D, \delta) \rightarrow (D', \delta')$ be a morphism in $(\mathcal{A} \downarrow E)$. Applying *Property 2.2.4* to $p_!(g)$, we obtain that the following diagram is a pullback.

$$\begin{array}{ccc} E \times_B D & \xrightarrow{\text{proj}_2} & D \\ \downarrow p^* p_!(g) = \tilde{g} & & \downarrow g \\ E \times_B D' & \xrightarrow{\text{proj}'_2} & D' \end{array}$$

Considering the following commutative diagram, since the right diagram is a pullback, as we just proved, and the total diagram is a pullback, we can conclude that the left diagram is a pullback, as stated.

$$\begin{array}{ccccc} & & 1_D & & \\ & \nearrow & & \searrow & \\ D & \xrightarrow{\eta_{(D, \delta)}} & E \times_B D & \xrightarrow{\text{proj}_2} & D \\ \downarrow g & & \downarrow p^* p_!(g) = \tilde{g} & & \downarrow g \\ D' & \xrightarrow{\eta_{(D', \delta')}} & E \times_B D' & \xrightarrow{\text{proj}'_2} & D' \\ & \nwarrow & & \nearrow & \\ & & 1_{D'} & & \end{array}$$

Now we want to prove that ϵ is cartesian. Let $f : (A, \alpha) \rightarrow (A', \alpha')$ be a morphism in $(\mathcal{A} \downarrow B)$. By *Proposition 2.2.1* applied at the pullback in *Property 2.2.4* for f we obtain that the following diagram is a pullback.

$$\begin{array}{ccc} E \times_B A & \xrightarrow{\epsilon_{(A, \alpha)} = \text{proj}_2} & A \\ \downarrow p_! p^*(f) = \tilde{f} & & \downarrow f \\ E \times_B A' & \xrightarrow{\epsilon_{(A', \alpha')} = \text{proj}'_2} & A' \end{array}$$

□

Proposition 2.2.6. *Let $p : E \rightarrow B$ be a morphism in $\mathcal{A}$ with pullbacks. Then p^* preserves coequalizer of kernel pairs whose coequalizers are pullback stable regular epimorphisms.*

Proof. Consider the following diagram:

$$\begin{array}{ccccc}
 p^*(\text{Eq}(q)) & \xrightarrow[\tilde{p}_1]{\tilde{p}_2} & p^*(A) = A \times_B E & \xrightarrow{\tilde{q}} & p^*(\text{Coeq}(p_1, p_2)) \\
 \downarrow & & \downarrow & & \downarrow \\
 \text{Eq}(q) & \xrightarrow[p_1]{p_2} & A & \xrightarrow{q} & \text{Coeq}(p_1, p_2)
 \end{array}$$

Let q be the coequalizer of the kernel pair $\text{Eq}(q)$. We can affirm by *Property 2.2.4* applied to q that the diagram in the bottom-right corner is a pullback. Since, by hypothesis, q is a pullback stable regular epimorphism, we have that $\tilde{q}$ is a regular epimorphism.

We now aim to show that $\tilde{q}$ is also the coequalizer of $(\tilde{p}_1, \tilde{p}_2)$. Since $\tilde{q}$, $\tilde{p}_1$ and $\tilde{p}_2$ are obtained applying p^* to the kernel pair $\text{Eq}(q)$, and p^* preserves limits, it follows that $p^*(\text{Eq}(q))$ is the kernel pair of $\tilde{q}$. We already know that $\tilde{q}$ is a regular epi. Thus, by *Proposition A.0.6*, $\tilde{q}$ is also the coequalizer of $(\tilde{p}_1, \tilde{p}_2)$. $\square$

Later on it will be useful to study monads. Thus, we state the following proposition.

Property 2.2.7. *Let $p : E \rightarrow B$ be a morphism in $\mathcal{A}$ with pullbacks. Then the associated monad $(p^*p_!, \eta, p^*\epsilon p_!)$ is cartesian.*

Proof. We need to prove that the category $(\mathcal{A} \downarrow E)$ has pullback, the monad preserves pullbacks, and both the multiplication $p^*\epsilon p_!$ and the unit η are cartesian.

The category $(\mathcal{A} \downarrow E)$ has pullbacks, by *Proposition 2.1.3*, since $\mathcal{A}$ has. The monad $p^*p_!$ preserves pullbacks, in fact p^* is a right adjoint so it preserves limits, therefore preserves pullbacks, and $p_!$ preserves pullbacks by *Proposition 2.2.1*. We already proved that the unit η is cartesian, so it remains to prove that the multiplication $p^*\epsilon p_!$ is cartesian. Since ϵ is cartesian and p^* preserves pullbacks as shown above, we can conclude that the multiplication is cartesian. $\square$

2.3 Nullary and essentially nullary monads

In this section we look for conditions such that the monad associated to a morphism $p : E \rightarrow B$ is essentially nullary. This will be useful in the main

theorem of the ideally exact categories. This section is based on the work of [Jan24].

Definition 2.3.1. A monad (T, η, μ) on a category $\mathcal{X}$ with finite coproducts is said to be **nullary** if, for each object X in $\mathcal{X}$, the morphism $[T(!_X), \eta_X]$ is an iso, where $!_X$ is the unique morphism from the initial object of $\mathcal{X}$ to X . A monad on a category $\mathcal{X}$ with finite coproducts is said to be **essentially nullary** if, for each object X in $\mathcal{X}$, the morphism $[T(!_X), \eta_X]$ is a strong epi.

The following lemma, from [BB04], will be useful for our purposes.

Lemma 2.3.2. Let $\mathcal{X}$ be a protomodular category. Then for each commutative diagram of the form

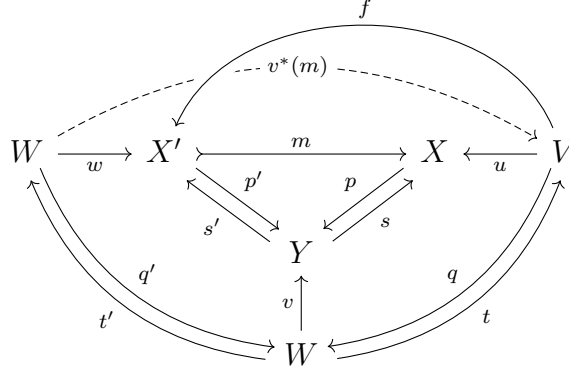
$$\begin{array}{ccc} V & \xrightarrow{u} & X \\ q \downarrow \uparrow t & & p \downarrow \uparrow s \\ W & \xrightarrow{v} & Y \end{array}$$

where p, q are split epimorphisms with respective sections s, t and the downward directed square is a pullback, it holds that the pair (u, s) is a strongly epimorphic pair.

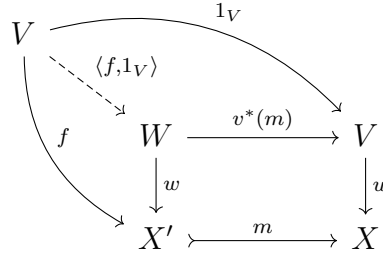
Proof. Assume $\mathcal{X}$ is protomodular. We want to prove that if u and s factorize through a mono m as $u = mf$ and $s = ms'$, then m is an iso, which means (u, s) is an extremally epimorphic pair. By a standard result, proved in *Proposition A.0.4*, this implies that (u, s) is a strongly epimorphic pair.

Define $p' = pm$, we have that $p's' = 1_Y$, in fact $p's' = pms' = ps = 1_Y$. Pulling back with the change-of-base functor v^* the mono m we obtain the following diagram. In this operation we also generate the morphisms q', t' and w . In the same way as in *Property 2.2.4* we have that the diagram

between m and $v^*(m)$ is a pullback.



Now we want to prove that $v^*(m)$ is an iso. Consider the following diagram. We know that the external diagram commutes by the hypothesis factorization.



Since the square is a pullback, it's obvious that $v^*(m)\langle f, 1_V \rangle = 1_V$. It remains to prove that $\langle f, 1_V \rangle v^*(m) = 1_W$.

Notice that $mf v^*(m) = uv^*(m) = mw$, so $f v^*(m) = w$ since m is a mono. Now consider the following compositions, and use what we just noticed.

- $v^*(m)(\langle f, 1_V \rangle v^*(m)) = (v^*(m)\langle f, 1_V \rangle)v^*(m) = 1_V v^*(m) = v^*(m)1_W$
- $w(\langle f, 1_V \rangle v^*(m)) = (w\langle f, 1_V \rangle)v^*(m) = f v^*(m) = w = w1_W$

Since the pair $(v^*(m), w)$ is jointly monic we have that $\langle f, 1_V \rangle v^*(m) = 1_W$. Considering this composition and the one above, it follows that $v^*(m)$ is an iso.

By the protomodularity of $\mathcal{A}$, since $v^*(m)$ is an iso, we have that m is an iso, as claimed. $\square$

We now aim to prove a sufficient condition for the monad associated to a morphism $p : E \rightarrow B$ to be essentially nullary. We could start with the

following proposition.

Theorem 2.3.3. *If $\mathcal{X}$ is a pointed protomodular category with finite coproducts and (T, η, μ) is a monad on $\mathcal{X}$ with cartesian η , then the monad is essentially nullary.*

Proof. For each object X in $\mathcal{X}$ we can consider the following commutative diagram:

$$\begin{array}{ccc} X & \xrightarrow{\eta_X} & T(X) \\ \downarrow !^X & \uparrow !_X & \downarrow T(!^X) \\ 0 & \xrightarrow{\eta_0} & T(0) \end{array}$$

By hypothesis η is cartesian, so the downward directed square is a pullback. Since $\mathcal{X}$ is protomodular then by *Lemma 2.3.2*, the pair $(\eta_X, T(!_X))$ is strongly epimorphic. By a standard result, proved in *Proposition A.0.5*, this yields that $[\eta_X, T(!_X)]$ is a strong epi. Thus the monad is essentially nullary. $\square$

In our case $(A \downarrow E)$ is pointed if and only if E is an initial object, in fact in this case the following initial and terminal objects coincide.

$$\begin{array}{ccc} E & \xrightarrow{!_A} & A \\ & \searrow & \downarrow \alpha \\ & & E \end{array} \quad \begin{array}{ccc} A & \xrightarrow{\alpha} & E \\ \downarrow \alpha & \swarrow 1_E & \\ E & & \end{array}$$

While this would work in that specific case, we would like to drop such condition. The following result fits our needs.

Theorem 2.3.4. *Let $p : E \rightarrow B$ be a morphism in $\mathcal{A}$ protomodular with finite coproducts. Then the monad associated to p is essentially nullary.*

Proof. Let (D, δ) be an object in $(\mathcal{A} \downarrow E)$. To prove that the monad is essentially nullary we need to study the morphism of the initial object. So first note the morphism for each object (D, δ) in $(\mathcal{A} \downarrow E)$ the morphism of the initial object $!_D$ is the same as the morphism of the initial object in

$\mathcal{A}$, in fact the following diagram commutes.

$$\begin{array}{ccc} 0 & \xrightarrow{!_D} & D \\ & \searrow & \downarrow \delta \\ & !_E & E \end{array}$$

Then consider the following diagram.

$$\begin{array}{ccc} E \times_B 0 & \xrightarrow{p^*p_!(!_D)} & E \times_B D \\ \text{proj}_2 \downarrow \uparrow \langle !_E, 1_0 \rangle & & \text{proj}'_2 \downarrow \uparrow \langle \delta, 1_D \rangle \\ 0 & \xrightarrow{!_D} & D \end{array}$$

The downward directed square is commutative by the definition of p^* , and it is a pullback by *Property 2.2.4*. The upward directed square commute by naturality of η , in fact $\eta_{(0,!_E)} = \langle !_E, 1_0 \rangle$ and $\eta_{(D,\delta)} = \langle \delta, 1_D \rangle$. So, by *Lemma 2.3.2*, the pair $(p^*(!_D), \langle !_E, 1_0 \rangle)$ is strongly epimorphic. By *Proposition A.0.5* this means that $[T(!_D), \eta_{(D,\delta)}] = [p^*p_!(!_D), \langle \delta, 1_D \rangle]$ is a strong epi. Thus the monad is essentially nullary. $\square$

Chapter 3

Lifting results on the E-M categories

3.1 Exactness and protomodularity lift

We present here some technical results on the Eilenberg-Moore categories. These will play a key role in the definition and characterization of the ideally exact categories.

Proposition 3.1.1. *Let T be a monad preserving regular epimorphisms and kernel pairs. If $\mathcal{X}$ is exact, then the Eilenberg-Moore category $\mathcal{X}^T$ is exact.*

Proof. Assume $\mathcal{X}$ is exact. With the given hypotheses on the monad T , we aim to prove that $\mathcal{X}^T$ is exact. We need to establish that it is finitely complete, that each equivalent relation is the kernel pair of some arrow, that regular epis are pullback stable and that each morphism admits a (regular epi, mono) factorization.

First we show that $\mathcal{X}^T$ is finitely complete. We recall that the forgetful functor $U^T : \mathcal{X}^T \rightarrow \mathcal{X}$ creates all the limits that $\mathcal{X}$ has. Since $\mathcal{X}$ is exact, it has all finite limits. Therefore, $\mathcal{X}^T$ has all finite limits.

We proceed to prove that each equivalent relation is the kernel pair of some arrow. Consider an equivalence relation R in $\mathcal{X}^T$ with ξ_R and ξ_A giving the structure of algebras, as in the diagram below. Then, under the forgetful functor U^T , R is an equivalence relation in $\mathcal{X}$. $\mathcal{X}$ is exact, so R is the kernel pair of some arrow. In particular, by *Proposition A.0.9*, we can suppose that R is the kernel pair of its coequalizer. Let call it $q : A \rightarrow C$.

Since T preserves kernel pairs, we have that $T(R)$ is the kernel pair of $T(q)$. Moreover T preserves regular epis, so $T(q)$ is a regular epi. By *Proposition A.0.6*, we can assume that $T(q)$ is the coequalizer of its kernel pair $(T(R), T(r_1), T(r_2))$. Since $q\xi_A T(r_1) = qr_1\xi_R = qr_2\xi_R = q\xi_A T(r_2)$, by the universal property of the coequalizer, there exists a unique morphism ξ_C such that $\xi_C T(q) = q\xi_A$. The morphism ξ_C is the algebra structure of $T(B)$. Since U^T creates all the limits that $\mathcal{X}$ has, we can conclude that R is the kernel pair in $\mathcal{X}^T$ of the arrow q .

$$\begin{array}{ccccc}
 T(R) & \xrightleftharpoons[T(r_2)]{T(r_1)} & T(A) & \xrightarrow{T(q)} & T(C) \\
 \downarrow \xi_R & & \downarrow \xi_A & & \downarrow \xi_C \\
 R & \xrightleftharpoons[r_2]{r_1} & A & \xrightarrow{q} & C
 \end{array}$$

Consider now a morphism $f : (A, \xi_A) \rightarrow (B, \xi_B)$ in $\mathcal{X}^T$, we aim to prove that it admits a (regular epi, mono) factorization. The morphism f , under the forgetful functor U^T , is a morphism in $\mathcal{X}$, which is exact and in particular regular. Thus f admits a (regular epi, mono) factorization in $\mathcal{X}$. Let it be $f = me$ with $m : Q \rightarrow B$ mono and $e : A \rightarrow Q$ regular epi, thus the coequalizer of some pair of parallel arrows. A and B already have an algebra structure. We want to provide Q and the domain of the parallel arrows with a structure of algebra, so that we can lift the factorization to $\mathcal{X}^T$.

$$\begin{array}{ccccc}
 T(X) & \xrightleftharpoons{\quad} & T(A) & \xrightarrow{T(f)} & T(B) \\
 & & \downarrow \xi_A & \searrow T(e) & \nearrow T(m) \\
 & & & T(Q) & \\
 & & & \downarrow \xi_B & \\
 X & \xrightleftharpoons{\quad} & A & \xrightarrow{f} & B \\
 & & \searrow e & & \nearrow m \\
 & & & Q &
 \end{array}$$

In order to do so, we first need to show that T preserves coequalizers in $\mathcal{X}$ of kernel pairs arising from $\mathcal{X}^T$, once the algebra structure is forgotten, and considering only the pair of the whole kernel pair. Let (Eq, r_1, r_2) be a kernel pair in $\mathcal{X}$, arising from a kernel pair in $\mathcal{X}^T$ of a morphism, and let p be the coequalizer in $\mathcal{X}$ of (r_1, r_2) . By *Proposition A.0.7*, we can affirm

that (r_1, r_2) is the kernel pair of p . Given that (Eq, r_1, r_2) is a kernel pair arising from $\mathcal{X}^T$, it follows that $(T(\text{Eq}), T(r_1), T(r_2))$ is the kernel pair of some arrow. Since T preserves regular epi and kernel pair we can assert that $T(p)$ is a regular epi and, by *Proposition A.0.6*, it is the coequalizer of the kernel pair $(T(\text{Eq}), T(r_1), T(r_2))$, as claimed.

$$\begin{array}{ccccc}
 T(\text{Eq}) & \xrightleftharpoons[T(r_2)]{T(r_1)} & T(A) & \xrightarrow{T(p)} & T(B) \\
 \downarrow \xi_{\text{Eq}} & & \downarrow \xi_A & & \downarrow \xi_B \\
 \text{Eq} & \xrightleftharpoons[r_2]{r_1} & A & \xrightarrow{p} & B
 \end{array}$$

Picking up where we left off, consider f and its kernel pair $(\text{Eq}(f), r_1, r_2)$. Since U^T creates all the limits that $\mathcal{X}$ has, we can lift the kernel pair $(\text{Eq}(f), r_1, r_2)$ to a kernel pair in $\mathcal{X}^T$. Let ξ_{Eq} be the structure of algebra of Eq .

We know that U^T creates all the colimits that $\mathcal{X}$ has and T preserves. Hence, U^T creates all coequalizers that $\mathcal{X}$ has of kernel pairs arising from $\mathcal{X}^T$, once the algebra structure is forgotten. Observe that e is such a coequalizer, in fact it is a regular epi and by *Proposition A.0.10* f and e share the same kernel pair, because of the factorization $f = me$ with m mono. Thus $(\text{Eq}(f), r_1, r_2)$ is also the kernel pair of e . Hence, by *Proposition A.0.6*, e is the coequalizer of $(\text{Eq}(f), r_1, r_2)$. In particular by the creation of such colimit we proved that Q has a structure of algebra, called ξ_Q , and that e can be lifted to a coequalizer in $\mathcal{X}^T$.

$$\begin{array}{ccccc}
 T(\text{Eq}(f)) & \xrightleftharpoons[T(r_2)]{T(r_1)} & T(A) & \xrightarrow{T(f)} & T(B) \\
 \downarrow \xi_{\text{Eq}(f)} & & \downarrow \xi_A & \searrow T(e) & \nearrow T(m) \\
 & & & T(Q) & \\
 & & & \downarrow \xi_Q & \\
 \text{Eq}(f) & \xrightleftharpoons[r_2]{r_1} & A & \xrightarrow{f} & B \\
 & & \searrow e & & \nearrow m \\
 & & & Q &
 \end{array}$$

In order to prove the factorization of f in $\mathcal{X}^T$, it remains to establish that m is a morphism in $\mathcal{X}^T$ and that it is a mono. Observe that m is a

mono in $\mathcal{X}^T$; in fact it is a mono in $\mathcal{X}$ and, furthermore, it is merely a morphism $\mathcal{X}$ that satisfies the algebra condition. We still need to establish that m is a morphism in $\mathcal{X}^T$.

$$\begin{aligned} f = me &\implies f\xi_A = me\xi_A \\ &\implies \xi_B T(f) = m\xi_Q T(e) \\ &\implies \xi_B T(m)T(e) = m\xi_Q T(e) \end{aligned}$$

We know that $T(e)$ is a regular epi, thus it is a epi. So we can cancel it from both sides and we obtain the condition for m to be a morphism in $\mathcal{X}^T$:

$$\xi_B T(m) = m\xi_Q$$

We proved that m and p are respectively a mono and a regular epi in $\mathcal{X}^T$. This allows the lifting of the factorization $f = me$ to $\mathcal{X}^T$.

It remains to prove the last condition of a regular category, that is the pullback stability of regular epi. Let $q : A \rightarrow B$ be a regular epi in $\mathcal{X}^T$, ie a coequalizer of some arrows in $\mathcal{X}^T$. In particular, q is a regular epi in $\mathcal{X}$. Since $\mathcal{X}$ is exact, q is pullback stable. Let $\tilde{q}$ be the pullback in $\mathcal{X}^T$ of q along some morphism. As just noticed, $\tilde{q}$ is a regular epi in $\mathcal{X}$. Let $(\text{Eq}(\tilde{q}), r_1, r_2)$ be the kernel pair of $\tilde{q}$. By *Proposition A.0.6*, $\tilde{q}$ is the coequalizer of such kernel pair. We want to lift $\tilde{q}$ to a regular epi in $\mathcal{X}^T$. By construction, the pullback considered is a pullback in $\mathcal{X}^T$, thus it comes with algebra structures of $A \times_B C$ and C . Let $\xi_{A \times_B C}$ and ξ_C be the algebra structures of $A \times_B C$ and C respectively. Moreover, $\tilde{q}$ is a morphism in $\mathcal{X}^T$. We know that the forgetful functor U^T creates all the limits that $\mathcal{X}$ has, so we can lift the kernel pair $(\text{Eq}(\tilde{q}), r_1, r_2)$ to a kernel pair in $\mathcal{X}^T$. Let $\xi_{\text{Eq}(\tilde{q})}$ be the algebra structure of $\text{Eq}(\tilde{q})$. Then we know that U^T creates all the colimits that $\mathcal{X}$ has and the monad T preserves, and we already proved that T preserves coequalizers of kernel pairs. Thus,

since $\tilde{q}$ is such a coequalizer, we can lift it to a coequalizer in $\mathcal{X}^T$.

$$\begin{array}{ccccccc}
 & & T(K) & \rightrightarrows & T(A) & \xrightarrow{T(q)} & T(B) \\
 & & \downarrow & & \downarrow & & \downarrow \\
 T(\text{Eq}(\tilde{q})) & \xrightarrow{T(r_1)} & & \nearrow & T(A \times_B C) & \xrightarrow{T(\tilde{q})} & T(C) \\
 & \searrow T(r_2) & & & \downarrow & & \downarrow \\
 & & K & \rightrightarrows & A & \xrightarrow{q} & B \\
 & & \downarrow \xi_{A \times_B C} & & \downarrow \xi_C & & \downarrow \\
 \text{Eq}(\tilde{q}) & \xrightarrow[r_2]{r_1} & A \times_B C & \xrightarrow{\tilde{q}} & C & &
 \end{array}$$

□

Proposition 3.1.2. *Let $T : \mathcal{X} \rightarrow \mathcal{X}$ be a monad. If $\mathcal{X}$ is protomodular, then the Eilenberg-Moore category $\mathcal{X}^T$ is protomodular.*

Proof. Since the forgetful functor $\mathcal{X}^T \rightarrow \mathcal{X}$ creates all the limits that $\mathcal{X}$ has, we can assert that $\mathcal{X}^T$ is finitely complete.

Then, let $f : (X, \xi) \rightarrow (Y, \xi')$ be a morphism in $\mathcal{X}^T$. The following diagram of functors is commutative. This essentially follows from the fact that the forgetful functor $\mathcal{X}^T \rightarrow \mathcal{X}$ creates all the limits that $\mathcal{X}$ has, in particular the pullbacks, which are used to define the change-of-base functor f^* .

$$\begin{array}{ccc}
 \text{Pt}_{\mathcal{X}^T}(Y, \xi') & \xrightarrow{f^*} & \text{Pt}_{\mathcal{X}^T}(X, \xi) \\
 \downarrow U^T & & \downarrow U^T \\
 \text{Pt}_{\mathcal{X}}(Y) & \xrightarrow{f^*} & \text{Pt}_{\mathcal{X}}(X)
 \end{array}$$

Let α be a morphism in $\text{Pt}_{\mathcal{X}^T}(Y, \xi')$. Suppose that $f^*(\alpha)$ is iso, we want to prove that α is iso. Since $f^*(\alpha)$ is iso, we have that $U^T(f^*(\alpha))$ is iso. By the commutativity of the diagram, we have that $U^T(f^*(\alpha)) = f^*(U^T(\alpha))$. Thus $f^*(U^T(\alpha))$ is an iso. Since $\mathcal{X}$ is protomodular, $U^T(\alpha)$ is an iso. Finally, U^T reflects isomorphisms, so α is an iso. □

3.2 Products and coproducts lift

In this section we aim construct products and coproducts in the Eilenberg–Moore category. Products are quite easy to lift, and we implicitly already saw it in the exactness lift, whereas coproducts need additional conditions.

Even if we already know that products can be easily lifted, since the forgetful functor U^T creates all the limits, it will be useful to have the explicit construction.

Remark 3.2.1. *Given A, B objects in a category with products and and T a monad on such category, we can construct the product in the Eilenberg–Moore category $\mathcal{X}^T$ as follows. Let $\pi_1 : A \times B \rightarrow A$ and $\pi_2 : A \times B \rightarrow B$ be the projections of the product in $\mathcal{X}$, then we define the product in $\mathcal{X}^T$ as follows.*

$$\begin{array}{ccccc}
 T(A) & \xleftarrow{T(\pi_1)} & T(A \times B) & \xrightarrow{T(\pi_2)} & T(B) \\
 \downarrow \xi_A & & \downarrow \langle T(\pi_1), T(\pi_2) \rangle & & \downarrow \xi_B \\
 & & T(A) \times T(B) & & \\
 & & \downarrow \xi_A \times \xi_B & & \\
 A & \xleftarrow{\pi_1} & A \times B & \xrightarrow{\pi_2} & B
 \end{array}$$

The algebra structure coincides with the one given by the creations of limits by U^T , in fact, in that case the algebra structure is the unique making the diagram commutes. The universal property holds by the universal property of the product in $\mathcal{X}$, in fact one can prove that the unique morphism induced by the product in $\mathcal{X}$ is also a morphism in $\mathcal{X}^T$.

This leads to the following.

Proposition 3.2.2. *Let $T : \mathcal{X} \rightarrow \mathcal{X}$ be a monad. If $\mathcal{X}$ has finite products, then $\mathcal{X}^T$ has finite products.*

We now proceed with results that will allow us to construct coproducts in the Eilenberg–Moore category, assuming they exist in the base category.

The construction of the coproducts in the Eilenberg–Moore category $\mathcal{X}^T$ will looks quite intricate at first glance and it may appear overly complicated. However, this construction is needed because, given two object with an algebra structure each, the coproduct of the two algebra structures is not an algebra structure in general. To make this clearer consider the following example.

Example 3.2.3. Let $\mathbf{Mon}$ be the category of monoids and let $U : \mathbf{Mon} \rightarrow \mathbf{Set}$ be the forgetful functor. One can define the free functor $F : \mathbf{Set} \rightarrow \mathbf{Mon}$ sending each set X to the free monoid on X

$$F(X) = \{(x_1, \dots, x_n) \mid n \in \mathbb{N}, \forall i = 1, \dots, n, x_i \in X\}$$

with the operation of concatenation. This induces a monad $T = UF$ on $\mathbf{Set}$ and the functor turns out to be monadic. The coproduct in $\mathbf{Mon}$ is the free product, while in $\mathbf{Set}$ it is the disjoint union. The free product of two monoids, as sets, is in general a bigger set than the disjoint union of the two set underlying the monoids. Thus the forgetful functor U in general does not preserve coproducts. Instead, if the coproduct of the algebra structures, which correspond to the operation of the monoids, were itself an algebra structure, the forgetful functor would preserve coproducts.

$$\begin{array}{ccc} \mathbf{Mon} & \xrightarrow{\cong} & \mathbf{Set}^T \\ F \uparrow & \searrow U & \\ \mathbf{Set} & & \end{array}$$

The following will be rather intricate and technical; we aim to get to the final outcome *Proposition 3.2.7*. The following results are based on the work of [Lin69].

For the sake of clarity, we first present a technical definition.

Definition 3.2.4. Let $U : \mathcal{X} \rightarrow \mathcal{A}$ be a functor, let X be an object in $\mathcal{X}$ and let $\{(f_i, g_i)\}_{i \in I}$ be a family of pair of parallel arrows in $\mathcal{A}$, where $f_i, g_i : A_i \rightarrow U(X)$. A morphism $q : X \rightarrow Q$ in $\mathcal{X}$ is said to be a *U -coequalizer* of the pair $\{(f_i, g_i)\}_{i \in I}$ if

- $U(q)f_i = U(q)g_i \forall i \in I$ in $\mathcal{A}$;
- If $h : X \rightarrow Y$ is a morphism in $\mathcal{X}$ such that $U(h)f_i = U(h)g_i \forall i \in I$ in $\mathcal{A}$, then there exists a unique morphism $t : Q \rightarrow Y$ in $\mathcal{X}$ such that $tq = h$.

$$\begin{array}{ccc} A & \xrightarrow[f_i]{g_i} & U(X) \xrightarrow{U(q)} U(Q) \\ & & \searrow U(h) \quad \downarrow U(t) \\ & & U(Y) \end{array} \qquad \begin{array}{ccc} X & \xrightarrow{q} & Q \\ & \searrow h & \downarrow t \\ & & Y \end{array}$$

Before continuing, we recall the universal property of the unit in an adjunction. Consider a functor $U : \mathcal{A} \rightarrow \mathcal{B}$ right adjoint to a functor F . Then for each object A in $\mathcal{A}$, B in $\mathcal{B}$, and $f : B \rightarrow U(A)$ in $\mathcal{B}$ there exists a unique morphism $g : F(B) \rightarrow A$ in $\mathcal{A}$ such that $f = U(g)\eta_B$. We denote $g = \bar{f}$.

$$\begin{array}{ccc} B & \xrightarrow{\eta_B} & UF(B) \\ & \searrow f & \downarrow U(g) \\ & & U(A) \end{array} \qquad \begin{array}{c} F(B) \\ \downarrow g \\ A \end{array}$$

Also recall that a monad (T, μ, η) induces a functor U^T and a functor F^T defined as follows:

$$\begin{array}{ccc} \mathcal{X}^T & & (X, \xi) \xrightarrow{U^T} X \\ \uparrow F^T \downarrow U^T & & \\ \mathcal{X} & & X \xrightarrow{F^T} (T(X), \mu_X) \end{array}$$

Lemma 3.2.5. *Let $U : \mathcal{X} \rightarrow \mathcal{A}$ be a functor right adjoint to a functor $F : \mathcal{A} \rightarrow \mathcal{X}$, let X be an object in $\mathcal{X}$ and let $\{(f_i, g_i)\}_{i \in I}$ be a family of parallel arrows in $\mathcal{X}$, where $f_i, g_i : A_i \rightarrow U(X)$. Consider the family $\{(\bar{f}_i, \bar{g}_i)\}_{i \in I}$ of unique morphisms induced by the universal property of the unit, such that $f_i = U(\bar{f}_i)\eta_{A_i}$ and $g_i = U(\bar{g}_i)\eta_{A_i}$. If $q : X \rightarrow Q$ is the coequalizer of the family $\{(\bar{f}_i, \bar{g}_i)\}_{i \in I}$ in $\mathcal{X}$, then q is the U -coequalizer of the family $\{(f_i, g_i)\}_{i \in I}$ in $\mathcal{A}$.*

$$\begin{array}{ccc} A_i & \xrightarrow{\eta_{A_i}} & UF(A_i) \\ & \searrow f_i & \downarrow U(\bar{g}_i) \\ & & U(X) \end{array} \qquad \begin{array}{ccc} F(A_i) & \xrightarrow[\bar{g}_i]{\bar{f}_i} & X \\ & & \downarrow q \\ & & Q \end{array}$$

$$\begin{array}{ccc} A_i & \xrightarrow[f_i]{g_i} & U(X) \\ & & \downarrow U(q) \\ & & U(Q) \end{array}$$

Proof. Assume $q : X \rightarrow Q$ is the coequalizer of the family $\{(\bar{f}_i, \bar{g}_i)\}_{i \in I}$ in $\mathcal{X}$, we want to prove that q is the U^T -coequalizer of the family $\{(f_i, g_i)\}_{i \in I}$ in $\mathcal{A}$. First we show, for each i , that $U(q)f_i = U(q)g_i$. Assume that

$$q\bar{f}_i = q\bar{g}_i.$$

$$\begin{aligned} q\bar{f}_i = q\bar{g}_i &\implies U(q)U(\bar{f}_i) = U(q)U(\bar{g}_i) \\ &\implies U(q)U(\bar{f}_i)\eta_{A_i} = U(q)U(\bar{g}_i)\eta_{A_i} \\ &\implies U(q)f_i = U(q)g_i \end{aligned}$$

We now aim to prove the universal property of the U^T -coequalizer. Let $h : X \rightarrow Y$ be a morphism such that $U(h)f_i = U(h)g_i$. By the universal property of the unit applied to $U(h)f_i = U(h)g_i : A_i \rightarrow U(Y)$ there exist a unique morphism $k : F(A_i) \rightarrow Y$ such that $U(k)\eta_{A_i} = U(h)f_i = U(h)g_i$. Both $h\bar{f}_i$ and $h\bar{g}_i$ satisfies such condition; in fact $U(h\bar{f}_i)\eta_{A_i} = U(h)U(\bar{f}_i)\eta_{A_i} = U(h)f_i$ and similarly for $h\bar{g}_i$. Thus by the uniqueness $h\bar{f}_i = k = h\bar{g}_i$.

$$\begin{array}{ccc} A_i & \xrightarrow{\eta_{A_i}} & UF(A_i) \\ & \searrow f_i & \downarrow U(\bar{g}_i) \downarrow U(\bar{f}_i) \\ & \searrow g_i & U(X) \\ & & \downarrow U(h) \\ & & U(Y) \end{array} \quad \begin{array}{ccc} F(A_i) & & \\ \bar{f}_i \downarrow \bar{g}_i & & \\ X & & \\ \downarrow h & & \\ Y & & \end{array}$$

$U(h)f_i = U(h)g_i$ $U(k)$ k

Then, by the universal property of the coequalizer q in $\mathcal{X}^T$, there exists a unique morphism $t : Q \rightarrow Y$ such that $tq = h$. $\square$

Proposition 3.2.6. *Let $\mathcal{X}$ be a category, let (T, μ, η) be a monad on $\mathcal{X}$ and let $\{(A_i, \xi_i)\}_{i \in I}$ be a family of objects in $\mathcal{X}^T$. Assume that the coproducts $\coprod_i A_i$ and $\coprod_i T(A_i)$ exist in $\mathcal{X}$ with respective canonical injections $\{\iota_i\}_{i \in I}$ and $\{\iota'_i\}_{i \in I}$. If the following parallel arrows in $\mathcal{X}^T$ admits a coequalizer $q : T(\coprod_i) \rightarrow (Q, \xi_Q)$*

$$\begin{array}{ccc} & (T^2(\coprod_i A_i), \mu) & \\ T[\dots, T(\iota_i), \dots] \nearrow & & \searrow \mu_{\coprod_i A_i} \\ (T(\coprod_i T(A_i)), \mu) & \xrightarrow{T(\coprod_i \xi_i)} & (T(\coprod_i A_i), \mu) \end{array} \quad (3.1)$$

then both the following diagrams of parallel arrow in $\mathcal{X}$ have q as U^T -

coequalizer $\forall j \in I$.

$$\begin{array}{ccc}
 \Pi_i T(A_i) & \xrightarrow{[\dots, T(\iota_i), \dots]} & T(\Pi_i A_i) \\
 \searrow \Pi_i \xi_i & & \nearrow \eta_{\Pi_i A_i} \\
 & \Pi_i A_i &
 \end{array} \quad (3.2)$$

$$\begin{array}{ccc}
 T(A_j) & \xrightarrow{T(\iota_j)} & T(\Pi_i A_i) \\
 \searrow \xi_j & & \nearrow \eta_{\Pi_i A_i} \\
 & A_j \xrightarrow{\iota_j} \Pi_i A_i &
 \end{array} \quad (3.3)$$

Proof. Consider the diagram (3.1) of parallel arrows in $\mathcal{X}^T$, and p its coequalizer, here below explicitly.

$$\begin{array}{ccccc}
 & & T^3(\Pi_i A_i) & & \\
 & \nearrow T^2[\dots, T(\iota_i), \dots] & \downarrow & \searrow T(\mu_{\Pi_i A_i}) & \\
 T^2(\Pi_i T(A_i)) & \xrightarrow{T(\Pi_i \xi_i)} & T^2(\Pi_i A_i) & \xrightarrow{\mu_{\Pi_i A_i}} & T^2(\Pi_i A_i) \\
 \downarrow \mu_{\Pi_i T(A_i)} & & \downarrow \mu_{T(\Pi_i A_i)} & & \downarrow \mu_{\Pi_i A_i} \\
 T(\Pi_i T(A_i)) & \xrightarrow{T[\dots, T(\iota_i), \dots]} & T^2(\Pi_i A_i) & \xrightarrow{\mu_{\Pi_i A_i}} & T(\Pi_i A_i) \\
 & \xrightarrow{T(\Pi_i \xi_i)} & & &
 \end{array}$$

The top and bottom arrows from $(T(\Pi_i T(A_i)), \mu)$ to $(T(\Pi_i A_i), \mu)$ of the diagram (3.1) correspond, by the universal property of the unit of the adjunction, respectively to the top and bottom arrows of the diagram (3.2). In order to prove so, by the uniqueness of the morphism induced by the universal property of the unit, it suffices to show that the image of the diagram (3.1) through U^T , composed with the unit, is the diagram

(3.2). Such composition results to the following diagram.

$$\begin{array}{ccccc}
 & T(\coprod_i T(A_i)) & \xrightarrow{T[\dots, T(\iota_i), \dots]} & T^2(\coprod_i A_i) & \\
 \eta_{\coprod_i T(A_i)} \nearrow & & & & \searrow \mu_{\coprod_i A_i} \\
 \coprod_i T(A_i) & & (\spadesuit) & & T(\coprod_i A_i) \\
 \eta_{\coprod_i T(A_i)} \searrow & & & & \nearrow T(\coprod_i \xi_i) \\
 & T(\coprod_i T(A_i)) & & &
 \end{array}$$

Consider now the top arrow in the diagram $(\spadesuit)$ from $\coprod_i T(A_i)$ to $T(\coprod_i A_i)$. Using the naturality of η , we have that

$$\begin{array}{ccc}
 \coprod_i T(A_i) & \xrightarrow{\eta_{\coprod_i T(A_i)}} & T(\coprod_i T(A_i)) \\
 \downarrow [\dots, T(\iota_i), \dots] & & \downarrow T[\dots, T(\iota_i), \dots] \\
 T(\coprod_i A_i) & \xrightarrow{\eta_{T(\coprod_i A_i)}} & T^2(\coprod_i A_i)
 \end{array}$$

$$\begin{aligned}
 \mu_{\coprod_i A_i} T[\dots, T(\iota_i), \dots] \eta_{\coprod_i T(A_i)} &= \mu_{\coprod_i A_i} \eta_{T(\coprod_i A_i)} [\dots, T(\iota_i), \dots] \\
 &= 1_{T(\coprod_i A_i)} [\dots, T(\iota_i), \dots] \\
 &= [\dots, T(\iota_i), \dots]
 \end{aligned}$$

which is the top arrow in (3.2). Then, for the bottom arrow from $\coprod_i T(A_i)$ to $T(\coprod_i A_i)$ in $(\spadesuit)$, by the naturality of η and the property of the monad, we have that

$$\begin{array}{ccc}
 \coprod_i T(A_i) & \xrightarrow{\eta_{\coprod_i T(A_i)}} & T(\coprod_i T(A_i)) \\
 \downarrow \coprod_i \xi_i & & \downarrow T(\coprod_i \xi_i) \\
 \coprod_i A_i & \xrightarrow{\eta_{\coprod_i A_i}} & T(\coprod_i A_i)
 \end{array}$$

$$T(\coprod_i \xi_i) \eta_{\coprod_i T(A_i)} = \eta_{\coprod_i A_i} (\coprod_i \xi_i)$$

which is the bottom arrow in (3.2). By *Lemma 3.2.5*, it follows that the diagram (3.2) has p as U^T -coequalizer.

It remains to prove that the diagram (3.3) has p as U^T -coequalizer. We already know that (3.2) has p as U^T -coequalizer. Note that these two diagrams are related by the canonical injection $\iota'_j : T(A_j) \rightarrow \coprod_i T(A_i)$, as

shown in the diagram below. In particular, by the universal property of the coproduct, it holds that $(\coprod_i \xi_i) \iota'_j = \iota_j \xi_j$ and $[\dots, T(\iota_i), \dots] \iota'_j = T(\iota_j)$.

$$\begin{array}{ccccc}
 & & T(\iota_j) & & \\
 & \nearrow & \text{---} & \searrow & \\
 T(A_j) & \xrightarrow{\iota'_j} & \coprod_i T(A_i) & \xrightarrow{[\dots, T(\iota_i), \dots]} & T(\coprod_i A_i) \\
 & \searrow & \downarrow \coprod_i \xi_i & \nearrow \eta_{\coprod_i A_i} & \\
 & & \coprod_i A_i & &
 \end{array}$$

$\iota_j \xi_j$

The first condition of the U^T -coequalizer for (3.3) is obviously satisfied, just composing with ι'_j the condition of the U^T -coequalizer of (3.2):

$$\begin{aligned}
 U^T(q)[\dots, T(\iota_i), \dots] &= U^T(q) \eta_{\coprod_i A_i} \coprod_i \xi_i \implies \\
 \implies U^T(q)[\dots, T(\iota_i), \dots] \iota'_j &= U^T(q) \eta_{\coprod_i A_i} \coprod_i \xi_i \iota'_j \\
 \implies U^T(q) T(\iota_j) &= U^T(q) \iota_j \xi_j
 \end{aligned}$$

The universal property of the U^T -coequalizer of (3.3) also holds. In fact, let h be a morphism in $\mathcal{X}^T$ such that $U^T(h) T(\iota_j) = U^T(h) \iota_j \xi_j \quad \forall j \in I$, then

$$\begin{aligned}
 U^T(h) T(\iota_j) &= U^T(h) \iota_j \xi_j \quad \forall j \in I \implies \\
 \implies U^T(h)[\dots, T(\iota_i), \dots] \iota'_j &= U^T(h) \eta_{\coprod_i A_i} \coprod_i \xi_i \iota'_j \quad \forall j \in I \\
 \implies U^T(h)[\dots, T(\iota_i), \dots] &= U^T(h) \eta_{\coprod_i A_i} \coprod_i \xi_i
 \end{aligned}$$

and by the universal property of the U^T -coequalizer of (3.2) there exists a unique morphism k such that $kq = h$. $\square$

The following gives a criteria to construct coproducts in the Eilenberg–Moore category, assuming they exist in the base category. This is possible if we assume the existence of reflexive coequalizers in the Eilenberg–Moore category.

Proposition 3.2.7. *Let (T, μ, η) be a monad on $\mathcal{X}$. If $\mathcal{X}$ has finite coproducts and the Eilenberg–Moore category $\mathcal{X}^T$ has reflexive coequalizers, then $\mathcal{X}^T$ has finite coproducts. Moreover $\mathcal{X}^T$ is finitely cocomplete.*

Proof. Let (T, μ, η) be the given monad, and suppose that $\mathcal{X}$ has finite coproducts. Let (A, ξ_A) and (B, ξ_B) be two objects in $\mathcal{X}^T$. First, we aim to prove that (A, ξ_A) and (B, ξ_B) have a coproduct in $\mathcal{X}^T$. All the coproducts in the diagram below are taken in $\mathcal{X}$, and therefore they exist. Let ι_1 and ι_2 be the canonical injections in $\mathcal{X}$ of A and B into $A + B$, respectively. Our goal is to prove that the diagram below is composed of a reflexive pair in $\mathcal{X}^T$. Then, by assumption, it admits a coequalizer in $\mathcal{X}^T$, denoted by q . This diagram correspond to the diagram (3.1) in Proposition 3.2.6.

$$\begin{array}{ccccc}
 & & T^3(A + B) & & \\
 & \nearrow T^2[T(\iota_1), T(\iota_2)] & \downarrow & \searrow T(\mu_{A+B}) & \\
 T^2(T(A) + T(B)) & \xleftarrow{T^2(\eta_A + \eta_B)} & T^2(A + B) & \xrightarrow{T(q)} & T(Q) \\
 & \xrightarrow{T^2(\xi_A + \xi_B)} & & & \\
 \downarrow \mu_{T(A) + T(B)} & & \downarrow \mu_{T(A+B)} & & \downarrow \xi_Q \\
 & \nearrow T[T(\iota_1), T(\iota_2)] & T^2(A + B) & \searrow \mu_{A+B} & \\
 T(T(A) + T(B)) & \xleftarrow{T(\eta_A + \eta_B)} & T(A + B) & \xrightarrow{q} & Q \\
 & \xrightarrow{T(\xi_A + \xi_B)} & & &
 \end{array}$$

The parallel arrows in $\mathcal{X}^T$ to be coequalized are

- $T(\xi_A + \xi_B)$
- $\mu_{A+B}T[T(\iota_1), T(\iota_2)]$

We proceed to show that they form a reflexive pair by exhibiting a common section. $T(\eta_A + \eta_B)$ is such a common section, as shown below. It follows the first composition

$$T[\xi_A, \xi_B]T(\eta_A + \eta_B) = T([\xi_A, \xi_B](\eta_A + \eta_B)) = T(1_{A+B}) = 1_{T(A+B)}$$

and the second composition, using the naturality of η .

$$\begin{aligned}
\mu_{A+B}T[T(\iota_1), T(\iota_2)]T(\eta_A + \eta_B) &= \mu_{A+B}T([T(\iota_1), T(\iota_2)](\eta_A + \eta_B)) \\
&= \mu_{A+B}T[T(\iota_1)\eta_A, T(\iota_2)\eta_B] \\
&= \mu_{A+B}T[\eta_{A+B}\iota_1, \eta_{A+B}\iota_2] \\
&= \mu_{A+B}T(\eta_{A+B}[\iota_1, \iota_2]) \\
&= \mu_{A+B}T(\eta_{A+B}1_{A+B}) \\
&= \mu_{A+B}T(\eta_{A+B}) = 1_{T(A+B)}
\end{aligned}$$

Thus it is a reflexive pair.

Let $q : T(A + B) \rightarrow Q$ be the coequalizer in $\mathcal{X}$, and ξ_Q be the algebra structure of Q . We aim to prove that (Q, ξ_Q) is the coproduct of (A, ξ_A) and (B, ξ_B) in $\mathcal{X}^T$. First we need to exhibit the canonical injections $\iota_{Q,1}$ and $\iota_{Q,2}$. They are respectively defined as the following compositions.

$$A \xrightarrow{\iota_1} A + B \xrightarrow{\eta_{A+B}} T(A + B) \xrightarrow{q} Q$$

$$B \xrightarrow{\iota_2} A + B \xrightarrow{\eta_{A+B}} T(A + B) \xrightarrow{q} Q$$

We still need to prove that they are morphisms^a in $\mathcal{X}^T$. We prove the first one, the second one is similar. Consider the following diagram. All diagrams, except the left and the external ones, commute. We aim to prove that the left diagram commutes, so that the external one commutes as well. This holds by *Proposition 3.2.6*, since it corresponds to the diagram (3.3).

$$\begin{array}{ccccccc}
T(A) & \xrightarrow{T(\iota_1)} & T(A + B) & \xrightarrow{T(\eta_{A+B})} & T^2(A + B) & \xrightarrow{T(q)} & T(Q) \\
\downarrow \xi_A & & & \searrow & \downarrow \mu_{A+B} & & \downarrow \xi_Q \\
& & & & T(A + B) & & \\
& & & & \searrow q & & \\
A & \xrightarrow{\iota_1} & A + B & \xrightarrow{\eta_{A+B}} & T(A + B) & \xrightarrow{q} & Q
\end{array}$$

Once we have the canonical injections, we need to show that they satisfy the universal property of coproducts in $\mathcal{X}^T$. Let $f_i : (A_i, \xi_i) \rightarrow (K, \xi_K)$ for $i = 1, 2$ in $\mathcal{X}^T$. We aim to prove that there exists a unique morphism $h : (Q, \xi_Q) \rightarrow (K, \xi_K)$ in $\mathcal{X}^T$ such that $h\iota_{Q,i} = f_i$ for $i = 1, 2$. Refer to

the diagram at the end of the proof for a visualization of the following arguments.

First, we show that there exists a unique morphism $g : (T(A + B), \mu_{A+B}) \rightarrow (K, \xi_K)$ in $\mathcal{X}^T$ such that $g(\eta_{A+B})\iota_i = f_i$ for $i = 1, 2$. Define $g = \xi_K T[f_1, f_2]$. It is a morphism in $\mathcal{X}^T$, in fact the following diagram commutes.

$$\begin{array}{ccccc}
 T^2(A + B) & \xrightarrow{T[f_1, f_2]} & T^2(K) & \xrightarrow{\mu_K} & T(K) \\
 \downarrow \mu_{A+B} & & \downarrow \mu_K & & \downarrow \xi_K \\
 T(A + B) & \xrightarrow{T[f_1, f_2]} & T(K) & \xrightarrow{\xi_K} & K
 \end{array}$$

Suppose now that there exists an other morphism $g' : (T(A + B), \mu_{A+B}) \rightarrow (K, \xi_K)$ such that $g'(\eta_{A+B})\iota_i = f_i$ for $i = 1, 2$. Then we have that

$$\begin{aligned}
 g(\eta_{A+B})\iota_i &= g'(\eta_{A+B})\iota_i \quad i = 1, 2 \\
 \implies g(\eta_{A+B}) &= g'(\eta_{A+B}) \\
 \implies U^T(g)(\eta_{A+B}) &= U^T(g')(\eta_{A+B}) \\
 \implies g &= g'
 \end{aligned}$$

where we used the uniqueness in the universal property of the unit and the fact that g and g' are morphisms in $\mathcal{X}^T$. So g is unique. Moreover, we have that $gT(\iota_i) = g(\eta_{A+B})\iota_i \xi_i$ for $i = 1, 2$. In fact, we have that

$$\begin{array}{ccccccc}
 & & T(A + B) & & & & \\
 & \nearrow T(\iota_1) & & \searrow T[f_1, f_2] & & & \\
 T(A) & \xrightarrow{T(f_1)} & T(K) & \xrightarrow{\xi_K} & K & & \\
 \downarrow \xi_A & & \nearrow f_1 & & \uparrow \xi_K & & \\
 & A & & & & & \\
 & \nwarrow \xi_A & & \nearrow \eta_A & & & \\
 A & \xrightarrow{\iota_1} & A + B & \xrightarrow{\eta_{A+B}} & T(A + B) & \xrightarrow{T[f_1, f_2]} & T(K) \\
 & & & & \downarrow T(\iota_1) & & \\
 & & & & T(A) & \xrightarrow{T(f_1)} & T(K)
 \end{array}$$

$$\begin{aligned}
gT(\iota_1) &= \xi_K T[f_1, f_2] T(\iota_i) = \xi_K T([f_1, f_2] \iota_i) \\
&= \xi_K T(f_1) = f_1 \xi_A = f_1 1_A \xi_A \\
&= f_1 \xi_A \eta_A \xi_A = \xi_K T(f_1) \eta_A \xi_A \\
&= \xi_K T[f_1, f_2] T(\iota_1) \eta_A \xi_A \\
&= \xi_K T[f_1, f_2] (\eta_{A+B}) \iota_1 \xi_A \\
&= g(\eta_{A+B}) \iota_1 \xi_A
\end{aligned}$$

and the same holds for B . Therefore g has equal compositions with the parallel arrows corresponding to (3.3), which has, by *Proposition 3.2.6*, q as U^T -coequalizer. Then by the universal property of the U^T -coequalizer, it follows that there exists a unique morphism $h : (Q, \xi_Q) \rightarrow (K, \xi_K)$ such that $g = hq$ in $\mathcal{X}^T$. Thus, we have that

$$h\iota_{Q,i} = hq(\eta_{A+B})\iota_i = g(\eta_{A+B})\iota_i = f_i \quad i = 1, 2$$

and by the uniqueness of g , we have that h is the unique morphism in $\mathcal{X}^T$ satisfying such condition.



In order to prove that also finite coproducts exists, it suffices to show that the initial object exists. Let 0 be the initial object in $\mathcal{X}$. It holds that $(T(0), \mu_0)$ is the initial object in $\mathcal{X}^T$. In fact, let (A, ξ_A) be an object in $\mathcal{X}^T$. Since 0 is initial in $\mathcal{X}$, there exists a unique morphism $!^A : 0 \rightarrow A$ in $\mathcal{X}$. Then, the morphism $f = \xi_A T(!^A)$ is a morphism in $\mathcal{X}$ from $(T(0), \mu_0)$ to (A, ξ_A) .

$$T(0) \xrightarrow{T(!^A)} T(A) \xrightarrow{\xi_A} A$$

We aim to prove that it is a morphism in $\mathcal{X}^T$ and that it is the unique morphism in $\mathcal{X}^T$ from $(T(0), \mu_0)$ to (A, ξ_A) . It is a morphism in $\mathcal{X}^T$, in fact, using property of algebra $\xi_A T(\xi_A) = \xi_A \mu_A$ and the naturality of μ , we have that

$$\begin{aligned} \xi_A T(f) &= \xi_A T(\xi_A T(!^A)) \\ &= \xi_A T(\xi_A) T^2(!^A) \\ &= \xi_A \mu_A T^2(!^A) \\ &= \xi_A T(!^A) \mu_0 \\ &= f \mu_0 \end{aligned}$$

$$\begin{array}{ccc} T^2(A) & \xrightarrow{T(\xi_A)} & T(A) \\ \downarrow \mu_A & & \downarrow \xi_A \\ T(A) & \xrightarrow{\xi_A} & A \end{array} \qquad \begin{array}{ccc} T^2(0) & \xrightarrow{T(!^A)} & T^2(A) \\ \downarrow \mu_0 & & \downarrow \mu_A \\ T(0) & \xrightarrow{T(!^A)} & T(A) \end{array}$$

Moreover, it is the unique morphism in $\mathcal{X}^T$ from $(T(0), \mu_0)$ to (A, ξ_A) . In fact, let $f' : (T(0), \mu_0) \rightarrow (A, \xi_A)$ be an other morphism in $\mathcal{X}^T$. Then, it holds that

- $f \mu_0 = \xi_A T(f)$
- $f' \mu_0 = \xi_A T(f')$

and, composing both sides with $T(\eta_0)$, we have that

- $f = f 1_{T(0)} = f \mu_0 T(\eta_0) = \xi_A T(f) T(\eta_0) = \xi_A T(f \eta_0) = \xi_A T(!^A)$
- $f' = f' 1_{T(0)} = f' \mu_0 T(\eta_0) = \xi_A T(f') T(\eta_0) = \xi_A T(f' \eta_0) = \xi_A T(!^A)$

since $!^A$ is the unique morphism from 0 to A . Thus $f = f'$.

It remains to prove that $\mathcal{X}^T$ is finitely cocomplete. We already know that it has finite coproducts, so we only need to show that it has coequalizers. This follows from the fact that it has finite coproducts and reflexive coequalizers; in fact each coequalizer can be constructed as a reflexive coequalizer from the coproduct, as shown below.

$$\begin{array}{ccc} A & \xrightarrow[f]{g} & B \\ A + B & \xleftarrow{[f, 1_B]} \xrightarrow{[g, 1_B]} & B \end{array}$$

□

^aNote that ι_1 and η_{A+B} are not morphisms in $\mathcal{X}^T$. This is not a problem, however, since we will show that the whole composition $q\eta_{A+B}\iota_1$ is a morphism in $\mathcal{X}^T$

We now want to study a sufficient condition for a category to admit reflexive coequalizers, so that we can apply the proposition above.

Recall that a pair (f, g) from A to B in a regular category induces a relation with the following construction. Consider the morphism induced by the universal property $\langle f, g \rangle$. Since the category is regular, we can take the factorization $\langle f, g \rangle = \langle r_1, r_2 \rangle e$, with $\langle r_1, r_2 \rangle$ mono and e regular epi. Then $\langle r_1, r_2 \rangle$ is the relation induced.

$$\begin{array}{ccc} A & \xrightarrow{\langle f, g \rangle} & B \times B \\ & \searrow e & \nearrow \langle r_1, r_2 \rangle \\ & R & \end{array}$$

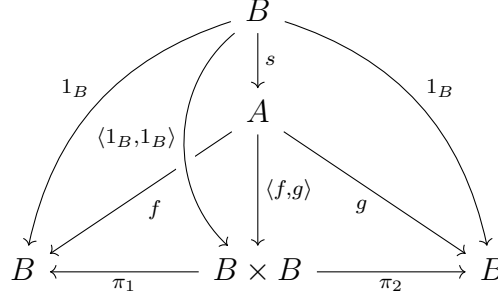
Proposition 3.2.8. *The internal relation induced by a reflexive pair (f, g) in a regular category is reflexive.*

Proof. Consider the pair (f, g) from A to B and the section s with $fs = 1_B$ and $gs = 1_B$.

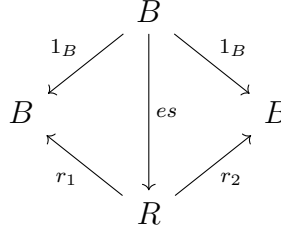
$$A \xrightarrow[f]{s} B$$

Let $\langle f, g \rangle = \langle r_1, r_2 \rangle e$ be the (regular epi, mono) factorization. Moreover,

by the universal property we have that $\langle f, g \rangle s = \langle 1_B, 1_B \rangle$.



Thus $\langle r_1, r_2 \rangle es = \langle 1_B, 1_B \rangle$. The morphism es makes the relation reflexive.



□

Proposition 3.2.9. *Let $\mathcal{A}$ be an exact Mal'tsev category. Then $\mathcal{A}$ has reflexive coequalizers.*

Proof. Assume $\mathcal{A}$ is exact and Mal'tsev. We want to prove that $\mathcal{A}$ has reflexive coequalizer. Consider a reflexive pair (f, g) from A to B . By *Proposition 3.2.8* the internal relation induced $\langle r_1, r_2 \rangle$ is reflexive. Since $\mathcal{A}$ is Mal'tsev, then each internal reflexive relation is an equivalence relation. Hence $\langle r_1, r_2 \rangle$ is an equivalence relation. $\mathcal{A}$ is also exact, then, by *Proposition A.0.9* the equivalence relation (r_1, r_2) has a coequalizer k .

We want to prove that k is also the coequalizer of (f, g) . First we note that $kf = kr_1e = kr_2e = kg$. It remains to prove the universal property. Let $h : B \rightarrow C$ be a morphism such that $hf = hg$. Then $hr_1e = hr_2e$. Since e is a regular epi, hence by *Proposition A.0.8* an epi, we have that

$hr_1 = hr_2$. Thus there exists a unique morphism t such that $th = k$.

$$\begin{array}{ccccc}
 A & & & & \\
 \downarrow e & \searrow f & & & \\
 & g & & & C \\
 R & \xrightarrow[r_2]{r_1} & B & \xrightarrow[k]{} & \text{Coeq}(r_1, r_2) \\
 & & \nearrow h & & \downarrow \exists! t
 \end{array}$$

□

Chapter 4

Effective descent morphisms

In this chapter we prove a monadicity theorem for the pullback functor. The standard terminology in the literature is *descent morphism* for a morphism whose pullback functor is premonadic, and *effective descent morphism* for a morphism whose pullback functor is monadic. We do not aim to develop descent theory; therefore, we will stick to the monadic theory nomenclature.

This theorem is usually stated in the literature in the descent theory framework, and then by a comparison theorem it holds in the monadic theory. The reader can refer to [JST03] for the first approach. We aim to prove this monadicity theorem directly in monadic theory, without the need of the comparison theorem. This part is based on the work of Sobral and Tholen [ST92] and some ideas drawn from Janelidze and Tholen [JT91], but we introduce some substantial changes to suit our context. In fact, [ST92] assumes that the framework category has all coequalizers, which is not the case in our setting. Our approach adapts their construction while relaxing this assumption.

First, we state the theorem.

Theorem. *In an exact category, if $p : E \rightarrow B$ is a regular epi, then p^* is a monadic functor.*

The reader may refer to *Section 2.2* for the preliminaries of this discussion. We recall that a right adjoint is monadic if the comparison functor $K : (\mathcal{A} \downarrow B) \rightarrow (\mathcal{A} \downarrow E)^T$ is an equivalence of categories, where $T = p^*p_!$ is the monad

associated to p .

$$\begin{array}{ccc}
 (\mathcal{A} \downarrow B) & \xrightarrow{K} & (\mathcal{A} \downarrow E)^T \\
 \uparrow p! & \nearrow F^T & \\
 (\mathcal{A} \downarrow E) & \xleftarrow{U^T} &
 \end{array}$$

Let $\mathcal{A}$ be a category with pullbacks and let $p : E \rightarrow B$ be a morphism in $\mathcal{A}$. Let $(C, \gamma; \xi)$ be an object in $(\mathcal{A} \downarrow E)^T$ with $\xi : T(C) \rightarrow C$ an algebra.

Consider the pullback in $\mathcal{A}$.

$$\begin{array}{ccc}
 E \times_B C & \xrightarrow{\text{proj}_{2,C}} & C \\
 \text{proj}_{1,C} \downarrow & \nearrow \gamma & \downarrow p\gamma \\
 E & \xrightarrow{p} & B
 \end{array}$$

Thus, we have the span in $\mathcal{A}$:

$$\begin{array}{ccc}
 & E \times_B C & \\
 \text{proj}_{2,C} \swarrow & & \searrow \xi \\
 C & & C
 \end{array}$$

Since ξ is also a morphism in $(\mathcal{A} \downarrow E)$, we have that $\gamma\xi = \text{proj}_{1,C}$

$$\begin{array}{ccc}
 E \times_B C & \xrightarrow{\xi} & C \\
 \text{proj}_{1,C} \searrow & & \swarrow \gamma \\
 & E &
 \end{array}$$

In the following for the p^* -image of a morphism f in $(\mathcal{A} \downarrow B)$ we will use both notation $p^*(f)$ and $1_E \times_C f$.

We recall that the unit η of the monad T in an object (C, γ) in $(\mathcal{A} \downarrow E)$ is the morphism $\langle \gamma, 1_C \rangle$ induced by the universal property of the pullback of $p\gamma$ along p and the counit ϵ in an object (D, ζ) in $(\mathcal{A} \downarrow B)$ is the morphism $\text{proj}_{2,D} : E \times_B D \rightarrow D$ given by the pullback of p along ζ .

Lemma 4.0.1. *Let $\mathcal{A}$ be a finitely complete category and let $p : E \rightarrow B$ be a morphism in $\mathcal{A}$. For any object $(C, \gamma; \xi)$ in $(\mathcal{A} \downarrow E)^T$, with T the monad associated to p , the morphism ξ is the coequalizer of $(1_E \times_B \xi, 1_E \times_B \text{proj}_{2,C})$.*

Proof. Let $(C, \gamma; \xi)$ be an object in $(\mathcal{A} \downarrow E)^T$. First we aim to show that $\xi(1_E \times_B \xi) = \xi(1_E \times_B \text{proj}_{2,C})$. This holds because ξ is an algebra for the monad $T = (p^*p_!, p^*\epsilon p_!, \eta)$. Observe that the multiplication in an object (C, γ) is

$$p^*\epsilon p_!(C, \gamma) = p^*(\epsilon_{(C, p\gamma)}) = p^*(\text{proj}_{2,C}) = 1_E \times_B \text{proj}_{2,C}$$

Then we have the algebra property

$$\begin{array}{ccc} E \times_B (E \times_B C) & \xrightarrow{1_E \times_B \xi} & E \times_B C \\ \downarrow 1_E \times_B \text{proj}_{2,C} & & \downarrow \xi \\ E \times_B C & \xrightarrow{\xi} & C \end{array}$$

$$\xi(1_E \times_B \text{proj}_{2,C}) = \xi(1_E \times_B \xi)$$

It remains to prove the universal property of the coequalizer. First we observe that ξ is a split epi, in fact $\xi\eta_{(C,\gamma)} = 1_C$. Thus ξ is the coequalizer of $(1_E \times_B C, \eta_{(C,\gamma)}\xi)$. We aim to prove that ξ is also the coequalizer of $(1_E \times_B \xi, 1_E \times_B \text{proj}_{2,C})$. We already know that the universal property holds for ξ with $(1_E \times_B C, \eta_{(C,\gamma)}\xi)$, so it suffices to show that there exist a morphism $t : E \times_B C \rightarrow E \times_B (E \times_B C)$ such that

- $(1_E \times_B \xi)t = \eta_{(C,\gamma)}\xi$
- $(1_E \times_B \text{proj}_{2,C})t = 1_E \times_B C$

Then given a morphism f such that $f(1_E \times_B \text{proj}_{2,C}) = f(1_E \times_B \xi)$, we have, composing with t ,

$$\begin{aligned} f(1_E \times_B \text{proj}_{2,C})t &= f(1_E \times_B \xi)t \\ \implies f1_E \times_B C &= f\eta_{(C,\gamma)}\xi \end{aligned}$$

By the universal property for ξ with $(1_E \times_B C, \eta_{(C,\gamma)}\xi)$ we obtain the universal property for ξ with $(1_E \times_B \xi, 1_E \times_B \text{proj}_{2,C})$.

$$\begin{array}{ccccc} & & E \times_B C & & \\ & \swarrow t & \downarrow 1_E \times_B C & \downarrow \eta_{(C,\gamma)}\xi & \\ E \times_B (E \times_B C) & \xrightarrow[1_E \times_B \text{proj}_{2,C}]{1_E \times_B \xi} & E \times_B C & \xrightarrow{f} & F \\ & & \downarrow \xi & \nearrow \exists! g & \\ & & C & & \end{array}$$

We now have to prove that such a morphism t actually exists. Define $t = \langle \text{proj}_{1,C}, 1_{E \times_B C} \rangle$. We aim to prove that this morphism satisfies the two conditions above, ie the following diagram of pairs commutes.

$$\begin{array}{ccc}
 & E \times_B C & \\
 \langle \text{proj}_{1,C}, 1_{E \times_B C} \rangle \swarrow & \downarrow 1_{E \times_B C} \quad \eta_{(C,\gamma)}\xi & \\
 E \times_B (E \times_B C) & \xrightarrow[1_{E \times_B \text{proj}_{2,C}}]{1_{E \times_B \xi}} & E \times_B C
 \end{array}$$

In order to prove that we use the jointly monic pair $(\text{proj}_{1,C}, \text{proj}_{2,C})$. The following compositions hold.

- $\text{proj}_{2,C}(1_E \times_B \xi) \langle \text{proj}_{1,C}, 1_{E \times_B C} \rangle = \xi \text{proj}_{2,E \times_B C} \langle \text{proj}_{1,C}, 1_{E \times_B C} \rangle = \xi 1_{E \times_B C} = \xi$
- $\text{proj}_{2,C} \eta_{(C,\gamma)} \xi = \text{proj}_{2,C} \langle \gamma, 1_C \rangle \xi = 1_C \xi = \xi$
- $\text{proj}_{2,C}(1_E \times_B \text{proj}_{2,C}) \langle \text{proj}_{1,C}, 1_{E \times_B C} \rangle = \text{proj}_{2,C} \text{proj}_{2,E \times_B C} \langle \text{proj}_{1,C}, 1_{E \times_B C} \rangle = \text{proj}_{2,C} 1_{E \times_B C} = \text{proj}_{2,C}$
- $\text{proj}_{2,C} 1_{E \times_B C} = \text{proj}_{2,C}$
- $\text{proj}_{1,C}(1_E \times_B \xi) \langle \text{proj}_{1,C}, 1_{E \times_B C} \rangle = \text{proj}_{1,E \times_B C} \langle \text{proj}_{1,C}, 1_{E \times_B C} \rangle = \text{proj}_{1,C}$
- $\text{proj}_{1,C} \eta_{(C,\gamma)} \xi = \text{proj}_{1,C} \langle \gamma, 1_C \rangle \xi = \gamma \xi = \text{proj}_{1,C}$
- $\text{proj}_{1,C}(1_E \times_B \text{proj}_{2,C}) \langle \text{proj}_{1,C}, 1_{E \times_B C} \rangle = \text{proj}_{1,E \times_B C} \langle \text{proj}_{1,C}, 1_{E \times_B C} \rangle = \text{proj}_{1,C}$
- $\text{proj}_{1,C} 1_{E \times_B C} = \text{proj}_{1,C}$.

Thus $t = \langle \text{proj}_{1,C}, 1_{E \times_B C} \rangle$ satisfies the conditions required. $\square$

Lemma 4.0.2. *Let $\mathcal{A}$ be a finitely complete category and let $p : E \rightarrow B$ be a morphism in $\mathcal{A}$. For any object $(C, \gamma; \xi)$ in $(\mathcal{A} \downarrow E)^T$, with T the monad associated to p , the pair $(\text{proj}_{2,C}, \xi)$ is an equivalence relation in $\mathcal{A}$.*

Proof. First we show that $(\text{proj}_{2,C}, \xi)$ is an internal relation. We need to show that $\langle \text{proj}_{2,C}, \xi \rangle$ is a mono. Consider the following diagram of solid arrows¹. It commutes, in fact

- $(p\gamma)\text{proj}_{2,C} = 1_B(p\gamma)\text{proj}_{2,C} = p\text{proj}_{1,C} = p\gamma\xi$
- $1_B(p\gamma)p_2 = 1_B(p\gamma)p_1 = p\gamma p_1$

Thus we can consider the morphism h' induced by the universal property of the pullback from p_2 and γp_1 , and h similarly induced from $\text{proj}_{2,C}$ and ξ .



In order to prove that $\langle \text{proj}_{2,C}, \xi \rangle$ is a mono, we first need to show that h is a mono. Since $(\text{proj}_{1,C}, \text{proj}_{2,C})$ is a jointly monic pair, by the following compositions

- $\text{proj}_{1,C}h'h = \gamma p_1 h = \gamma\xi = \text{proj}_{1,C} = \text{proj}_{1,C}1_{E \times_B C}$
- $\text{proj}_{2,C}h'h = p_2 h = \text{proj}_{2,C} = \text{proj}_{2,C}1_{E \times_B C}$

we have that $h'h = 1_{E \times_B C}$. This implies that h is a mono, in fact suppose

¹[ST92] assumes the existence of all coequalizers, which is not the case in our setting. However, this is not a problem: at this point, we can use the morphism $p\gamma$ instead of the coequalizer of $(\text{proj}_{2,C}, \xi)$, as it serves our purpose. Then, *a posteriori*, this proposition ensures, under the assumption of an exact category, that the pair $(\text{proj}_{2,C}, \xi)$ has a coequalizer, which we will make use of, following the approach of the aforementioned work.

that f and g are two morphisms such that $hf = hg$, then we have

$$\begin{aligned}
 hf &= hg \\
 \implies h'hf &= h'hg \\
 \implies 1_{E \times_B C} f &= 1_{E \times_B C} g \\
 \implies f &= g
 \end{aligned}$$

Consider now the canonical morphism $i : C \times_B C \rightarrow C \times C$ induced by the universal property of the product. We have that i is a mono, in fact if f and g are two morphisms such that $if = ig$, then we have

$$\begin{array}{ccccc}
 & & & C & \\
 & & & \uparrow p'_1 & \\
 A & \xrightarrow[f]{g} & C \times_B C & \xrightarrow{i} & C \times C \\
 & & & \downarrow p'_2 & \\
 & & & C &
 \end{array}$$

p_1 (arrow from $C \times_B C$ to C), p_2 (arrow from $C \times_B C$ to C)

- $if = ig \implies p'_1 if = p'_1 ig \implies p_1 f = p_1 g$
- $if = ig \implies p'_2 if = p'_2 ig \implies p_2 f = p_2 g$

Since (p_1, p_2) is a jointly monic pair, we have that $f = g$. Thus, i is a mono. Since i and h are both monos, we have that ih is a mono.

By the following commutative diagram

$$\begin{array}{ccccc}
 & & \xi & & \\
 & & \curvearrowright & & \\
 E \times_B C & \xrightarrow{h} & C \times_B C & \xrightarrow{i} & C \times C \\
 & & & & \uparrow p'_1 \\
 & & & & C \\
 & & & & \downarrow p'_2 \\
 & & & & C \\
 & & \text{proj}_{2,C} & &
 \end{array}$$

p_1 (arrow from $C \times_B C$ to C), p_2 (arrow from $C \times_B C$ to C)

we conclude that $ih = \langle \text{proj}_{2,C}, \xi \rangle$, so $\langle \text{proj}_{2,C}, \xi \rangle$ is a mono.

We proved that $(\text{proj}_{2,C}, \xi)$ is an internal relation, now we need to show that it is an equivalence, ie it is reflexive, symmetric and transitive.

- Claim: $(\text{proj}_{2,C}, \xi)$ is reflexive.

Called η the unit of the monad, we have that $\langle \text{proj}_{2,C}, \xi \rangle \eta_{(C,\gamma)} = \langle 1_C, 1_C \rangle$. In fact $\xi \eta_{(C,\gamma)} = 1_C$ and by the definition of the unit of the monad, we have that $\text{proj}_{2,C} \eta_{(C,\gamma)} = \text{proj}_{2,C} \langle \gamma, 1_C \rangle = 1_C$.

$$\begin{array}{ccccc}
 & & C & & \\
 & \swarrow 1_C & \downarrow \eta_{(C,\gamma)} & \searrow 1_C & \\
 C & \xleftarrow{\text{proj}_{2,C}} & E \times_B C & \xrightarrow{\xi} & C
 \end{array}$$

- Claim: $(\text{proj}_{2,C}, \xi)$ is symmetric.

Consider the following diagram, where $\text{proj}_{1,E \times_B C}$ and $\text{proj}_{2,E \times_B C}$ are the projections of $E \times_B (E \times_B C)$. Let l be the morphism induced by the universal property of the pullback from $1_{E \times_B C}$ and $\gamma \text{proj}_{2,C}$, in particular it is the unique such that

$$\gamma \text{proj}_{2,C} = \text{proj}_{1,E \times_B C} l = \text{proj}_{1,C} (1_E \times \text{proj}_{2,C}) l$$

and

$$\begin{array}{c}
 1_{E \times_B C} = \text{proj}_{2,E \times_B C} l \\
 \begin{array}{ccccc}
 & & E \times_B C & & \\
 & \searrow 1_{E \times_B C} & & \searrow & \\
 & & E \times_B (E \times_B C) & \xrightarrow{\text{proj}_{2,E \times_B C}} & E \times_B C \\
 & \swarrow l & \downarrow 1_E \times_B \text{proj}_{2,C} & \downarrow 1_E \times_B \xi & \downarrow \text{proj}_{2,C} \\
 E \times_B C & \xrightarrow{\text{proj}_{2,C}} & C & \xrightarrow{\xi} & C \\
 \downarrow \text{proj}_{2,C} & \swarrow \eta_{(C,\gamma)} & \downarrow \text{proj}_{1,C} & \downarrow \text{proj}_{1,E \times_B C} & \downarrow p\gamma \\
 C & \xrightarrow{\gamma} & E & \xrightarrow{p} & B
 \end{array}
 \end{array}$$

Define $\sigma = (1_E \times_B \xi) l$. Observe that

$$\text{proj}_{2,C} \sigma = \text{proj}_{2,C} (1_E \times_B \xi) l = \xi \text{proj}_{2,E \times_B C} l = \xi 1_{E \times_B C} = \xi$$

It remains to show that $\xi\sigma = \text{proj}_{2,C}$. First, by the following compositions and the fact that $(\text{proj}_{1,C}, \text{proj}_{2,C})$ is a jointly monic pair

- $\text{proj}_{1,C}(1_E \times_B \text{proj}_{2,C})l = \text{proj}_{1,E \times_B C}l = \gamma \text{proj}_{2,C}$
- $\text{proj}_{1,C}\eta_{(C,\gamma)}\text{proj}_{2,C} = \text{proj}_{1,C}\langle \gamma, 1_C \rangle \text{proj}_{2,C} = \gamma \text{proj}_{2,C}$
- $\text{proj}_{2,C}(1_E \times_B \text{proj}_{2,C})l = \text{proj}_{2,C}\text{proj}_{2,E \times_B C}l = \text{proj}_{2,C}1_{E \times_B C} = \text{proj}_{2,C}$
- $\text{proj}_{2,C}\eta_{(C,\gamma)}\text{proj}_{2,C} = \text{proj}_{2,C}\langle \gamma, 1_C \rangle \text{proj}_{2,C} = 1_C \text{proj}_{2,C} = \text{proj}_{2,C}$

we have that $\eta_{(C,\gamma)}\text{proj}_{2,C} = (1_E \times_B \text{proj}_{2,C})l$. Hence, since by *Lemma 4.0.1* ξ coequalizes $(1_E \times_B \xi, 1_E \times_B \text{proj}_{2,C})$, it holds that

$$\xi\sigma = \xi(1_E \times_B \xi)l = \xi(1_E \times_B \text{proj}_{2,C})l = \xi\eta_{(C,\gamma)}\text{proj}_{2,C} = 1_C \text{proj}_{2,C} = \text{proj}_{2,C}$$

as claimed. Thus the following diagram commutes.

$$\begin{array}{ccccc}
 & & E \times_B C & & \\
 & \swarrow \xi & \downarrow \sigma = (1_E \times_B \xi)l & \searrow \text{proj}_{2,C} & \\
 C & & & & C \\
 & \nwarrow \text{proj}_{2,C} & & \nearrow \xi & \\
 & & E \times_B C & &
 \end{array}$$

- Claim: $(\text{proj}_{2,C}, \xi)$ is transitive.

By *Property 2.2.4* the pullback of ξ and $\text{proj}_{2,C}$ is the square

$$\begin{array}{ccc}
 E \times_B (E \times_B C) & \xrightarrow{1_E \times_B \xi} & E \times_B C \\
 \downarrow \text{proj}_{2,E \times_B C} & & \downarrow \text{proj}_{2,C} \\
 E \times_B C & \xrightarrow{\xi} & C
 \end{array}$$

We aim to prove that there exists a morphism $\tau : E \times_B (E \times_B C) \rightarrow E \times_B C$ such that

- $\xi(1_E \times_B \xi) = \xi\tau$
- $\text{proj}_{2,C}\text{proj}_{2,E \times_B C} = \text{proj}_{2,C}\tau$

Define $\tau = (1_E \times_B \text{proj}_{2,C})$, we have that

- $\xi(1_E \times_B \xi) = \xi(1_E \times_B \text{proj}_{2,C})$
- $\text{proj}_{2,C} \text{proj}_{2,E \times_B C} = \text{proj}_{2,C}(1_E \times_B \text{proj}_{2,C})$

In particular the first condition holds by *Lemma 4.0.1*, and the second one holds by the definition of $(1_E \times_B \text{proj}_{2,C})$

□

In an exact category each equivalence relation is a kernel pair and by *Proposition A.0.9* has a coequalizer. Thus in $\mathcal{A}$ exact category $(\text{proj}_{2,C}, \xi)$, by *Lemma 4.0.2* has a coequalizer. Let $q : C \rightarrow Q$ be such coequalizer. Observe that, considering $\text{proj}_{2,C}$ and ξ as morphism in $(\mathcal{A} \downarrow B)$, then q is a morphism in $(\mathcal{A} \downarrow B)$ as well by the universal property. So, called δ the morphism induced by the universal property of the coequalizer from $\text{proj}_{2,C}$ and ξ , we have that q with codomain (Q, δ) is also the coequalizer of the pair in $(\mathcal{A} \downarrow B)$.

$$\begin{array}{ccccc}
 E \times_B C & \xrightarrow{\xi} & C & \xrightarrow{q} & Q \\
 & \searrow \text{proj}_{2,C} & \downarrow p\gamma & \swarrow \delta & \\
 & & B & &
 \end{array}$$

$p\text{proj}_{1,C}$ (arrow from $E \times_B C$ to B)

Consider the comparison functor $K : (\mathcal{A} \downarrow B) \rightarrow (\mathcal{A} \downarrow E)^T$ defined as $K(D, \zeta) = (p^*(D, \zeta), p^*(\epsilon_{(D, \zeta)}))$, where ϵ is the counit of the adjunction, and $\epsilon_{(D, \zeta)}$ is defined as the projection from $E \times_B D$ to D of the pullback of p along ζ .

In our context of exact category, turns out that the comparison functor K has a left adjoint $L : (\mathcal{A} \downarrow E)^T \rightarrow (\mathcal{A} \downarrow B)$ defined as $L(C, \gamma; \xi) = (Q, \delta)$, where (Q, δ) is the codomain of coequalizer of the pair $(\text{proj}_{2,C}, \xi)$ in $(\mathcal{A} \downarrow B)$.

$$\begin{array}{ccc}
 (\mathcal{A} \downarrow B) & \xrightleftharpoons[L]{K} & (\mathcal{A} \downarrow E)^T \\
 \uparrow p! & \nearrow F^T & \\
 \downarrow p^* & \nwarrow U^T & \\
 (\mathcal{A} \downarrow E) & &
 \end{array}$$

Let us introduce the counit ρ and counit α of the adjunction $K \dashv L$. For each object (D, ζ) in $(\mathcal{A} \downarrow B)$ the counit is a morphism of the kind $\rho_{(D, \zeta)} : LK(D, \zeta) \rightarrow (D, \zeta)$. First we compute $K(D, \zeta)$.

$$K(D, \zeta) = (E \times_B D, \text{proj}_{1,D}; p^*(\text{proj}_{2,D}))$$

$$\begin{array}{ccc}
E \times_B D & \xrightarrow{\text{proj}_{2,D}} & D \\
\downarrow \text{proj}_{1,D} & & \downarrow \zeta \\
E & \xrightarrow{p} & B
\end{array}$$

Then, call $q' : E \times_B D \rightarrow Q'$ the coequalizer of the equivalence relation $(\text{proj}_{2,E \times_B D}, p^*(\text{proj}_{2,D}))$, as done above we make it a coequalizer in $(\mathcal{A} \downarrow B)$ via the universal property. We obtain $q' : (E \times_B D, p\text{proj}_{1,D}) \rightarrow (Q', \delta')$. Then compute $LK(D, \zeta)$. By definition of the pullback functor, the top left square in the diagram below commutes. Thus we can define, as shown in the diagram, the morphism $\rho_{(D, \zeta)}$ by the universal property of the coequalizer q' .

$$\begin{array}{ccccc}
E \times_B (E \times_B D) & \xrightarrow[p\text{proj}_{2,E \times_B D}]{p^*(\text{proj}_{2,D})} & E \times_B D & \xrightarrow{q'} & Q' \\
\downarrow \text{proj}_{2,E \times_B D} & & \downarrow \text{proj}_{2,D} & \nearrow \rho_{(D, \zeta)} & \\
E \times_B D & \xrightarrow{\text{proj}_{2,D}} & D & & \\
\downarrow \text{proj}_{1,D} & \nearrow \text{proj}_{1,D} & \downarrow \zeta & \searrow \delta' & \\
E & \xrightarrow{p} & B & &
\end{array} \tag{4.1}$$

Observe that δ' was also defined as the morphism induced by the universal property of the coequalizer, considering the object $E \times_B D$ as an object in $(\mathcal{A} \downarrow B)$ via $p\text{proj}_{1,D}$. Thus $\rho_{(D, \zeta)}$ is a morphism in $(\mathcal{A} \downarrow B)$, ie $\zeta \rho_{(D, \zeta)} = \delta'$.

For each object $(C, \gamma; \xi)$ in $(\mathcal{A} \downarrow E)^T$ the unit is a morphism of the kind $\alpha_{(C, \gamma; \xi)} : (C, \gamma; \xi) \rightarrow KL(C, \gamma; \xi)$. We compute $KL(C, \gamma; \xi)$. Let $q : C \rightarrow Q$ be the coequalizer of the equivalence relation $(\text{proj}_{2,C}, \xi)$ in $\mathcal{A}$. By the universal property of the coequalizer, we have that q is a morphism in $(\mathcal{A} \downarrow B)$ via $p\gamma$. Thus, we can consider the coequalizer of the equivalence relation $(\text{proj}_{2,C}, \xi)$ in $(\mathcal{A} \downarrow B)$, which we denote by $q : (C, p\gamma) \rightarrow (Q, \delta)$. By functoriality, we have that $p^*(q)p^*(\xi) = p^*(q)p^*(\text{proj}_{2,C})$. Moreover, it holds that ξ is the coequalizer of $(p^*(\xi), p^*(\text{proj}_{2,C}))$ by *Lemma 4.0.1*. Thus we can define $\alpha_{(C, \gamma; \xi)}$ as the morphism induced by the universal property of the coequalizer q , as done in

the diagram below. One can check that $\alpha_{(C,\gamma;\xi)}$ is a morphism in $(\mathcal{A} \downarrow E)^T$.

$$\begin{array}{ccccc}
 E \times_B (E \times_B C) & \xrightarrow[p^*(\text{proj}_{2,C})]{p^*(\xi)} & E \times_B C & \xrightarrow{p^*(q)} & E \times_B Q \\
 \downarrow \text{proj}_{2,E \times_B C} & & \downarrow \text{proj}_{2,C} & \searrow \alpha_{(C,\gamma;\xi)} & \downarrow \text{proj}_{2,Q} \\
 E \times_B C & \xrightarrow[\text{proj}_{2,C}]{\xi} & C & \xrightarrow{q} & Q \\
 \downarrow \text{proj}_{1,C} & \nearrow \gamma & \downarrow p\gamma & \nwarrow \delta & \\
 E & \xrightarrow{p} & B & &
 \end{array} \tag{4.2}$$

One can show that ρ and α are natural transformations, and that they satisfy the universal property of the counit / unit. This follows from the universal property of the coequalizer of the equivalence relation considered.

Our goal is to show that p^* is a monadic functor, ie that the comparison functor K is an equivalence of categories. We prove that K is an adjoint equivalence, which is the same. In order to do so, we aim to prove that the unit α and the counit ρ of the adjunction $L \dashv K$ are pointwise isomorphisms.

Proposition 4.0.3. *In an exact category if $p : E \rightarrow B$ is a regular epi, then the counit ρ of the adjunction $L \dashv K$ is a pointwise isomorphism.*

Proof. We aim to prove that for each object (D, ζ) in $(\mathcal{A} \downarrow B)$, the morphism $\rho_{(D,\zeta)} : LK(D, \zeta) \rightarrow (D, \zeta)$ is an isomorphism. Consider the diagram (4.1) of the definition of the counit $\rho_{(D,\zeta)}$. An exact category is also regular, in particular regular epis are pullback stable. If p is a regular epi, then $\text{proj}_{2,D}$ is a regular epi as well. The top left square is a pullback by *Property 2.2.4*, thus $(p^*(\text{proj}_{2,D}), \text{proj}_{2,E \times_B D})$ is the kernel pair of $\text{proj}_{2,D}$. Since $\text{proj}_{2,D}$ is a regular epi, then, by *Proposition A.0.6*, it is the coequalizer of its kernel pair $(p^*(\text{proj}_{2,D}), \text{proj}_{2,E \times_B D})$. We have that q' and $\text{proj}_{2,D}$ are coequalizers of the same pair, thus the morphism $\rho_{(D,\zeta)}$ defined by the universal property of the coequalizer q is actually an isomorphism. $\square$

Proposition 4.0.4. *In an exact category the unit α of the adjunction $L \dashv K$ is a pointwise isomorphism.*

Proof. We aim to prove that for each object (C, γ, ξ) in $(\mathcal{A} \downarrow E)^T$, the morphism $\alpha_{(C, \gamma; \xi)} : (C, \gamma; \xi) \rightarrow KL(C, \gamma; \xi)$ is an isomorphism. Consider the diagram (4.2). An exact category is also regular, in particular regular epis are pullback stable. Since q is a regular epi, then $p^*(q)$ is a regular epi as well. Moreover, since q is the coequalizer of $(\xi, \text{proj}_{2,C})$ which is a kernel pair, then by *Proposition A.0.7* $(\xi, \text{proj}_{2,C})$ is the kernel pair of q . Then, since p^* is a right adjoint, it preserves kernel pairs, thus $(p^*(\xi), p^*(\text{proj}_{2,C}))$ is the kernel pair of $p^*(q)$. We already proved that $p^*(q)$ is a regular epi, then by *Proposition A.0.6* it is the coequalizer of its kernel pair $(p^*(\xi), p^*(\text{proj}_{2,C}))$. By *Lemma 4.0.1* also ξ is the coequalizer of $(p^*(\xi), p^*(\text{proj}_{2,C}))$. We have that $p^*(q)$ and ξ are coequalizers of the same pair, thus the morphism $\alpha_{(C, \gamma; \xi)}$ defined by the universal property of the coequalizer ξ is actually an isomorphism. $\square$

Finally, we have the main result we were aiming to prove.

Theorem 4.0.5. *In an exact category, if $p : E \rightarrow B$ is a regular epi, then p^* is a monadic functor.*

Proof. The exactness of the category implies that the comparison functor $K : (\mathcal{A} \downarrow B) \rightarrow (\mathcal{A} \downarrow E)^T$ has a left adjoint $L : (\mathcal{A} \downarrow E)^T \rightarrow (\mathcal{A} \downarrow B)$, as shown. Since p is a regular epi, the unit α and the counit ρ of the adjunction $L \dashv K$ are pointwise isomorphisms, by *Proposition 4.0.3* and *Proposition 4.0.4*. Thus, $L \dashv K$ is an adjoint equivalence. This implies that K is an equivalence of categories. So $p^* : (\mathcal{A} \downarrow B) \rightarrow (\mathcal{A} \downarrow E)$ is a monadic functor. $\square$

Chapter 5

Ideally exact categories

5.1 Definition and characterization

We are now ready to prove the theorem, which will lead to the definition of the ideally exact categories. The following is based on the work of Janelidze [Jan24]. The next theorem presents a slight variation of the original result.

Theorem 5.1.1. *Let $\mathcal{A}$ be a category. The following are equivalent:*

1. $\mathcal{A}$ is exact and protomodular, has finite coproducts and the morphism $! : 0 \rightarrow 1$ is a regular epimorphism.
2. $\mathcal{A}$ is exact¹ and there exists $\mathcal{X}$ semiabelian category with a monadic functor $U : \mathcal{A} \rightarrow \mathcal{X}$.
3. There exists $\mathcal{X}$ semiabelian category with a monadic functor $U : \mathcal{A} \rightarrow \mathcal{X}$ and F left adjoint to U such that the associated monad $T = UF$ preserves regular epimorphisms and kernel pairs.

$$\begin{array}{ccc}
 \mathcal{A} & \xrightarrow{\cong} & \mathcal{X}^{UF} \\
 \uparrow F & & \nearrow \\
 \downarrow U & & \\
 \mathcal{X} & &
 \end{array}$$

¹Janelidze in condition 2 requires that $\mathcal{A}$ has finite coproducts. As we will see in the proof, this requirement is not necessary, since one can still construct finite coproducts in $2 \implies 1$ in the same way as Janelidze does in $3 \implies 2$. Nevertheless, this is equivalent to the author's claim: condition 1 indeed implies that $\mathcal{A}$ has finite coproducts.

Definition 5.1.2. A category $\mathcal{A}$ is said to be **ideally exact** if it satisfies one of the equivalent conditions of Theorem 5.1.1.

Proof of Theorem 5.1.1. (1 $\implies$ 3) Suppose $! : 0 \rightarrow 1$ is a regular epi. Since $\mathcal{A}$ is exact, the pullback functor $(!)^* : (\mathcal{A} \downarrow 1) \rightarrow (\mathcal{A} \downarrow 0)$ is monadic by the main result of the previous chapter: *Theorem 4.0.5*. Note that $(\mathcal{A} \downarrow 1) \cong \mathcal{A}$. Now consider $\mathcal{X} = (\mathcal{A} \downarrow 0)$, we want to prove that $\mathcal{X}$ is semiabelian.

$(\mathcal{A} \downarrow 0)$ is pointed, in fact if (A, α) is an object in $(\mathcal{A} \downarrow 0)$, we have the following initial and terminal objects, which coincide:

$$\begin{array}{ccc} 0 & \xrightarrow{!_A} & A \\ & \searrow 1_0 & \downarrow \alpha \\ & & 0 \end{array} \qquad \begin{array}{ccc} A & \xrightarrow{\alpha} & 0 \\ \downarrow \alpha & \swarrow 1_0 & \\ 0 & & \end{array}$$

$\mathcal{A}$ has finite coproducts, so $(\mathcal{A} \downarrow 0)$ has finite coproducts. In fact consider $(A, \alpha), (B, \beta)$ in $(\mathcal{A} \downarrow 0)$ and $A + B$ in $\mathcal{A}$, with the canonical injections $\iota_1 : A \rightarrow A + B$ and $\iota_2 : B \rightarrow A + B$. Let (C, γ) be an object in $(\mathcal{A} \downarrow 0)$ with $f : A \rightarrow C$ and $g : B \rightarrow C$ morphisms in $(\mathcal{A} \downarrow 0)$. Providing to $A + B$ the morphism $[\alpha, \beta]$ we have that $(A + B, [\alpha, \beta])$ is the coproduct of (A, α) and (B, β) in $(\mathcal{A} \downarrow 0)$, in fact $\gamma[f, g] = [\alpha, \beta]$, so $[f, g]$ is also a morphism in $(\mathcal{A} \downarrow 0)$.

$$\begin{array}{ccccc} A & \xrightarrow{\iota_1} & A + B & \xleftarrow{\iota_2} & B \\ & \searrow \alpha & \downarrow [\alpha, \beta] & \swarrow \beta & \\ & & 0 & & \\ & \searrow [f, g] & \uparrow \gamma & \swarrow g & \\ & & C & & \end{array}$$

Now we want to prove that $(\mathcal{A} \downarrow 0)$ is exact, from the exactness of $\mathcal{A}$. First we prove it is finitely complete. It suffices to show that it has terminal a object and pullbacks. The terminal object is the one just shown above. It also has pullbacks, by *Proposition 2.1.3*, since pullbacks are connected limits and $\mathcal{A}$ has them.

Consider now an equivalence relation (R, r_1, r_2) in $(\mathcal{A} \downarrow 0)$, endowed with the morphism ρ and α as shown in the diagram below. We want to prove that such equivalence relation is the kernel pair of some arrow.

Since $\mathcal{A}$ is exact, it is a kernel pair in $\mathcal{A}$. By *Proposition A.0.9* we can suppose that it is the kernel pair of its coequalizer.

$$\begin{array}{ccccc}
 R & \xrightarrow{r_1} & A & \xrightarrow{q} & B \\
 & \searrow r_2 & \downarrow \alpha & \swarrow \exists! \beta & \\
 & \rho & 0 & &
 \end{array}$$

Since r_1 and r_2 are morphism in $(\mathcal{A} \downarrow 0)$, we have that $\alpha r_1 = \rho = \alpha r_2$. Thus, by the universal property of the coequalizer, there exists a unique morphism $\beta : B \rightarrow 0$ such that $\beta q = \alpha$. Providing to B the morphism β , we have that (B, β) is the coequalizer of (R, r_1, r_2) in $(\mathcal{A} \downarrow 0)$.

We proceed by proving that $(\mathcal{A} \downarrow 0)$ is regular. Both the conditions of the definition of regular category are easily extendable from $\mathcal{A}$ to $(\mathcal{A} \downarrow 0)$. In fact since in $\mathcal{A}$ each morphism admits a (regular epi, mono) factorization, then the same holds in $(\mathcal{A} \downarrow 0)$. Let $f = me$ be such a factorization through Q , with f morphism in $(\mathcal{A} \downarrow 0)$. Endow Q with the morphism t induced by the universal property of the coequalizer. Now we want to prove that e and m are morphism in $(\mathcal{A} \downarrow 0)$. Then they will be respectively regular epi and mono in $(\mathcal{A} \downarrow 0)$. Notice first that e is obviously a morphism in $(\mathcal{A} \downarrow 0)$, by the condition of t . Moreover, we have that $te = \alpha = \beta f = \beta me$ and since e is a regular epi, hence by *Proposition A.0.8* an epi, it follows that m is a morphism in $(\mathcal{A} \downarrow 0)$.

$$\begin{array}{ccccc}
 & & Q & & \\
 & \nearrow e & \downarrow \text{---} & \nwarrow m & \\
 A & \xrightarrow{f} & & \xrightarrow{\quad} & B \\
 & \searrow \alpha & \downarrow \exists! t & \swarrow \beta & \\
 & & 0 & &
 \end{array}$$

In order to prove that regular epis are pullback stable in $(\mathcal{A} \downarrow 0)$ we can use the fact that they are in $\mathcal{A}$. In fact, we just endow the regular epi with the morphism induced by universal property of the coequalizer.

We must now establish that the $(\mathcal{A} \downarrow 0)$ is protomodular. Since the construction of the change-of-base functor involves pullbacks, which are connected limits, then by *Proposition 2.1.3* the construction in $(\mathcal{A} \downarrow 0)$ is the same as in $\mathcal{A}$. Thus $(\mathcal{A} \downarrow 0)$ inherits protomodularity from $\mathcal{A}$, noting that the definition of iso in $\mathcal{A}$ and $(\mathcal{A} \downarrow 0)$ is essentially the same, except for the morphism condition. Hence, since $\mathcal{X} = (\mathcal{A} \downarrow 0)$ is exact,

protomodular, pointed and has binary coproducts, we can conclude that it is semiabelian.

It remains to prove that the monad $(!)^*(!)_!$ preserves regular epimorphisms and kernel pairs. Consider a regular epi $q : A \rightarrow B$ in $(\mathcal{A} \downarrow 0)$, it is the coequalizer of some parallel arrows. Since the coequalizer is a colimit, it is preserved by the left adjoint functor $(!)_!$. So it suffices to prove that, with $\mathcal{A}$ regular, $(!)^*$ preserves regular epis. Consider the following diagram:

$$\begin{array}{ccc}
 0 \times_1 A & \xrightarrow{(!)^*(q)} & 0 \times_1 B \\
 \downarrow & & \downarrow \\
 A & \xrightarrow{q} & B \\
 & \searrow & \swarrow \\
 & 1 &
 \end{array}$$

By *Property 2.2.4* the square is a pullback. Since $\mathcal{A}$ is regular, and hence so is $(\mathcal{A} \downarrow 0)$, it follows that $(!)^*(q)$ is a regular epi due to the stability of regular epis under pullbacks.

Finally, we want to establish that the monad $(!)^*(!)_!$ preserves kernel pairs. This is obvious: $(!)_!$ preserves connected limits, such as kernel pairs, by *Proposition 2.2.1*, while $(!)^*$ preserves kernel pairs since it is a right adjoint. Thus $(!)^*(!)_!$ preserves kernel pairs.

(3 $\implies$ 2) Suppose there exists $\mathcal{X}$ semiabelian category and a monadic functor $U : \mathcal{A} \rightarrow \mathcal{X}$. Assume that the associated monad T preserves regular epimorphisms and kernel pairs.

We need to verify that $\mathcal{A}$ is exact. Since U is monadic, we have that $\mathcal{A} \cong \mathcal{X}^T$. So it suffices to show that $\mathcal{X}^T$ is exact. We know that $\mathcal{X}$ is semiabelian, so it is exact. Then, by *Proposition 3.1.1*, $\mathcal{X}^T$ is exact.

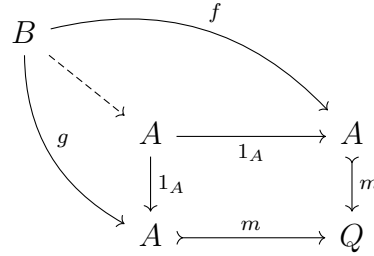
(2 $\implies$ 1) Suppose that $\mathcal{A}$ is exact and there exists a semiabelian category $\mathcal{X}$ and a monadic functor $U : \mathcal{A} \rightarrow \mathcal{X}$.

Since $\mathcal{X}$ is protomodular, so is $\mathcal{X}^T$ by *Proposition 3.1.2*. Again, by monadicity, $\mathcal{A}$ inherits this properties.

In order to prove that $\mathcal{A}$ has finite coproducts, it is first necessary to establish that $\mathcal{X}^T$ is protomodular. This is as already noted. Since any protomodular category is Mal'tsev, then $\mathcal{X}^T$ is Mal'tsev. Since $\mathcal{A}$ is exact, then, by $\mathcal{A} \cong \mathcal{X}^T$, $\mathcal{X}^T$ is also exact. By *Proposition 3.2.9*, it holds that $\mathcal{X}^T$ has reflexive coequalizers. Since $\mathcal{X}$ has finite coproducts and $\mathcal{X}^T$ has reflexive coequalizers, then by *Proposition 3.2.7*, $\mathcal{X}^T$ has finite coproducts. Finally, since $\mathcal{A} \cong \mathcal{X}^T$, it follows that $\mathcal{A}$ has finite coproducts.

We are left to show that the morphism $! : 0 \rightarrow 1$ is a regular epi. Since $\mathcal{A}$ is exact by assumption, the morphism $!$ admits a (regular epi, mono) factorization $! = me$.

Now we aim to prove that if a morphism m is a mono, then $U(m)$ acts as a mono in the morphisms in $\mathcal{X}$ between object coming from $\mathcal{A}$, ie if $f, g : U(A) \rightarrow U(B)$ for some A, B in $\mathcal{A}$, then $U(m)f = U(m)g \implies f = g$. First, we prove that if m is a mono, then $U^T(m)$ is a mono. Let m be a mono and A its domain. Then $(A, 1_A, 1_A)$ is the kernel pair of m , in fact, given f and g morphisms such that $mf = mg$, we obtain the universal property by the mono.



Since U^T is a right adjoint, it preserves limit, such as kernel pairs. Thus also the following diagram is a kernel pair.

$$\begin{array}{ccc}
 U^T(A) & \xrightarrow{1_{U^T(A)}} & U^T(A) \\
 \downarrow 1_{U^T(A)} & & \downarrow U(m) \\
 U^T(A) & \xrightarrow{U(m)} & Q
 \end{array}$$

This implies that $U^T(m)$ is a mono by the universal property of the kernel pair.

It remains to prove that the image of a mono through the comparison functor $K : \mathcal{A} \rightarrow \mathcal{X}^T$ acts as a mono in the way described above. Then, since $U = U^T K$, we have that U does the same. By monadicity, the functor K is fully faithful, this is sufficient to prove that the image of a mono through K acts as a mono. In fact, let m be a mono in $\mathcal{A}$ and let f, g be morphisms in $\mathcal{X}^T$ between object coming from $\mathcal{A}$ such that $K(m)f = K(m)g$. Let f' and g' be the morphisms in $\mathcal{A}$ such that

$K(f') = f$ and $K(g') = g$ by the fully property of K . Then we have that

$$\begin{aligned}
 K(m)f &= K(m)g \\
 \implies K(m)K(f') &= K(m)K(g') \\
 \implies K(mf') &= K(mg') \\
 \implies mf' &= mg' \\
 \implies f' &= g' \\
 \implies K(f') &= K(f') \\
 \implies f &= g
 \end{aligned}$$

Thus $K(m)$ acts as a mono in the morphisms in $\mathcal{X}^T$ between object coming from $\mathcal{A}$. By the above discussion, $U(m)$ acts as a mono in the morphisms in $\mathcal{X}$ between object coming from $\mathcal{A}$.

Picking up where we left off, consider m the mono arising from the (regular epi, mono) factorization of $! : 0 \rightarrow 1$ in $\mathcal{A}$. By the discussion above, $U(m)$ acts as a mono in the morphisms in $\mathcal{X}$ between object coming from $\mathcal{A}$. In particular, since the final object 1 is a limit, it is preserved by U , then $U(m)$ acts as mono and has codomain 1 in $\mathcal{X}$. Moreover, $\mathcal{X}$ is semiabelian, hence pointed, so $0 = 1$ in $\mathcal{X}$. This means that $U(m)$ acts as a mono and has codomain 0 in $\mathcal{X}$. Finally, this implies that $U(m)$ is an iso, in fact, suppose that $U(m)$ acts as a mono with codomain 0 and let $U(C)$ be its domain. Denote by $!_{U(C)}$ the unique morphism from $U(C)$ to 0 , then $!_{U(C)}$ is the inverse of $U(m)$. In fact $!_{U(C)}U(m) = !_0 = 1_0$. Then composing with $U(m)$ we obtain $U(m)!_{U(C)}U(m) = U(m)$, thus, observing that $U(m)!_{U(C)}$ and $1_{U(C)}$ are a morphisms between objects arising from $\mathcal{A}$, we also have that $U(m)!_{U(C)} = 1_{U(C)}$. Thus $U(m)$ is an iso.

$$U(C) \xrightarrow{U(m)} 0 \quad 0 \xrightarrow{!_{U(C)}} U(C)$$

Now we want to prove that U reflects iso. As done before we prove that for the comparison functor K and for U^T , then it follows for U . K reflect iso because, by monadicity, it is fully faithful. In fact, if $K(f) : K(X) \rightarrow K(Y)$ is an iso and $g : K(Y) \rightarrow K(X)$ is its inverse in $\mathcal{X}^T$, then $K(f)g = 1_{K(Y)}$ and $gK(f) = 1_{K(X)}$. Let g' be the morphism in $\mathcal{A}$ such that $K(g') = g$ by

the fully property of K . So we have that

$$\begin{aligned}
 1_{K(Y)} &= K(f)g \\
 \implies 1_{K(Y)} &= K(f)K(g') \\
 \implies K(1_Y) &= K(f)K(g') \\
 \implies K(1_Y) &= K(fg') \\
 \implies 1_Y &= fg'
 \end{aligned}$$

and similarly $1_X = g'f$. Thus f is an iso in $\mathcal{A}$.

Also U^T reflects iso. In fact, let $U^T(k) : U^T(V) \rightarrow U^T(W)$ be an iso in $\mathcal{X}^T$ and $h : U^T(W) \rightarrow U^T(V)$ its inverse. We aim to prove that h is also a morphism in $\mathcal{X}^T$, so that it is an inverse for k in $\mathcal{X}^T$. Let ξ_V and ξ_W be respectively the algebra structures of $U^T(V)$ and $U^T(W)$. Then, since k is a morphism in $\mathcal{X}^T$, we have that $\xi_W T(k) = k\xi_V$, denoting $U^T(k)$ just as k . Composing with h and $T(h)$ we obtain

$$\begin{array}{ccc}
 T(V) & \xrightleftharpoons[T(h)]{T(k)} & T(W) \\
 \downarrow & & \downarrow \\
 V & \xrightleftharpoons[T(h)]{k} & W
 \end{array}$$

$$\begin{aligned}
 \xi_W T(k) &= k\xi_V \\
 \implies h\xi_W T(k)T(h) &= hk\xi_V T(h) \\
 \implies h\xi_W &= \xi_V T(h)
 \end{aligned}$$

Thus h is a morphism in $\mathcal{X}^T$, and k is an iso in $\mathcal{X}^T$. This implies, as discussed above, that also U reflects iso.

Finally we want to prove that $! : 0 \rightarrow 1$ is a regular epi in $\mathcal{A}$. We already know that $U(m)$ is an iso. Since U reflects iso, then m is an iso. Thus $! = me$ is a regular epi.

This holds because an iso, as m , is also a strong epimorphism, and e is a regular epimorphism, hence it is also a strong epimorphism; therefore, the composition me is a strong epimorphism. Since the $\mathcal{A}$ is regular, then by a standard result $! = me$ is a regular epi, as claimed. $\square$

Proposition 5.1.3. *An ideally exact category is finitely cocomplete.*

Proof. An ideally exact category is exact and protomodular, thus Mal'tsev. So by *Proposition 3.2.9* it has reflexive coequalizers. Then *Proposition 3.2.7* implies that the category has all the finite colimits. $\square$

Proposition 5.1.4. *Let $\mathcal{A}$ be an ideally exact category. There exist $\mathcal{X}$ semiabelian category with a monadic functor $U : \mathcal{A} \rightarrow \mathcal{X}$ and the corresponding monad is essentially nullary. Moreover, the functor U is the pullback functor associated to $! : 0 \rightarrow 1$*

Proof. We can consider $(!)^*$ to be such functor, as shown in the proof of *Theorem 5.1.1: (1 $\implies$ 3)*. Then by *Theorem 2.3.4* we have that the monad associated $(!)^*(!)_!$ is essentially nullary. $\square$

5.2 The morphism-induced scenario

As illustrated in *Theorem 5.1.1: (1 $\implies$ 3)*, in the definition of an ideally exact category, the monadic functor involved can be taken to be the pullback functor induced by the unique morphism $! : 0 \rightarrow 1$. However, it may be more convenient to consider in the definition a different monadic functor U satisfying certain properties, as in *Theorem 5.1.1:(2),(3)*. That said, of course the monadic functor associated to $! : 0 \rightarrow 1$ is always there. A natural question arises: is this monadic functor U , up to equivalence of categories, the same as the pullback functor associated to the unique morphism $! : 0 \rightarrow 1$? In this section, we aim to address this question.

Given an adjunction $F \dashv U$ between categories finitely complete, with initial and terminal objects, and $\mathcal{X}$ pointed

$$\mathcal{A} \begin{array}{c} \xrightarrow{U} \\ \xleftarrow{F} \end{array} \mathcal{X}$$

we can consider the adjunction induced on the initial objects $F^0 \dashv U^0$, defined as follows. Let X be an object in $\mathcal{X}$ and let (A, α) be an object in $(\mathcal{A} \downarrow 0)$. Since $\mathcal{X}$ is pointed, then $\mathcal{X} \cong (\mathcal{X} \downarrow 0)$. The category of the initial objects 0 in $\mathcal{X}$ and $\mathcal{A}$ is implicit. Denote $!^X : X \rightarrow 0$ the unique morphism in the zero

object.

$$F^0(X) = (F(X), F(!^X))$$

$$U^0(A, \alpha) = 0 \times_{UF(0)} U(A) = 0 \times_{U(0)} U(A) = \text{Ker}(U(\alpha))$$

$$\begin{array}{ccc} 0 \times_{U(0)} U(A) & \xrightarrow{\text{proj}_2} & U(A) \\ \downarrow \text{proj}_1 & & \downarrow U(\alpha) \\ 0 & \xrightarrow{!_{U(0)}} & U(0) \end{array}$$

Since F is a left adjoint, it preserves colimits. So we can assume $F(0_{\mathcal{X}}) = 0_{\mathcal{A}}$.

$$(\mathcal{A} \downarrow 0) \cong (\mathcal{A} \downarrow F(0)) \xleftarrow[F^0]{U^0} (\mathcal{X} \downarrow 0) \cong \mathcal{X} \quad (5.1)$$

One can prove that this is an adjunction. Given η and ϵ the unit and counit of $F \dashv U$, then the unit of $F^0 \dashv U^0$ in an object X in $\mathcal{X}$ is η_X^0

$$\begin{array}{ccc} X & \xrightarrow{\eta_X} & U(F(X)) \\ \searrow \eta_X^0 & & \downarrow U(F(!^X)) \\ 0 \times_{UF(0)} UF(X) & \xrightarrow{\text{proj}_2} & U(F(X)) \\ \downarrow \text{proj}_1 = !^{U^0 F^0(X)} & & \downarrow U(F(!^X)) \\ X & \xrightarrow{!^X} & 0 \xrightarrow{\eta_0 = !_{UF(0)}} UF(0) \end{array} \quad (5.2)$$

where the outer diagram commutes because of the naturality of η , and the counit of $F^0 \dashv U^0$ in an object (A, α) in $(\mathcal{A} \downarrow 0)$ is

$$\epsilon_{(A, \alpha)}^0 = \epsilon_A F(\text{proj}_2)$$

$$\begin{array}{ccccc} F(U^0(A, \alpha)) = F(0 \times_{UF(0)} U(A)) & \xrightarrow{F(\text{proj}_2)} & FU(A) & \xrightarrow{\epsilon_A} & A \\ & \searrow F(\text{proj}_1) = F(!^{F(U^0(A, \alpha))}) & & \swarrow \alpha & \\ & & F(0) & & \end{array}$$

One can prove that the counit is a morphism in $(\mathcal{A} \downarrow 0)$ and that both η^0 and ϵ^0 are natural transformations.

Proposition 5.2.1. *Given an adjunction $F \dashv U$*

$$\mathcal{A} \xrightleftharpoons[F]{U} \mathcal{X}$$

in which $\mathcal{X}$ is pointed and protomodular, U reflects isomorphisms, and the unit of the adjunction η is cartesian. Then the adjunction induced in 0 $F^0 \dashv U^0$ is an adjoint equivalence.

Proof. Consider the induced adjunction $F^0 \dashv U^0$ in (5.1). Observe that $(\mathcal{A} \downarrow 0) \cong \text{Pt}_{\mathcal{A}}(0)$ via the unique morphism from the initial object.

First, we aim to prove that U^0 reflects isos. Let $f : (A, \alpha) \rightarrow (A', \alpha')$ be a morphism in $\text{Pt}_{\mathcal{A}}(0)$. Observe that $U^0(f) = (!_{U(0)})^*(U(f))$.

$$\begin{array}{c}
 \begin{array}{ccccc}
 & & U^0(f) = (!_{U(0)})^*(U(f)) & & \\
 & \swarrow \text{dashed} & & \searrow \text{dashed} & \\
 0 \times_{U(0)} U(A) & \xrightarrow{\text{proj}_2} & U(A) & \xrightarrow{U(f)} & U(A') & \xleftarrow{\text{proj}_2'} & 0 \times_{U(0)} A' \\
 & \swarrow U(\alpha) & \swarrow U(\alpha') & \searrow U(\alpha') & \searrow U(\alpha) & \\
 & U(!_A) & & & U(!_{A'}) & \\
 & \searrow & U(0) & \swarrow & \searrow & \\
 & \text{proj}_1 & \uparrow !_{U(0)} & \text{proj}_1' & & \\
 & \swarrow & 0 & \searrow & &
 \end{array}
 \end{array}$$

Since $\mathcal{A}$ is protomodular, then $U(f)$ is an iso. By assumption U reflects iso, thus f is an iso, as claimed.

In order to prove that $F^0 \dashv U^0$ is an adjoint equivalence we need to prove that its unit η^0 and its counit ϵ^0 are pointwise isos. By assumption η is cartesian, thus, referring to (5.2), for each X in $\mathcal{X}$ η_X^0 is an iso. Then by the triangular identity, for each (A, α) in $(\mathcal{A} \downarrow 0)$, we have that

$$U^0(\epsilon_{(A, \alpha)}^0) \eta_{U^0((A, \alpha))}^0 = \text{Id}_{U^0(A, \alpha)}$$

Hence $U^0(\epsilon_{(A, \alpha)}^0)$ is an iso. We already proved that U^0 reflects iso, thus for each (A, α) in $(\mathcal{A} \downarrow 0)$ also $\epsilon_{(A, \alpha)}^0$ is an iso. $\square$

Theorem 5.2.2. *Let $\mathcal{A}$ be an ideally exact category, and let $U : \mathcal{A} \rightarrow \mathcal{X}$ be the monadic functor associated to the definition, coming with a left adjoint F . The functor U is, up to category equivalence, the pullback functor $!^*$ if and only if the unit associated to U is cartesian.*

$$\begin{array}{ccc}
 (\mathcal{A} \downarrow 1) & \xrightarrow{\cong} & \mathcal{A} \\
 \uparrow \scriptstyle ! & & \uparrow \scriptstyle F \\
 \downarrow \scriptstyle !^* & & \downarrow \scriptstyle U \\
 (\mathcal{A} \downarrow 0) & \xrightarrow{\cong} & \mathcal{X}
 \end{array} \tag{5.3}$$

Proof. We already proved that $(\mathcal{A} \downarrow 1) \cong \mathcal{A}$, by the unique morphism to the terminal object. Suppose that the unit η associated to U is cartesian, we aim prove that $(\mathcal{A} \downarrow 0) \cong \mathcal{X}$ and that the diagram commutes up to natural isomorphisms.

By the definition of ideally exact category we have that $\mathcal{X}$ is semia-belian, thus $\mathcal{X}$ is pointed and protomodular. Moreover, since U is monadic, U reflects isos. By assumption we also have that η is cartesian. Then, by *Proposition 5.2.1* the adjunction $F^0 \dashv U^0$ in (5.1) is an adjoint equivalence, thus an equivalence of categories.

We need to prove that such equivalence of categories makes the diagram commute up to natural iso.

$$\begin{array}{ccc}
 (\mathcal{A} \downarrow 1) & \xrightarrow{\cong} & \mathcal{A} \\
 \uparrow \scriptstyle ! & & \uparrow \scriptstyle F \\
 \downarrow \scriptstyle !^* & & \downarrow \scriptstyle U \\
 (\mathcal{A} \downarrow 0) & \xrightleftharpoons[U^0]{F^0} & \mathcal{X}
 \end{array}$$

The upward-left directed square commutes up to natural iso by definition of F^0 . Then, up to category equivalence, $! \cong F$. Since $F \dashv U$ and $! \dashv !^*$, then, since the right adjoint is unique up to natural iso, $!^* \cong U$. Hence, also the downward-right directed square commutes up to natural iso.

Conversely, suppose that U is, up to natural iso, the pullback functor of $!$, then, by *Property 2.2.5*, the unit is cartesian. $\square$

Remark 5.2.3. One can also prove directly that downward-right directed square (5.3) commutes up to natural iso. So, we aim to prove that $U^{0!*}$ corresponds to the functor U . In order to do that we compute $U^{0!*}$ in an object A in $\mathcal{A}$. Consider the object $(A, !^A)$, and its pullback

$$\begin{array}{ccc} 0 \times_1 A & \xrightarrow{\text{proj}_2} & A \\ \downarrow \text{proj}_1 & & \downarrow !^A \\ 0 & \xrightarrow{!} & 1 \end{array}$$

Since $!^*$ is a right adjoint, also the following square is a pullback.

$$\begin{array}{ccc} U(0) \times_1 U(A) & \xrightarrow{U(\text{proj}_2)} & A \\ \downarrow U(\text{proj}_1) & & \downarrow U(!^A) \\ U(0) & \xrightarrow{U(!)} & 1 \end{array}$$

Then, apply U^0 to $(U(0) \times_1 U(A), U(\text{proj}_1))$. By definition (5.1), we need to compute the pullback of $U(\text{proj}_1)$ along $!_{U(0)}$. Since the right square is a pullback, as we just proved, and the whole diagram is a pullback, then the left square is the pullback of $U(\text{proj}_1)$ along $!_{U(0)}$. Thus the result of $U^{0!*}$ in an object A in $\mathcal{A}$ is the pullback $0 \times_1 U(A)$.

$$\begin{array}{ccccc} 0 \times_1 U(A) & \longrightarrow & U(0) \times_1 U(A) & \xrightarrow{U(\text{proj}_2)} & A \\ \downarrow & & \downarrow U(\text{proj}_1) & & \downarrow U(!^A) \\ 0 & \xrightarrow{!_{U(0)}} & U(0) & \xrightarrow{U(!)} & 1 \end{array}$$

Finally we prove that the pullback $0 \times_1 U(A)$ is just $U(A)$. Observe that the following square is the pullback of $U(!^A)$ along $!$. In fact, given two morphism $f_1 : X \rightarrow U(A)$, $f_2 : X \rightarrow 0$ such that the external diagram commutes, there exists a unique morphism such the two triangles commutes. Such unique morphism is f_2 .

$$\begin{array}{ccccc} X & & & & \\ & \searrow f_1 & & \searrow f_1 & \\ & & U(A) & \xrightarrow{1_{U(A)}} & U(A) \\ & \searrow f_2 & \downarrow !^{U(A)} & & \downarrow \\ & & 0 = 1 & \longrightarrow & 1 \end{array}$$

Thus the pullback $0 \times_1 U(A)$ of $U(!^A)$ along $!$ is just A . This means that also the downward-right directed square commutes up to natural isomorphisms.

This previous theorem answer the question we posed at the beginning of this section. The following example shows that we already came across to such a scenario.

Example 5.2.4. In the first chapter we saw the same theorem in the particular case of $\mathcal{A} = \mathbf{Ring}$ and $\mathcal{X} = \mathbf{Ring}_*$, where $1 = \{0\}$ and $0 = \mathbb{Z}$.

$$\begin{array}{ccc}
 (\mathbf{Ring} \downarrow 1) & \xrightarrow{\cong} & \mathbf{Ring} \\
 \uparrow \scriptstyle ! & & \uparrow \scriptstyle F \\
 \downarrow \scriptstyle !^* & & \downarrow \scriptstyle U \\
 (\mathbf{Ring} \downarrow \mathbb{Z}) & \xrightleftharpoons[\text{Ker}]{(F, \pi_{\mathbb{Z}})} & \mathbf{Ring}_*
 \end{array}$$

In particular the bottom category equivalence constructed in that case exactly corresponds to (5.1)

5.3 Ideals and quotients

In this section we will investigate the notion of ideals in an ideally exact category and their corresponding quotients. We aim to emulate the classical definition of ideals in a ring, and the corresponding quotients. As already mentioned, recall that a proper ideal in a ring is a non-unital ring. Note that the notion of normal subobjects is well defined in a semiabelian category as the kernel of a morphism. Also recall that a non-unital ring is semiabelian, while a unital ring fails to be, as mentioned in the first chapter. Given the previous observations, one can model the notion of ideals as the normal subobjects through a functor, which is the forgetful functor in rings, having a semiabelian category as codomain. Note that this framing is consistent with the characterization of ideally exact categories given in *Theorem 5.1.1*. In particular in $\mathbf{Ring}$ we have the following structure.

$$\mathbf{Ring} \xrightarrow{U} \mathbf{Ring}_*$$

In general in an ideally exact category $\mathcal{A}$ we have the following structure, with semiabelian $\mathcal{X}$.

$$\mathcal{A} \xrightarrow{U} \mathcal{X}$$

We start with a technical lemma.

Lemma 5.3.1. *Let $\mathcal{X}$ be a category with finite coproducts and let T be an essentially nullary monad in $\mathcal{X}$. Given (X_1, ξ_1) and (X_2, ξ_2) T -algebras in $\mathcal{X}$, X an object in $\mathcal{X}$, $m : X_1 \rightarrow X$ a morphism and $n : X \rightarrow X_2$ a mono, such that their composition is a T -algebra morphism, then there exists a unique T -algebra structure ξ on X making m and n T -algebra morphisms.*

$$\begin{array}{ccccc}
 T(X_1) & \xrightarrow{T(m)} & T(X) & \xrightarrow{T(n)} & T(X_2) \\
 \downarrow \xi_1 & & \downarrow \xi & & \downarrow \xi_2 \\
 X_1 & \xrightarrow{m} & X & \xrightarrow{n} & X_2
 \end{array} \tag{5.4}$$

Proof. First we show that if there exists a T -algebra structure ξ making n a T -algebra morphism, then necessarily m is a T -algebra morphism. In fact, since n is a mono, and both the outer and the right diagram in (5.4) commute, then the left diagram commutes.

$$\begin{aligned}
 \xi_2 T(n) T(m) &= nm \xi_1 \\
 \implies n \xi T(m) &= nm \xi_1 \\
 \implies \xi T(m) &= m \xi_1
 \end{aligned}$$

Now we aim to prove that there exists a unique T -algebra structure ξ on X making n a T -algebra morphism. Since T is an essentially nullary monad, then $[T(!_X), \eta_X]$ is a strong epi. Consider the following diagram of solid arrows.

$$\begin{array}{ccc}
 T(0) + X & \xrightarrow{[T(!_X), \eta_X]} & T(X) \\
 \downarrow [m\xi_1 T(!_{X_1}), 1_X] & \searrow \xi & \downarrow T(n) \\
 X & \xrightarrow{n} & X_2 \\
 & & \downarrow \xi_2 \\
 & & T(X_2)
 \end{array}$$

The following compositions with the canonical injections of $T(0) + X$ prove that the diagram commutes.

- $\xi_2 T(n) T(!_X) = \xi_2 T(n !_X) = \xi_2 T(!_X)$
- $nm \xi_1 T(!_X) = \xi_2 T(nm) T(!_X) = \xi_2 T(nm !_X) = \xi_2 T(!_X)$

- $\xi_2 T(n) \eta_X = \xi_2 \eta_{X_2} n = n$
- $n 1_X = n$

Since $[T(!_X), \eta_X]$ is a strong epi, there exists a unique morphism ξ making the diagram commute. ξ turns out to be a algebra for X .

It remains to prove that it is unique. Suppose that there exists another algebra structure ξ' on X making n a T -algebra morphism. Then

$$\begin{aligned} n\xi &= \xi_2 T(n) = n\xi' \\ \implies n\xi &= n\xi' \\ \implies \xi &= \xi' \end{aligned}$$

□

Corollary 5.3.2. *Let $\mathcal{X}$ be a category with finite coproducts and let T be an essentially nullary monad in $\mathcal{X}$. Given a T -algebra (X, ξ) in $\mathcal{X}$, then every reflexive relation R on $\mathcal{X}$*

$$R \begin{array}{c} \xrightarrow{r_1} \\ \xleftarrow{r_2} \end{array} X$$

admits a unique T -algebra structure making R a reflexive relation on $\mathcal{X}^T$.

Proof. Let s be the common section of r_1 and r_2 . Then

$$\begin{array}{ccc} & \langle r_1, r_2 \rangle s = \langle 1_X, 1_X \rangle & \\ X & \xrightarrow{s} R \xrightarrow{\langle r_1, r_2 \rangle} X \times X & \\ & \searrow \langle 1_X, 1_X \rangle & \end{array}$$

Observe that $\langle 1_X, 1_X \rangle$ is a T -algebra morphism, in fact given the product

structure from *Remark 3.2.1* of $X \times X$ on the category $\mathcal{X}^T$

$$\begin{array}{ccccc}
 T(X) & \xleftarrow{T(\pi_1)} & T(X \times X) & \xrightarrow{T(\pi_2)} & T(X) \\
 \downarrow \xi & & \downarrow \langle T(\pi_1), T(\pi_2) \rangle & & \downarrow \xi \\
 & & T(X) \times T(X) & & \\
 & & \downarrow \xi \times \xi & & \\
 X & \xleftarrow{\pi_1} & X \times X & \xrightarrow{\pi_2} & X
 \end{array}$$

we have that the following diagram commutes, easily composing with the canonical projections π_1 and π_2 .

$$\begin{array}{ccc}
 T(X) & \xrightarrow{T(\langle 1_X, 1_X \rangle)} & T(X \times X) \\
 \downarrow \xi & & \downarrow \langle T(\pi_1), T(\pi_2) \rangle \\
 & & T(X) \times T(X) \\
 & & \downarrow \xi \times \xi \\
 X & \xrightarrow{\langle 1_X, 1_X \rangle} & X \times X
 \end{array}$$

Thus we have s and $\langle r_1, r_2 \rangle$ morphisms whose composition is the T -algebra morphism $\langle 1_X, 1_X \rangle$. Then, by *Lemma 5.3.1*, there exists a unique T -algebra structure on R making s and $\langle r_1, r_2 \rangle$ T -algebra morphisms. $\square$

In the next results we consider the following lattice. Let A be an object in a category $\mathcal{A}$.

- $\text{Quot}(A)$: regular epis, up to iso, with domain A ;
- $\text{ERel}(A)$: equivalence relations, up to iso, on A ;
- $\text{RRel}(A)$: reflexive relations, up to iso, on A .

Finally, if the category is semiabelian, we consider also the lattice consisting of normal subobjects $\text{NSub}(A)$, containing the kernels, up to iso, of some morphisms with domain A .

The following theorem shows the correspondence between the lattices of quotients and normal subobjects in a semiabelian category. This is a generalization of the classical result in **Grp**.

Theorem 5.3.3. *Let $\mathcal{X}$ be a semiabelian category. Then for each object X in $\mathcal{X}$ the following lattices are isomorphic.*

$$\text{Quot}(X) \cong \text{NSub}(X)$$

Proof. Consider the map

$$\text{Quot}(X) \rightarrow \text{NSub}(X)$$

that associates to each regular epi $q : X \rightarrow Q$ its kernel

$$\text{Ker}(q) \longrightarrow X \xrightarrow{q} Q$$

and the map

$$\text{NSub}(X) \rightarrow \text{Quot}(X)$$

that associates to each kernel pair of some morphism $f : X \rightarrow Y$ the regular epi p underlying the corresponding (regular epi, mono) factorization of f .

$$\begin{array}{ccccc} & & P & & \\ & \nearrow p & & \searrow & \\ \text{Ker}(f) & \longrightarrow & X & \xrightarrow{f} & Y \end{array}$$

Consider a regular epi q , and its kernel. Then consider the factorization of q . Since q is a regular epi, we can assume, up to iso, that the regular epi underlying the factorization is q itself with the identity morphism as mono. Thus the composition

$$\text{Quot}(X) \rightarrow \text{NSub}(X) \rightarrow \text{Quot}(X)$$

is the identity morphism.

Consider a kernel pair of some morphism $f : X \rightarrow Y$. Then consider the regular epi underlying the factorization of f . By *Proposition A.0.11* the regular epi of the factorization and f share the same kernel. Thus the composition

$$\text{NSub}(X) \rightarrow \text{Quot}(X) \rightarrow \text{NSub}(X)$$

is the identity morphism. We proved that the two maps are inverses of each other, hence the isomorphism holds. $\square$

Now we aim to generalize the previous theorem to an ideally exact category.

Theorem 5.3.4. *Let $\mathcal{A}$ be an ideally exact category and let $U : \mathcal{A} \rightarrow \mathcal{X}$ be, up to equivalence, the underlying pullback functor associated to $! : 0 \rightarrow 1$. Then for each object A in $\mathcal{A}$ the following lattices are isomorphic.*

$$\text{Quot}(A) \cong \text{NSub}(U(A))$$

Proof. Let A be an object in $\mathcal{A}$. First we have that

$$\text{Quot}(A) \cong \text{ERel}(A)$$

in fact $\mathcal{A}$ is exact, hence kernel pairs and equivalence relations correspond to each others, and by *Proposition A.0.6* and *Proposition A.0.7* we have the given isomorphism. Furthermore

$$\text{ERel}(A) \cong \text{RRel}(A)$$

because $\mathcal{A}$ is protomodular, hence Mal'tsev. By *Proposition 5.1.4* the unit of the adjunction associated to U is essentially nullary, then, by *Corollary 5.3.2*, we have that

$$\text{RRel}(A) \cong \text{RRel}(U^T(A))$$

Since U is monadic we also have that

$$\text{RRel}(A) \cong \text{RRel}(U(A))$$

By definition, the codomain of U is semiabelian, thus exact and Mal'tsev, then as above

$$\text{RRel}(U(A)) \cong \text{ERel}(U(A)) \cong \text{Quot}(U(A))$$

Finally, by the previous result *Theorem 5.3.3* we have that

$$\text{Quot}(U(A)) \cong \text{NSub}(U(A))$$

□

Given an ideally exact category $\mathcal{A}$, we call the elements of $\text{NSub}(U(A))$ the U -ideals of $\mathcal{A}$. This theorem gives the correspondence between U -ideals and regular epis, which play the role of quotients.

Remark 5.3.5. *Note that in an ideally exact category $\mathcal{A}$, given an object A in $\mathcal{A}$, the lattice $\text{NSub}(A)$ is not well defined, since $\mathcal{A}$ is in general non-pointed.*

Remark 5.3.6. *In $\mathbf{Ring}$, given an object A in $\mathcal{A}$, the lattice $\text{NSub}(U(A))$ consists of ideals. As shown in the first chapter, the forgetful functor $U : \mathbf{Ring} \rightarrow \mathbf{Ring}_*$ is, up to equivalence, the pullback functor associated to $! : 0 \rightarrow 1$. We have the following corresponding morphisms, composed by an ideal I , ie a kernel, and its corresponding quotient map, ie a regular epi, π .*

$$I \hookrightarrow R \xrightarrow{\pi} R/I$$

With this final theorem we proved that in an ideally exact category quotients and normal subobjects correspond to each others. This is exactly what happens in $\mathbf{Ring}$ as mentioned in the first chapter. In fact in $\mathbf{Ring}$ the ideals, which are the normal subobjects in $\mathbf{Ring}_*$ through the forgetful functor, correspond to quotients, which are the regular epis. Observe that in $\mathbf{Grp}$ the result holds directly from *Theorem 5.3.3*, since $\mathbf{Grp}$ is a semiabelian category, and normal subgroups are objects of $\mathbf{Grp}$. Thus in this case we do not need to consider an underlying functor, or, equivalently, we can just consider the identity functor.

Chapter 6

Cotopoi as a fundamental example

6.1 The general case

In this section, we aim to show that any cotopos $\mathcal{E}^{\text{op}}$, the dual of a topos $\mathcal{E}$, is an ideally exact category. Then, since the category of **Set** is a topos, we will analyze the case of $\mathbf{Set}^{\text{op}}$. This section is based on [Bou04], [Jo77], [B94II] and [B94III].

First, we recall the definition of a topos.

Definition 6.1.1. *Let $\mathcal{E}$ be a finitely complete category. A **subobject classifier** of $\mathcal{E}$ is a pair (Ω, t) , where Ω is an object of $\mathcal{E}$ and $t : 1 \rightarrow \Omega$ is a monomorphism, such that for any object A of $\mathcal{E}$ and any monomorphism $m : S \rightarrow A$, there exists a unique morphism $\chi_S : A \rightarrow \Omega$ making the diagram*

$$\begin{array}{ccc} S & \longrightarrow & 1 \\ m \downarrow & & \downarrow t \\ A & \xrightarrow{\chi_S} & \Omega \end{array}$$

a pullback.

Definition 6.1.2. A category $\mathcal{E}$ is said to be an **elementary topos**, in the following just a **topos**, if

- it is finitely complete;
- it is cartesian closed;
- it has a subobject classifier.

A topos turns out to have a rich structure. First, we need some properties. Recall that the exponential is given by the cartesian closedness. For all objects A , B , and C in $\mathcal{E}$, it holds that

$$\mathrm{Hom}_{\mathcal{E}}(A, \Omega^B) \cong \mathrm{Hom}_{\mathcal{E}}(A \times B, \Omega) \cong \mathrm{Hom}_{\mathcal{E}}(B, \Omega^A)$$

Proposition 6.1.3. Given a topos, the exponential functor

$$\Omega^{(-)} : \mathcal{E}^{\mathrm{op}} \rightarrow \mathcal{E}$$

is monadic.

The proof is contained in [B94III] *Theorem 5.7.3* and it is based on the Beck monadicity theorem.

Proposition 6.1.4. A topos is finitely cocomplete.

Proof. Consider a topos $\mathcal{E}$. Since the functor $\Omega^{(-)} : \mathcal{E}^{\mathrm{op}} \rightarrow \mathcal{E}$, by *Proposition 6.1.3*, is monadic, and $\mathcal{E}$ is finitely complete, it follows that $\mathcal{E}^{\mathrm{op}}$ is finitely complete. Thus, $\mathcal{E}$ is finitely cocomplete. $\square$

Proposition 6.1.5. Let $\mathcal{E}$ be a category with the following conditions:

- It has pullbacks;
- It has coequalizers of kernel pairs;
- Regular epimorphisms are pullback stable.

Then each morphism $f : A \rightarrow B$ in $\mathcal{E}$ admits a (regular epi, mono) factorization.

Moreover, if the category is also finitely complete, then it is a regular category.

Proof. Let $f : A \rightarrow B$ be a morphism. Consider the kernel pair (r_1, r_2) of f , and the coequalizer e of the kernel pair.

$$\begin{array}{ccccc} A \times_B A & \xrightarrow[r_2]{r_1} & A & \xrightarrow{f} & B \\ & & \searrow e & & \\ & & I & & \end{array}$$

Then, since $fr_1 = fr_2$, by the universal property of the coequalizer, there exists a unique morphism m such that $me = f$. We already know that e is a regular epi, since it is a coequalizer.

It remains to prove that m is a mono. Equivalently, we show that, letting $p_1, p_2 : I \times_B I \rightarrow I$ be the projections of the kernel pair, we have that $p_1 = p_2$. First, we consider the morphism $e \times_B e : A \times_B A \rightarrow I \times_B I$.

$$\begin{array}{ccccc} A \times_B A & \xrightarrow[r_2]{r_1} & A & \xrightarrow{f} & B \\ & \searrow e \times_B e & \searrow e & \nearrow m & \nearrow \\ & & I & & \\ & & \nearrow p_1 & \nearrow p_2 & \\ & & I \times_B I & & \end{array}$$

Since e is a coequalizer, it is a regular epi. Thus $e \times_B e$ is an epi. In fact, all the square in the following diagram are pullbacks, and $e \times_B e$ is the composition of the two regular epis, thus, by *Proposition A.0.8*, it is an epi.

$$\begin{array}{ccccc} & & r_1 & & \\ & & \curvearrowright & & \\ A \times_B A & \xrightarrow{\quad} & I \times_B A & \xrightarrow{\quad} & A \\ \downarrow & \searrow e \times_B e & \downarrow & & \downarrow e \\ A \times_B I & \xrightarrow{\quad} & I \times_B I & \xrightarrow{p_1} & I \\ \downarrow & & \downarrow p_2 & & \downarrow m \\ A & \xrightarrow{e} & I & \xrightarrow{m} & B \end{array}$$

Finally

$$\begin{aligned} p_1(e \times_B e) &= er_1 = er_2 = p_2(e \times_B e) \\ \implies p_1 &= p_2 \end{aligned}$$

□

Proposition 6.1.6. *In a topos $\mathcal{E}$, for every morphism $p : E \rightarrow B$, the pullback functor $p^* : (\mathcal{E} \downarrow B) \rightarrow (\mathcal{E} \downarrow E)$ preserves all finite colimits.*

The proof is contained in [B94II] *Proposition 5.9.1*, or [Jo77] *Corollary 1.43*, and it based on the existence of a right adjoint to the pullback functor.

Proposition 6.1.7. *Let $\mathcal{A}$ be a finitely complete category and let $q : X \rightarrow Y$ be a coequalizer in $\mathcal{A}$. Then $q : (X, q) \rightarrow (Y, 1_Y)$ is a coequalizer in $(\mathcal{A} \downarrow Y)$. Moreover, the following conditions are equivalent:*

- $q : X \rightarrow Y$ is a pullback stable coequalizer;
- The pullback functor along any morphism $f : E \rightarrow Y$ preserves the coequalizer $q : (X, q) \rightarrow (Y, 1_Y)$.

Proof. First, recall that the forgetful functor U from a slice category of $\mathcal{A}$ to $\mathcal{A}$ creates all the colimits that $\mathcal{A}$ has, by *Proposition 2.1.3*. Thus, given $q : X \rightarrow Y$ coequalizer in a category $\mathcal{A}$, it yields that $q : (X, q) \rightarrow (Y, 1_Y)$ is a coequalizer in the slice category $(\mathcal{A} \downarrow Y)$.

Let $f : E \rightarrow B$ be a morphism in $\mathcal{A}$. Let $q : X \rightarrow Y$ be the coequalizer of a pair (a, b) in $(\mathcal{A} \downarrow B)$. Suppose that q is a pullback stable coequalizer. Consider the action of the pullback functor $f^* : (\mathcal{A} \downarrow E) \rightarrow (\mathcal{A} \downarrow B)$ on q , a and b .

By *Property 2.2.4*, the top squares are pullbacks. Since the coequalizer q is pullback stable, then $f^*(q)$ is a coequalizer of $(f^*(a), f^*(b))$ in $\mathcal{A}$. Since

the forgetful functor U creates all the colimits that $\mathcal{A}$ has, it follows the statement in $(\mathcal{A} \downarrow E)$.

Conversely, we aim to prove that q is pullback stable, ie if q is the coequalizer of (a, b) then q' is the coequalizer of (a', b') in the pullback squares

$$\begin{array}{ccccc} E \times_Y X & \xrightleftharpoons[b']{a'} & E \times_Y X & \xrightarrow{q'} & E \\ \downarrow & & \downarrow & & \downarrow f \\ Z & \xrightleftharpoons[b]{a} & X & \xrightarrow{q} & Y \end{array}$$

We aim to show that the pullback of q along f is the same as the one resulting by *Property 2.2.4* between $f^*(q)$ and q , so that $f^*(q) = q'$. Then, consequently, the same happens for the left diagram, so that $f^*(a) = a'$ and $f^*(b) = b'$. In order to prove that those pullbacks of f are the same, we first need to focus on the pullback $E \times_Y Y$ of f along 1_Y . In particular, since one morphism is the identity, it turns out that $E \times_Y Y \cong E$ with the projections $\text{proj}_{1,Y} = 1_E$ and $\text{proj}_{2,Y} = f$. Thus, the pullback between q and $f^*(q)$ is the same as the one between q and f . We obtain the following diagram.

$$\begin{array}{ccccc} E \times_Y Z & \xrightleftharpoons[b']{f^*(a)} & E \times_Y X & \xrightarrow{f^*(q)} & E \\ \downarrow \text{proj}_{2,Z} & & \downarrow \text{proj}_{2,X} & & \downarrow f \\ Z & \xrightleftharpoons[b]{a} & X & \xrightarrow{q} & Y \\ & \text{proj}_{1,X} \downarrow & & & \uparrow 1_Y \\ & Y & & & \\ & \downarrow f & & & \\ & E & & & \end{array}$$

Curved arrows in the diagram include: $qa=qb$ from Z to Y , 1_E from E to E , $\text{proj}_{1,Z}$ from Z to E , and $\text{proj}_{2,X}$ from X to E .

Then, since the coequalizer $q : (X, q) \rightarrow (Y, 1_Y)$ of (a, b) is preserved by the pullback functor f^* , it follows that q' is the coequalizer of (a', b') in the slice category. Since $\mathcal{A}$ is finitely complete, one can prove that a coequalizer in the slice category is also a coequalizer in $\mathcal{A}$, using binary products to prove the universal property in $\mathcal{A}$. In fact, given a morphism $h : Y \rightarrow Z$ in $\mathcal{A}$, one can form the morphism $\langle h, q \rangle : (X, q) \rightarrow (Y \times Z, \text{proj}_2)$ and the

universal property for h follows from the one for $\langle h, q \rangle$. Hence q' is the coequalizer of (a', b') in $\mathcal{A}$, ie q is a pullback stable coequalizer. $\square$

Proposition 6.1.8. *A topos is a regular category.*

Proof. A topos is finitely complete. Moreover, by *Proposition 6.1.6* we have that coequalizers are preserved by the pullback functor along any morphism, then, by *Proposition 6.1.7*, this implies that coequalizers are pullback stable. Hence regular epis are pullback stable. By *Proposition 6.1.4*, each topos is also finitely cocomplete. Then, by *Proposition 6.1.5*, each morphism admits a (regular epi, mono) factorization. Thus, a topos is a regular category. $\square$

Now we aim to transfer the regularity of a topos to its dual, the cotopos.

Proposition 6.1.9. *In a topos, every monomorphism is an equalizer.*

Proof. Let $m : S \rightarrowtail A$ be a mono in a topos, let χ_S be its characteristic morphism and let $t : 1 \rightarrow \Omega$ be the morphism underlying the subobject classifier. We aim to prove that $m = \text{Eq}(\chi_S, t!^A)$.

$$S \xrightarrow{m} A \xrightarrow{!^A} 1 \xrightarrow{t} \Omega$$

$\searrow \chi_S \nearrow$

It holds that $\chi_S m = t!^S = t!^A m$. Thus, it remains to show the universal property of the equalizer. Let $n : X \rightarrow A$ be a morphism such that $\chi_S n = t!^A n$. By the universal property of the pullback of the subobject classifier, there exists a unique morphism $t : X \rightarrow S$ such that $!^S t = !^X n$ and $mt = n$, as wanted.

$$\begin{array}{ccccc} X & & & & \\ & \searrow t & & \searrow !^X & \\ & S & \xrightarrow{!^S} & 1 & \\ & \downarrow m & \searrow !^A & \downarrow t & \\ & A & \xrightarrow{\chi_S} & \Omega & \end{array}$$

$\nearrow n$

$\square$

Proposition 6.1.10. *In a topos, a morphism is an isomorphism if and only if it is both a monomorphism and an epimorphism.*

Proof. Clearly, every iso is both a mono and an epi. Conversely, let $f : X \rightarrow Y$ be a morphism which is both a mono and an epi. By *Proposition 6.1.9*, it is also an equalizer. Let (a, b) be pair such that $f = \text{Eq}(a, b)$. It holds $af = bf$. Since f is an epi, then $a = b$. On the other hand, $\text{Eq}(a, b) = \text{Eq}(a, a) = 1_Y$, so there exists an iso g such that $fg = 1_Y$. Hence f is an iso.

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \rightrightarrows Z \\ g \uparrow & \nearrow 1_Y & \downarrow b \\ Y & & \end{array}$$

□

Proposition 6.1.11. *In a topos, every epimorphism is a coequalizer.*

Proof. Let $e : A \twoheadrightarrow B$ be an epi in a topos. By *Proposition 6.1.8*, a topos is a regular category, thus e admits factorization $e = mr$ where r is a regular epi and m is a mono. It holds that m is also an epi. In fact, if a and b are morphisms such that $am = bm$, then $ae = amr = bmr = be$, and since e is an epi, $a = b$. We proved that m is both a mono and an epi, thus, by *Proposition 6.1.10*, it is an iso. Hence e is a regular epi, ie a coequalizer.

$$\begin{array}{ccc} A & \xrightarrow{e} & B \rightrightarrows D \\ & \searrow r & \nearrow m \\ & C & \end{array}$$

□

Proposition 6.1.12. *In a topos, monomorphisms are pushout stable.*

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ m \downarrow & & \downarrow n \\ Z & \xrightarrow{g} & P \end{array}$$

we aim to prove that, if m is a mono, then so is n . In order to do that, we construct two morphisms respectively from Z to an object A and from Y to the same object A , such that the external square commutes. Then, if the considered morphism from Y to A is a mono, so is n . In fact, if a morphism post-composed with an other is a mono, then the first one is also a mono.

$$\begin{array}{ccc}
X & \xrightarrow{f} & Y \\
m \downarrow & & \downarrow n \\
Z & \xrightarrow{g} & P
\end{array}
\begin{array}{c}
\searrow \\
\downarrow \text{dashed} \\
A
\end{array}$$

To construct such morphisms, we make use of the universal property of the counit of the exponential adjunction, given by the cartesian closedness of a topos, and the subobject classifier. Consider the morphisms $\Delta_Y : Y \rightarrowtail Y \times Y$ and $\langle f, m \rangle : X \rightarrowtail Y \times Z$. They are monos, in fact the post-compositions of each of them with their second projection are respectively 1_Y and m , which are monos. Thus, we can consider their characteristic morphisms χ_Y and $\chi_{\langle f, m \rangle}$ given by the subobject classifier (Ω, t) .

$$\begin{array}{ccc} Y & \longrightarrow & 1 \\ \Delta_Y \downarrow & & \downarrow t \\ Y \times Y & \xrightarrow{\chi_Y} & \Omega \end{array} \qquad \begin{array}{ccc} X & \longrightarrow & 1 \\ \langle f, m \rangle \downarrow & & \downarrow t \\ Y \times Z & \xrightarrow{\chi_{\langle f, m \rangle}} & \Omega \end{array}$$

Then, the universal property of the counit ϵ of the exponential adjunction gives us the unique morphisms

$$\alpha : Y \rightarrow \Omega^Y$$

$$\varphi : Z \rightarrow \Omega^Y$$

$$\begin{array}{ccc}
\Omega & \xleftarrow{\epsilon_\Omega} & Y \times (\Omega^Y) \\
& \searrow \chi_Y & \uparrow 1_Y \times \alpha \\
& & Y \times Y
\end{array}
\qquad
\begin{array}{ccc}
\Omega & \xleftarrow{\epsilon_\Omega} & Y \times (\Omega^Y) \\
& \searrow \chi_{\langle f, m \rangle} & \uparrow 1_Y \times \varphi \\
& & Y \times Z
\end{array}$$

such that respectively $\epsilon_\Omega(1_Y \times \alpha) = \chi_Y$ and $\epsilon_\Omega(1_Y \times \varphi) = \chi_{\langle f, m \rangle}$. These morphisms are the ones we were looking for.

We still need to show that the external square commutes, ie that $\alpha f = \varphi m$.

$$\begin{array}{ccc}
X & \xrightarrow{f} & Y \\
m \downarrow & & \downarrow n \\
Z & \xrightarrow{g} & P
\end{array}
\quad
\begin{array}{c}
\searrow \alpha \\
\varphi \nearrow \\
\Omega^Y
\end{array}$$

In order to prove that, we first prove that the following holds. Let χ_Y and $\chi_{\langle f, m \rangle}$ be as before, and $\chi_{\langle f, 1_X \rangle}$ be the characteristic morphism of the mono $\langle f, 1_X \rangle : X \rightarrowtail Y \times X$, then

$$\chi_{\langle f, m \rangle}(1_Y \times m) = \chi_{\langle f, 1_X \rangle} = \chi_Y(1_Y \times f)$$

We prove this equality using the subobject classifier. Consider the following commutative squares. They are pullbacks. In fact, for the left one, let $k : V \rightarrow X$ and $\langle l_1, l_2 \rangle : V \rightarrow Y \times X$ be morphisms such that $\langle f, m \rangle k = (1_Y \times m) \langle l_1, l_2 \rangle$, then $k : V \rightarrow X$ is the unique morphism such that $1_X k = k$ and $\langle f, 1_X \rangle k = \langle l_1, l_2 \rangle$. For the right one, let $b : W \rightarrow Y$ and $\langle a_1, a_2 \rangle : W \rightarrow Y \times X$ be morphisms such that $\langle 1_Y, 1_Y \rangle b = (1_Y \times f) \langle a_1, a_2 \rangle$, then $a_2 : W \rightarrow X$ is the unique morphism such that $\langle f, 1_X \rangle a_2 = \langle a_1, a_2 \rangle$ and $f a_2 = b$.

$$\begin{array}{ccc}
V & \xrightarrow{k} & X \\
\downarrow \langle l_1, l_2 \rangle & \searrow k & \downarrow 1_X \\
Y \times X & \xrightarrow{1_Y \times m} & Y \times Z
\end{array}
\quad
\begin{array}{ccc}
W & \xrightarrow{b} & Y \\
\downarrow \langle a_1, a_2 \rangle & \searrow a_1 & \downarrow f \\
Y \times X & \xrightarrow{1_Y \times f} & Y \times Y
\end{array}$$

Thus, the following squares are pullbacks.

$$\begin{array}{ccc}
 X & \xrightarrow{1_X} & X \xrightarrow{!^X} 1 \\
 \downarrow \langle f, 1_X \rangle & & \downarrow \langle f, m \rangle \quad \downarrow t \\
 Y \times X & \xrightarrow{1_Y \times m} & Y \times Z \xrightarrow{\chi_{\langle f, m \rangle}} \Omega
 \end{array}
 \quad
 \begin{array}{ccc}
 X & \xrightarrow{f} & Y \xrightarrow{!^Y} 1 \\
 \downarrow \langle f, 1_X \rangle & & \downarrow \langle 1_Y, 1_Y \rangle \quad \downarrow t \\
 Y \times X & \xrightarrow{1_Y \times f} & Y \times Y \xrightarrow{\chi_Y} \Omega
 \end{array}$$

$\chi_{\langle f, 1_X \rangle}$

Since (Ω, t) is a subobject classifier, given the mono $\langle f, 1_X \rangle$, there exists a unique morphism from $Y \times X$ to Ω making the squares pullbacks, thus

$$\chi_{\langle f, m \rangle}(1_Y \times m) = \chi_{\langle f, 1_X \rangle} = \chi_Y(1_Y \times f)$$

Now, we can finally prove that $\alpha f = \varphi m$. By the universal property of the counit, there exists a unique morphism $\psi : X \rightarrow \Omega^Y$ such that $\epsilon_\Omega(1_Y \times \psi) = \chi_{\langle f, 1_X \rangle}$. Observe that both αf and φm satisfy this property. In fact, by definition of α and φ

- $\epsilon_\Omega(1_Y \times \alpha)(1_Y \times f) = \chi_Y(1_Y \times f) = \chi_{\langle f, 1_X \rangle}$
- $\epsilon_\Omega(1_Y \times \varphi)(1_Y \times m) = \chi_{\langle f, m \rangle}(1_Y \times m) = \chi_{\langle f, 1_X \rangle}$

$$\begin{array}{ccc}
 \Omega \xleftarrow{\epsilon_\Omega} Y \times (\Omega^Y) & \Omega \xleftarrow{\epsilon_\Omega} Y \times (\Omega^Y) & \Omega \xleftarrow{\epsilon_\Omega} Y \times (\Omega^Y) \\
 \uparrow \chi_Y & \uparrow \chi_{\langle f, m \rangle} & \uparrow \chi_{\langle f, 1_X \rangle} \\
 Y \times Y & Y \times Z & Y \times X \\
 \uparrow 1_Y \times \alpha & \uparrow 1_Y \times \varphi & \uparrow 1_Y \times \psi \\
 Y \times X & Y \times X & Y \times X
 \end{array}$$

So, by uniqueness, $\alpha f = \psi = \varphi m$.

It remains to show that α is a mono, so that also n is a mono. Let $u, v : K \rightarrow Y$ be morphisms such that $\alpha u = \alpha v$. Then, by the universal property of the counit, we have that

$$\begin{aligned}
\chi_Y(1_Y \times u) &= \epsilon_\Omega(1_Y \times \alpha)(1_Y \times u) \\
&= \epsilon_\Omega(1_Y \times \alpha u) \\
&= \epsilon_\Omega(1_Y \times \alpha v) \\
&= \epsilon_\Omega(1_Y \times \alpha)(1_Y \times v) \\
&= \chi_Y(1_Y \times v)
\end{aligned}$$

$$\begin{array}{ccc}
\Omega & \xleftarrow{\epsilon_\Omega} & Y \times (\Omega^Y) \\
& \searrow \chi_Y & \uparrow 1_Y \times \alpha \\
& & Y \times Y \\
& & \uparrow 1_Y \times u \quad \uparrow 1_Y \times v \\
& & Y \times K
\end{array}$$

Consider the following diagram. The right square is a pullback by the subobject classifier. The two left square are pullbacks as already shown similarly before. Moreover, by the previous equality, the bottom compositions are equal. Thus, the whole diagram is a pullback.

$$\begin{array}{ccccc}
K & \xrightarrow[u]{u} & Y & \xrightarrow{!^Y} & 1 \\
\langle 1_K, u \rangle \downarrow \downarrow \langle 1_K, v \rangle & & \downarrow \langle 1_Y, 1_Y \rangle & & \downarrow t \\
K \times Y & \xrightarrow[v \times 1_Y]{u \times 1_Y} & Y \times Y & \xrightarrow{\chi_Y} & \Omega
\end{array}$$

By the universal property of the pullback, there exists a unique $\sigma : K \rightarrow K$ such that $\langle 1_K, v \rangle \sigma = \langle 1_K, u \rangle$.

$$\begin{array}{ccccc}
K & & & & \\
& \searrow \sigma & & \searrow !^K & \\
& & K & \xrightarrow{!^K} & 1 \\
& \searrow \langle 1_K, u \rangle & \downarrow \langle 1_K, v \rangle & & \downarrow t \\
& & K \times Y & \xrightarrow{\chi_Y(u \times 1_Y) = \chi_Y(v \times 1_Y)} & \Omega
\end{array}$$

So, $1_K \sigma = 1_K$, ie $\sigma = 1_K$, and $v \sigma = u$, ie $v = u$. Thus, α is a mono. This concludes the proof. $\square$

Proposition 6.1.13. *A cotopos is a regular category.*

Proof. Consider a cotopos. It is finitely complete, in fact, by *Proposition 6.1.4*, its dual is finitely cocomplete. In the underlying topos, we aim to prove that each morphism admits a (regular mono, epi) factorization.

Let $f : A \rightarrow B$ be a morphism and consider its factorization (regular epi, mono) $f = me$ given by *Proposition 6.1.8*. e is regular epi, thus, by *Proposition A.0.8*, an epi. m is a mono, thus, by *Proposition 6.1.9*, it is an equalizer, ie a regular mono. Finally, by *Proposition 6.1.12*, monos, which are the equalizers by *Proposition 6.1.9*, are pushout stable. Thus, the cotopos is a regular category. $\square$

We need to prove that a cotopos is also exact. First, we introduce the notion of subobject. A subobject is a monomorphism into an object. We can define the intersection and union of subobjects.

Given $a : X_1 \rightarrowtail X$ and $b : X_2 \rightarrowtail X$ subobjects, the intersection is given by the pullback

$$\begin{array}{ccc} X_1 \cap X_2 & \rightarrowtail & X_1 \\ \downarrow & & \downarrow a \\ X_2 & \xrightarrow{b} & X \end{array}$$

while the union is given by the monomorphism m underlying the (regular epi, mono) factorization from *Proposition 6.1.8* of $[a, b] : X_1 + X_2 \rightarrow X$.

$$\begin{array}{ccc} X_1 + X_2 & \xrightarrow{[a,b]} & X \\ & \searrow & \nearrow m \\ & X_1 \cup X_2 & \end{array}$$

The union is characterized in term of pushout.

Proposition 6.1.14. *Let $a : X_1 \rightarrowtail X$ and $b : X_2 \rightarrowtail X$ be subobjects of a cotopos $\mathcal{E}$ and their intersection given by*

$$\begin{array}{ccc} X_1 \cap X_2 & \xrightarrow{c} & X_1 \\ \downarrow d & & \downarrow a \\ X_2 & \xrightarrow{b} & X \end{array}$$

Then the union $X_1 \cup X_2 \rightarrowtail X$ is given by the pushout

$$\begin{array}{ccc} X_1 \cap X_2 & \xrightarrow{c} & X_1 \\ \downarrow d & & \downarrow \\ X_2 & \rightarrowtail & X_1 \cup X_2 \end{array}$$

The proof is contained in [Jo77] *Proposition 1.55*.

We are ready to prove the exactness of a cotopos

Proposition 6.1.15. *A cotopos is an exact category.*

Proof. Consider a cotopos $\mathcal{E}^{\text{op}}$. By *Proposition 6.1.13*, it is a regular category. It remains to prove that each equivalence relation is a kernel pair. We prove that each reflexive relation is a kernel pair, then, it follows the result. Consider a reflexive relation on an object X in $\mathcal{E}^{\text{op}}$, which is the following in $\mathcal{E}$

$$Y \begin{array}{c} \xleftarrow{r_2} \\ \xleftarrow{s} \rightarrow \\ \xleftarrow{r_1} \end{array} X$$

where s is a common retraction $sr_1 = sr_2 = 1_X$.

Since (Z, r_1, r_2) is a relation in $\mathcal{E}^{\text{op}}$, then (r_1, r_2) is a jointly monic pair in $\mathcal{E}^{\text{op}}$. Hence $[r_1, r_2] : X + X \rightarrow Y$ is an epi in $\mathcal{E}$. By *Proposition 6.1.11*, $[r_1, r_2]$ is a regular epi. This means we can assume that $[r_1, r_2]$ is the regular epi underlying the (regular epi, mono) factorization of $[r_1, r_2]$, where the mono is the identity morphism.

$$\begin{array}{ccc} X + X & \xrightarrow{[r_1, r_2]} & Y \\ & \searrow & \nearrow 1_Y \\ & Y & \end{array}$$

Thus Y is, up to iso, the union $X \cup X$. Now consider the intersection

$$\begin{array}{ccc} X \cap X & \xrightarrow{d} & X \\ \downarrow c & & \downarrow r_2 \\ X & \xrightarrow{r_2} & Y \end{array}$$

Observe that

$$c = 1_X c = sr_1 c = sr_2 d = 1_X d = d$$

Thus the intersection diagram, which is also a pushout in $\mathcal{E}$ by *Proposition 6.1.14*, becomes

$$\begin{array}{ccc} X \cap X & \xrightarrow{c} & X \\ \downarrow c & & \downarrow r_2 \\ X & \xrightarrow{r_2} & Y \end{array}$$

This diagram is a kernel pair in $\mathcal{E}^{\text{op}}$ and it is the kernel pair we were looking for. $\square$

While a topos fails to be protomodular, a cotopos actually has this structure.

Proposition 6.1.16. *A cotopos is a protomodular category.*

Proof. Consider a cotopos $\mathcal{E}^{\text{op}}$. By *Proposition 6.1.3* the functor

$$\Omega^{(-)} : \mathcal{E}^{\text{op}} \rightarrow \mathcal{E}$$

is monadic. In particular, one can show that this functor has image in the internal Heyting algebras in $\mathcal{E}$ which is, by [Bou96] *Proposition 18*, protomodular. This functor, restricted in the codomain, is still monadic. Then by *Proposition 3.1.2*, also $\mathcal{E}^{\text{op}}$ is protomodular. $\square$

It remains to prove that in a cotopos $\mathcal{E}^{\text{op}}$ the unique morphism

$$! : 0_{\mathcal{E}^{\text{op}}} \rightarrow 1_{\mathcal{E}^{\text{op}}}$$

is a regular epi. Then, the associated pullback functor may serve as underlying functor in the definition of an ideally exact category. In order to prove that $!$ is a regular epi in $\mathcal{E}^{\text{op}}$, we prove, equivalently, that

$$! : 0_{\mathcal{E}} \rightarrow 1_{\mathcal{E}}$$

is an equalizer.

Proposition 6.1.17. *In a topos, the initial object 0 is the subobject of every object.*

Proof. Let $\mathcal{E}$ be a topos, and let $f : E \rightarrow 0$ be a morphism in $\mathcal{E}$. It holds that $0_{(\mathcal{E} \downarrow 0)} = (0, 1_0)$. The pullback given by $f^*(0, 1_0)$ is

$$\begin{array}{ccc} E \times_0 0 \cong E & \xrightarrow{f} & 0 \\ \downarrow 1_0 & & \downarrow 1_0 \\ E & \xrightarrow{f} & 0 \end{array}$$

And since, by *Proposition 6.1.6*, f^* preserves colimits, it holds that $E \cong 0$ is initial. In particular f is an iso.

Now, consider an object A . We aim to prove that

$$!_A : 0 \rightarrow A$$

is a mono. Let $a, b : X \rightarrow 0$ such that $!_A a = !_A b$. By the observation above,

X is initial, thus, by the universal property of the initial object, $a = b$. Hence $!_A$ is a mono. $\square$

Corollary 6.1.18. *In a topos $\mathcal{E}$, the unique morphism*

$$! : 0_{\mathcal{E}} \rightarrow 1_{\mathcal{E}}$$

is an equalizer.

Proof. By Proposition 6.1.17, the morphism $! : 0_{\mathcal{E}} \rightarrow 1_{\mathcal{E}}$ is a mono. Then, by Proposition 6.1.9, it is an equalizer. $\square$

All the results above leads to the following result.

Proposition 6.1.19. *A cotopos is an ideally exact category. Moreover, the underlying functor is the pullback functor.*

Proof. We aim to prove that a cotopos $\mathcal{E}^{\text{op}}$ satisfies the condition Theorem 5.1.1:(1). By Proposition 6.1.15 it is exact, and by Proposition 6.1.16 it is protomodular. It has binary coproducts, in fact, the underlying topos is finitely complete. Finally, by Corollary 6.1.18 the morphism

$$! : 0_{\mathcal{E}} \rightarrow 1_{\mathcal{E}}$$

is an equalizer in $\mathcal{E}$. Thus the morphism

$$! : 0_{\mathcal{E}^{\text{op}}} \rightarrow 1_{\mathcal{E}^{\text{op}}}$$

is a coequalizer in $\mathcal{E}^{\text{op}}$, ie a regular epi. $\square$

Remark 6.1.20. *While a cotopos is an ideally exact category, a topos fails to be. A topos is exact and it has binary coproducts, that said, it does not satisfy the remaining conditions. For example one can consider the category **Set** which is a topos. It is not protomodular. Moreover, the unique morphism $! : \emptyset \rightarrow \{*\}$ is not a regular epi, in fact, it is not even*

an epi. In order to show that, consider the morphisms

$$\begin{aligned} f : \{*\} &\rightarrow \{a, b\} \\ * &\mapsto a \end{aligned}$$

$$\begin{aligned} g : \{*\} &\rightarrow \{a, b\} \\ * &\mapsto b \end{aligned}$$

Then $f! = g!$, since they are both the empty map, but $f \neq g$.

$$\emptyset \xrightarrow{!} \{*\} \xrightarrow[g]{f} \{a, b\}$$

6.2 The dual of sets

In the previous section we have seen that a cotopos is an ideally exact category by *Proposition 6.1.19*. An example of cotopos is the dual of the category of sets: $\mathbf{Set}^{\text{op}}$. In this section we aim to analyze this case.

As in the case of cotopos, we refer to *Theorem 5.1.1(1)* for the definition of ideally exact categories.

First we want to show directly that

$$! : \emptyset \rightarrow \{*\}$$

is an equalizer in the underlying topos. So that, it is a regular epi in the cotopos.

In $\mathbf{Set}$ the equalizer of a pair of parallel arrows $f, g : A \rightarrow B$ is

$$\text{Eq}(f, g) = \{x \in A \mid f(x) = g(x)\}$$

$$\text{Eq}(f, g) \xhookrightarrow{i} A \xrightarrow[g]{f} B$$

Consider in particular the morphisms

$$\begin{aligned} f : \{*\} &\rightarrow \{a, b\} \\ * &\mapsto a \end{aligned}$$

$$\begin{aligned} g : \{*\} &\rightarrow \{a, b\} \\ * &\mapsto b \end{aligned}$$

Clearly $f \neq g$, thus

$$\text{Eq}(f, g) = \emptyset$$

Since $!$ is the unique morphism from the initial object to any object, it corresponds in this case with the inclusion i . Thus $!$ is the equalizer of f and g .

$$\emptyset \xrightarrow{!} \{*\} \xrightleftharpoons[g]{f} \{a, b\}$$

In order to present $\mathbf{Set}^{\text{op}}$ as an ideally exact category and study the corresponding ideals, we aim to understand which is the underlying semiabelian category. Keep in mind that the morphism $!$, in $\mathbf{Set}^{\text{op}}$, is $! : \{*\} \rightarrow \emptyset$. As shown in *Theorem 5.1.1*: $(1 \implies 3)$ such semiabelian category is

$$\mathcal{X} = (\mathbf{Set}^{\text{op}} \downarrow \{*\})$$

where the objects are the object of the coslice category from the singleton to sets

$$\text{Ob}(\mathcal{X}) = \text{Ob}(\mathbf{Set}^{\text{op}} \downarrow \{*\}) = \text{Ob}(\{*\} \downarrow \mathbf{Set})$$

or, equivalently, of the category of pointed sets

$$\text{Ob}(\mathcal{X}) = \{(A, a) \mid A \in \mathbf{Set}, a \in A\}$$

and the morphism are the morphism of $\mathbf{Set}^{\text{op}}$ preserving the chosen element. Summing up

$$\text{Ob}(\mathcal{X}) = \{(A, a) \mid A \in \mathbf{Set}, a \in A\}$$

$$\text{Hom}_{\mathcal{X}}((A, a), (B, b)) = \{f : B \rightarrow A \mid f(b) = a\} \subseteq \text{Hom}_{\mathbf{Set}^{\text{op}}}$$

Thus, finally

$$\mathcal{X} \cong (\mathbf{Set}^{\bullet})^{\text{op}}$$

where $\mathbf{Set}^{\bullet}$ is the category of pointed sets, ie the category of sets with a chosen element.

This category is, actually, pointed in the categorical meaning, in fact, the singleton $(\{*\}, *) \cong \{*\}$ is both the initial and terminal object. Given an object $(A, a) \in \mathcal{X}$ we have the following unique morphism from the singleton and to the singleton, represented in $\mathbf{Set}$.

$$\begin{array}{ll} \{*\} \rightarrow (A, a) & (A, a) \rightarrow \{*\} \\ * \mapsto a & (x \in A) \mapsto * \end{array}$$

Moreover, the zero morphism in $\mathcal{X}$ turns out to be the constant morphism to the chosen element

$$\begin{aligned} 0_{(A,a),(B,b)} : (B, b) &\rightarrow (A, a) \\ (x \in B) &\mapsto a \end{aligned}$$

To understand the action of the underlying functor

$$U : \mathbf{Set}^{\text{op}} \rightarrow \mathcal{X} \cong (\mathbf{Set}^\bullet)^{\text{op}}$$

of the ideally exact category, we need to analyze the pullback functor of the morphism $!$. It is defined through the pullback $!$ along the morphism given with an object in $\mathbf{Set}^{\text{op}}$. Such pullback, described in the second chapter, corresponds, given an object A in $\mathbf{Set}^{\text{op}}$, to the pushout in $\mathbf{Set}$

$$\begin{array}{ccc} (\{*\} \sqcup A) / \sim & \xleftarrow{\iota_2} & A \\ \uparrow \iota_1 & & \uparrow \\ \{*\} & \xleftarrow{!} & \emptyset \end{array}$$

where $\sim$ is the smallest equivalence relation identifying the image of any element through the empty function $\emptyset \rightarrow A$ and $!$. The injection ι_1 and ι_2 are the canonical inclusions in the disjoint union, followed by the quotient map. In this case the definition of the $\sim$ is based in the empty set, thus $\sim$ is the empty relation and the pushout is just the disjoint union of the singleton and A . Thus we obtain

$$\begin{array}{ccc} \{*\} \sqcup A & \xleftarrow{\iota_2} & A \\ \uparrow \iota_1 & & \uparrow \\ \{*\} & \xleftarrow{!} & \emptyset \end{array}$$

Then, the pullback functor associates to an object A in $\mathbf{Set}^{\text{op}}$ the object $(\{*\} \sqcup A, \iota_1)$ in $\mathcal{X}$. By the identification above, we can write this as

$$\begin{aligned} U : \mathbf{Set}^{\text{op}} &\rightarrow (\mathbf{Set}^\bullet)^{\text{op}} \\ A &\mapsto (\{*\} \sqcup A, *) \end{aligned}$$

in fact $(\{*\} \sqcup A, \iota_1)$ corresponds to the pointed set $(\{*\} \sqcup A, *)$.

The functor U is monadic by the Beck monadicity theorem. In fact, consider (f, g) a pair of parallel morphism in $\mathbf{Set}^{\text{op}}$. Then, consider their coequalizer in $\mathbf{Set}^{\text{op}}$, which is actually the equalizer in $\mathbf{Set}$.

$$\{x \in A \mid f(x) = g(x)\} \hookrightarrow A \begin{array}{c} \xrightarrow{f} \\ \xrightarrow{g} \end{array} B$$

Its U -image is the equalizer of the pair $(1_{\{*\}} \sqcup f, 1_{\{*\}} \sqcup g)$.

$$\begin{aligned}
 & \left(\{*\} \sqcup \left\{ x \in A \mid f(x) = g(x) \right\}, * \right) \\
 & \quad = \\
 & \left(\left\{ x \in (\{*\} \sqcup A) \mid (1_{\{*\}} \sqcup f)(x) = (1_{\{*\}} \sqcup g)(x) \right\}, * \right) \\
 & \quad \downarrow \\
 & \left(\{*\} \sqcup A, * \right) \\
 & \quad \begin{array}{c} \downarrow \quad \downarrow \\ 1_{\{*\}} \sqcup g \quad 1_{\{*\}} \sqcup f \\ \downarrow \quad \downarrow \end{array} \\
 & \left(\{*\} \sqcup B, * \right)
 \end{aligned}$$

Thus the hypothesis of the Beck monadicity theorem is satisfied.

Now, we aim to understand what are the ideals given by *Theorem 5.3.4* in the case of $\mathbf{Set}^{\text{op}}$.

By the mentioned theorem, given an object A in $\mathbf{Set}^{\text{op}}$, we have that

$$\begin{array}{ccc}
 \text{Quot}(A) \cong \text{NSub}(U(A)) & & \\
 \left\{ \begin{array}{l} \text{Classes of iso of} \\ \text{regular epis in } \mathbf{Set}^{\text{op}} \\ \text{with domain } A \end{array} \right\} & \longleftrightarrow & \left\{ \begin{array}{l} \text{Classes of iso of} \\ \text{kernels in } (\mathbf{Set}^\bullet)^{\text{op}} \\ \text{with codomain } U(A) \end{array} \right\} \\
 \Psi & & \Psi \\
 \left[A \twoheadrightarrow B \right] & & \left[(C, c) \twoheadrightarrow U(A) \right]
 \end{array}$$

$\text{Quot}(A)$ are the regular epi in $\mathbf{Set}^{\text{op}}$, which are the equalizers in $\mathbf{Set}$. Equalizers in $\mathbf{Set}$ are all the monomorphisms, ie all the injective functions. In fact, every equalizer is a mono. Conversely, given a mono $m : B \rightarrowtail A$, we can consider

the pair of morphism

$$\begin{aligned} f : A &\rightarrow \{0, 1\} \\ x &\mapsto 1 \\ g : A &\rightarrow \{0, 1\} \\ x &\mapsto \begin{cases} 1 & \text{if } x \in \text{Im}(m) \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

Now, their equalizer is

$$\text{Eq}(f, g) = \{x \in A \mid f(x) = g(x)\} = \{x \in A \mid g(x) = 1\} = \text{Im}(m)$$

$$\text{Im}(m) \longrightarrow A \underset{g}{\overset{f}{\rightrightarrows}} \{0, 1\}$$

Then, since m induces an iso between $\text{Im}(m)$ and B , m is the equalizer of (f, g) .

$$B \xrightarrow{m} A \underset{g}{\overset{f}{\rightrightarrows}} \{0, 1\}$$

Thus $\text{Quot}(A)$ consists in $\mathbf{Set}$ of all the injective functions with codomain A .

On the other hand, $\text{NSub}(U(A))$ are the kernels of some morphism in $(\mathbf{Set}^\bullet)^{\text{op}}$ with domain $U(A)$. Recall that $U(A) = (\{*\} \sqcup A, *)$. Thus $\text{NSub}(U(A))$ are the cokernels of some morphism in $\mathbf{Set}^\bullet$ with codomain $(\{*\} \sqcup A, *)$.

We can conclude that the class of iso of injective functions with codomain A is in bijection with the class of iso of cokernels with domain $(\{*\} \sqcup A, *)$.

$$\left\{ \begin{array}{c} \text{Classes of iso of} \\ \text{injective functions in } \mathbf{Set} \\ \text{with codomain } A \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{c} \text{Classes of iso of} \\ \text{cokernels in } \mathbf{Set}^\bullet \\ \text{with domain } (\{*\} \sqcup A, *) \end{array} \right\}$$

Ψ

$$\left[B \twoheadrightarrow A \right]$$

Ψ

$$\left[(\{*\} \sqcup A, *) \twoheadrightarrow (C, c) \right]$$

Appendix

Some of the following results are due to [BB04].

Definition A.0.1. Let $\mathcal{C}$ be a category and $f : A \rightarrow B$. We say that f is a **strong epimorphism** if it for each commutative diagram of solid arrows

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ u \downarrow & \exists! d \swarrow & \downarrow v \\ C & \xrightarrow{m} & D \end{array}$$

where m is a mono, there exists a unique morphism d such that $df = u$ and $md = v$.

We say that f is an **extremal epimorphism** if given a factorizations $f = mf'$, with m mono, then m is an iso.

We say that f is a **regular epimorphism** if it is the coequalizer of some parallel arrows.

Proposition A.0.2. Let $\mathcal{C}$ be a category with pullbacks and $f : A \rightarrow B$. The following are equivalent:

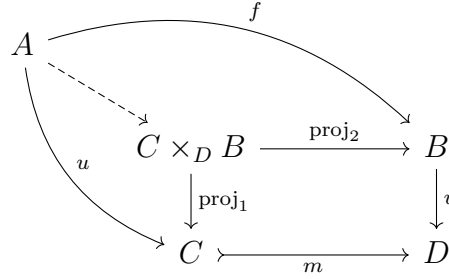
1. f is a strong epimorphism;
2. f is an extremal epimorphisms.

Proof. (1 $\implies$ 2) We want to prove that f is an extremal epi. Let $f = mg$ be a factorization with m mono and consider the square:

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ g \downarrow & d \swarrow & \downarrow 1_B \\ C & \xrightarrow{m} & B \end{array}$$

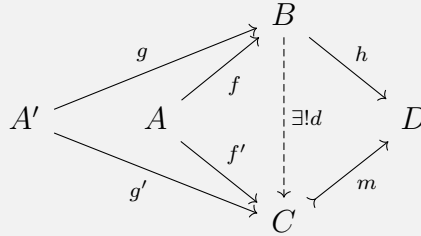
By (1) we have $md = 1_B$. Furthermore $mdm = m$ and, since m is mono, we have that $dm = 1_C$. Thus m is an iso.

(2 $\implies$ 1) We want to prove that f is a strong epi. Consider the pullback



Since m is a mono, the projection proj_2 is a mono. Now f factors through proj_2 , so by (2) proj_2 is an iso. We define $d = \text{proj}_1(\text{proj}_2)^{-1}$, which is the morphism we are looking for. $\square$

Definition A.0.3. Let $\mathcal{C}$ be a category and $f : A \rightarrow B$, $g : A' \rightarrow B$ be morphisms. We say that the pair (f, g) is a **strongly epimorphic pair** if for each commutative diagram of solid arrows



where m is a mono, there exists a unique morphism d such that $f' = df$, $g' = dg$ and $md = h$.

We say that the pair (f, g) is an **extremally epimorphic pair** if given factorizations $f = mf'$ and $g = mg'$, with m mono, then m is an iso.

Proposition A.0.4. Let $\mathcal{C}$ be a category with pullbacks and $f : A \rightarrow B$, $g : A' \rightarrow B$ be morphisms. The following are equivalent:

1. (f, g) is a strongly epimorphic pair;
2. (f, g) is a extremally epimorphic pair;

Proof. By adding the morphism g' and its conditions in the proof of Proposition A.0.2, we obtain the proof of the current proposition. $\square$

Proposition A.0.5. *Let $\mathcal{C}$ be a category with binary coproducts and $f : A \rightarrow B$, $g : A' \rightarrow B$ be morphisms. The following are equivalent:*

1. (f, g) is a strongly epimorphic pair;
2. $[f, g]$ is a strong epimorphism.

Proof. (1 $\implies$ 2) (f, g) is a strongly epimorphic pair. Consider the commutative diagram where m is a mono

$$\begin{array}{ccc} A + A' & \xrightarrow{[f,g]} & B \\ \downarrow l & & \downarrow h \\ C & \xrightarrow{m} & D \end{array}$$

Composing with the injection $\iota_1 : A \rightarrow A + A'$ and $\iota_2 : A' \rightarrow A + A'$, we have the following commutative diagram

$$\begin{array}{ccccc} & & B & & \\ & g \nearrow & & h \searrow & \\ A' & & A & & D \\ & \nwarrow \iota_2 & \searrow \iota_1 & & \\ & & C & & \end{array}$$

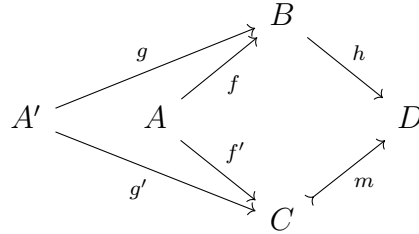
m (arrow from C to D)

Since (f, g) is strongly epimorphic, there exists a unique morphism $d : B \rightarrow C$ such that $\iota_1 d = f$, $\iota_2 d = g$ and $md = h$. The first two yield the equalities $d[f, g]\iota_1 = df = \iota_1 d$ and $[f, g]\iota_2 = dg = \iota_2 d$. So, by the universal property of the coproduct, we have that $d[f, g] = l$.

$$\begin{array}{ccccc} A & & & & A \\ \downarrow \iota_1 & & & & \downarrow \iota_1 \\ A + A' & \xrightarrow{[f,g]} & B & \xrightarrow{d} & C \\ \uparrow \iota_2 & & & & \uparrow \iota_2 \\ A' & & & & A' \end{array}$$

We already know that $md = h$, hence $[f, g]$ is a strong epimorphism.

(2 $\implies$ 1) $[f, g]$ is a strong epimorphism. Consider the following commutative diagram where m is a mono



Thus we have the commutative diagram

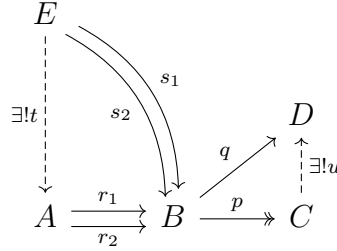
$$\begin{array}{ccc}
 A + A' & \xrightarrow{[f, g]} & B \\
 [f', g'] \downarrow & & \downarrow h \\
 C & \xrightarrow{m} & D
 \end{array}$$

Since $[f, g]$ is a strong epimorphism, there exists a unique morphism $d : B \rightarrow C$ such that $d[f, g] = [f', g']$ and $md = h$. Hence $f' = [f', g']\iota_1 = d[f, g]\iota_1 = df$ and $g' = [f', g']\iota_2 = d[f, g]\iota_2 = dg$. Thus (f, g) is a strongly epimorphic pair. $\square$

Proposition A.0.6. *If a regular epi has a kernel pair, then it is the coequalizer of its kernel pair.*

Proof. Let p be a regular epi, and let (r_1, r_2) be the kernel pair of p . Since p is a regular epi, it is the coequalizer of some parallel arrows. Let (s_1, s_2) be such pair as in the diagram below. We want to prove that p is the coequalizer of (r_1, r_2) .

We already know that $pr_1 = pr_2$, as (r_1, r_2) is the kernel pair of p . It remains to establish the universal property of the coequalizer. Let $q : B \rightarrow D$ be a morphism such that $qr_1 = qr_2$. By the universal property of the kernel pair, there exists a unique morphism $t : E \rightarrow A$ such that $r_1t = s_1$ and $r_2t = s_2$. We have that $qs_1 = qr_1t = qr_2t = qs_2$, so, by the universal property of the coequalizer for the pair (s_1, s_2) , there exists a unique morphism $u : C \rightarrow D$ such that $q = up$. Thus the universal property of the coequalizer also holds for the pair (r_1, r_2) .



□

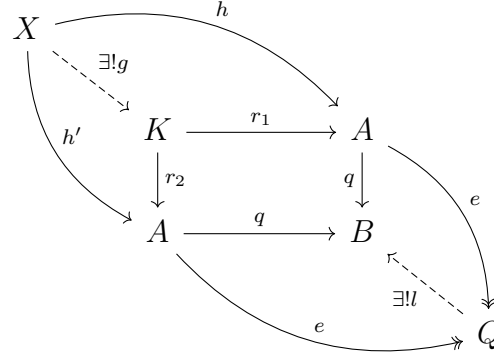
Proposition A.0.7. *If a kernel pair has a coequalizer, then it is the kernel pair of its coequalizer.*

Proof. Let (K, r_1, r_2) be a kernel pair of some arrow $q : A \rightarrow B$, and assume that such kernel pair has a coequalizer $e : B \rightarrow Q$. We want to prove that (r_1, r_2) is the kernel pair of its coequalizer e .

$$\begin{array}{ccccc}
 & & & & Q \\
 & & & \nearrow e & \\
 K & \xrightarrow[r_2]{r_1} & A & \xrightarrow{q} & B
 \end{array}$$

We already know that $er_1 = er_2$, since e is the coequalizer of (r_1, r_2) . It remains to show that the universal property of the kernel pair holds. Consider $h : X \rightarrow A$ and $h' : X \rightarrow A'$ morphisms such that $eh = eh'$. Since (r_1, r_2) is the kernel pair of q , it holds that $qr_1 = qr_2$. Thus by the universal property of the coequalizer e , there exists a unique morphism $l : Q \rightarrow B$ such that $q = le$. Composing with such morphism l we have that $leh = leh'$, hence $qh = qh'$. Since (r_1, r_2) is the kernel pair of q , by the universal property there exists a unique morphism $g : X \rightarrow K$ such that $r_1g = h$ and $r_2g = h'$. Hence (r_1, r_2) is the kernel pair of e , its

coequalizer.



□

Proposition A.0.8. *A regular epi is an epi.*

Proof. Let $p : A \rightarrow B$ be a regular epi. By the definition, it is the coequalizer of some parallel arrows (s_1, s_2) . Suppose that f and g are two morphisms such that $fp = gp$. We aim to prove that $f = g$.

Since p is the coequalizer of (s_1, s_2) , then $ps_1 = ps_2$. Moreover we have that $fps_1 = fps_2 = gps_2 = gps_1$.

$$S \begin{array}{c} \xrightarrow{s_1} \\ \xrightarrow{s_2} \end{array} A \xrightarrow{p} B \begin{array}{c} \xrightarrow{f} \\ \xrightarrow{g} \end{array} C$$

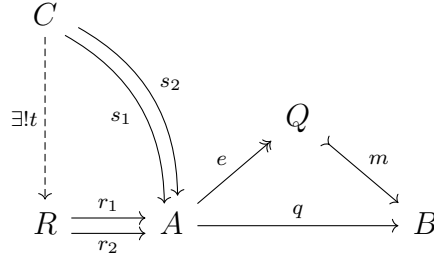
A dashed arrow $\exists!t$ points from B to C .

By the universal property of the coequalizer, there exists a unique morphism $t : B \rightarrow C$ such that $tp = fp = gp$. Hence, by uniqueness, $f = g$. □

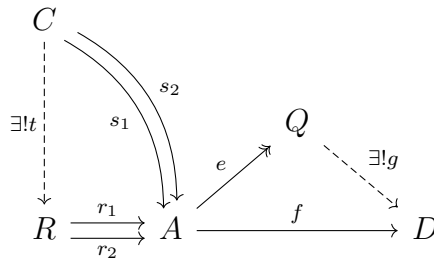
Proposition A.0.9. *In an exact category $\mathcal{A}$, each equivalence relation has a coequalizer and it is the kernel pair of its coequalizer.*

Proof. Consider an equivalence relation (R, r_1, r_2) in $\mathcal{A}$. Since $\mathcal{A}$ is exact, (R, r_1, r_2) is the kernel pair of some arrow $q : A \rightarrow B$. $\mathcal{A}$ is also regular, so q admits a factorization $q = me$, where m is a mono and e is a regular epi, ie the coequalizer of some arrows. Let s_1 and s_2 be such arrows, we have that $es_1 = es_2$. Composing with m , we have that $qs_1 = mes_1 = mes_2 = qs_2$. By the universal property of the kernel pair, there exists a unique morphism

$t : C \rightarrow R$ such that $r_1 t = s_1$ and $r_2 t = s_2$.



Now we want to prove that e is the coequalizer of (r_1, r_2) . We have that $mer_1 = fr_1 = fr_2 = mer_2$. Thus, since m is a mono, $er_1 = er_2$. It remains to prove the universal property of the coequalizer. Let $f : A \rightarrow D$ be a morphism such that $fr_1 = fr_2$. Composing with t we have $fr_1 t = fr_2 t$, hence $fs_1 = fs_2$. By the universal property of the coequalizer e , as coequalizer of s_1 and s_2 , there exists a unique morphism $g : Q \rightarrow D$ such that $ge = f$. Hence e is also the coequalizer of r_1 and r_2 .



Finally, we observe that (r_1, r_2) is the kernel pair of e , since by *Proposition A.0.7*, a kernel pair that admits a coequalizer is the kernel pair of its coequalizer. $\square$

Proposition A.0.10. *If a morphism f admits a factorization $f = mg$, with m mono, then f and g share the same kernel pair.*

Proof. Let (Eq, r_1, r_2) be the kernel pair of $f : A \rightarrow B$ and let $g : A \rightarrow Q$ be a morphism such that $f = mg$ with m mono. We have that $fr_1 = fr_2$, hence $mgr_1 = mgr_2$. Since m is a mono, we have that $gr_1 = gr_2$. Conversely, if (Eq, r_1, r_2) is the kernel pair of g , then we have that $gr_1 = gr_2$.

Just composing with m we have that $mgr_1 = mgr_2$. Hence $fr'_1 = fr'_2$.

$$\begin{array}{ccccc} \text{Eq} & \xrightleftharpoons[r_2]{r_1} & A & \xrightarrow{f} & B \\ & & \searrow g & & \nearrow m \\ & & Q & & \end{array}$$

Now we want to prove that if the universal property of the kernel pair holds for f , then it also holds for g . Let $s_1 : X \rightarrow A$ and $s_2 : X \rightarrow A$ be morphisms such that $gs_1 = gs_2$. Just composing with m , we have that $mgs_1 = mgs_2$. Hence $fs_1 = fs_2$. By the universal property of the kernel pair of f , there exists a unique morphism $k : X \rightarrow \text{Eq}$ such that $r_1k = s_1$ and $r_2k = s_2$. Conversely, consider $s_1 : X \rightarrow A$ and $s_2 : X \rightarrow A$ be morphisms such that $fs_1 = fs_2$, and suppose that the universal property holds for g . We have that $mgs_1 = mgs_2$. Since m is a mono, we have that $gs_1 = gs_2$. By the universal property of the kernel pair of g , there exists a unique morphism $k : X \rightarrow \text{Eq}$ such that $r_1k = s_1$ and $r_2k = s_2$.

$$\begin{array}{ccccccc} & & & & s_2 & & \\ & & & & \searrow & & \\ X & & & & & & A \\ & \searrow k & & & & & \downarrow f \\ & \text{Eq} & \xrightarrow{r_2} & & & & B \\ & \downarrow r_1 & & & & & \downarrow m \\ & A & \xrightarrow{f} & & & & Q \\ & \nearrow s_1 & & & & & \nearrow g \\ & & & & g & & \end{array}$$

□

Proposition A.0.11. *If a morphism f admits a factorization $f = mg$, with m mono, then f and g share the same kernel.*

Proof. Let (Ker, k) be the kernel pair of $f : A \rightarrow B$ and let $g : A \rightarrow Q$ be a morphism such that $f = mg$ with m mono. We have that $fk = 0$, hence $m g k = 0 = m0$. Since m is a mono, we have that $gk = 0$. Conversely, if (Ker, k) is the kernel of g , then we have that $gk = 0$. Just composing with

m we have that $mgk = m0$. Hence $fk = 0$.

$$\begin{array}{ccccc} \text{Ker} & \xrightarrow{k} & A & \xrightarrow{f} & B \\ & & \searrow g & & \nearrow m \\ & & & Q & \end{array}$$

Now we want to prove that if the universal property of the kernel holds for f , then it also holds for g . Let k' be morphisms such that $gk' = 0$. Just composing with m , we have that $mgk' = m0$. Hence $fk' = 0$. By the universal property of the kernel pair of f , there exists a unique morphism $t : X \rightarrow \text{Ker}$ such that $kt = k'$. Conversely, consider k' be a morphism such that $fk' = 0$, and suppose that the universal property holds for g . We have that $fk' = mgk' = 0 = m0$. Since m is a mono, we have that $gk' = 0$. By the universal property of the kernel pair of g , there exists a unique morphism $t : X \rightarrow \text{Ker}$ such that $kt = k'$.

$$\begin{array}{ccccc} \text{Ker} & \xrightarrow{k} & A & \xrightarrow{f} & B \\ \uparrow t & \nearrow k' & \searrow g & & \nearrow m \\ X & & & Q & \end{array}$$

□

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